

Equilibrium Uniqueness in Entry Games with Combinatorial Actions and Endogenous Pricing^{*}

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Abstract

Many entry settings—multinational production, retail chains, cloud infrastructure—share three features. Firms choose which subset of markets to enter; entry changes market structure; market structure feeds back into markups and entry incentives. This three-way circularity leaves equilibria potentially non-unique. We prove uniqueness of Bayesian Nash equilibrium in threshold strategies for combinatorial entry games with private fixed-cost shocks and endogenous markups, under a single primitive-level condition on the noise scale of private costs. The condition takes a *bathtub* shape: uniqueness tightens at both extremes of the noise spectrum and loosens only at intermediate scales where entry decisions are most coupled. The large-noise tail follows directly from density dampening, while the near-complete-information tail is established on localized non-marginal regions. We verify the condition in two canonical environments: Fréchet–CES demand with Atkeson–Burstein markups and Cournot pricing, and nested logit with Bertrand pricing. Applied to global cloud infrastructure (five hyperscalers, 42 countries), endogenous variable markups sustain 13 additional firm–market pairs and redistribute roughly one-third of producer surplus from leaders to mid-ranked rivals relative to a constant-markup benchmark. Three falsification tests discipline the calibration: suppressing the firm-productivity wedge costs 7.7 percentage points of in-sample fit; reassigning firm heterogeneity attenuates the endogenous-markup mechanism by 15% but preserves its sign; a firm-drop hold-out achieves a 94.3% activation hit rate.

Keywords: entry games, equilibrium uniqueness, combinatorial discrete choice, Bayesian Nash equilibrium, variable markups, supermodular games

JEL Codes: C72, C62, C63, L13, F12

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1. INTRODUCTION

Equilibrium multiplicity is a central obstacle in entry games. When entry decisions are strategic substitutes, multiple configurations of active firms can be mutually consistent, creating well-known difficulties for estimation and counterfactual analysis. Three literatures have made progress on different facets of this problem. [Espin-Sanchez et al. \(2023\)](#) prove uniqueness for *binary* entry games with private costs, but cannot handle within-firm complementarities across actions. The combinatorial discrete choice (CDCP) literature ([Jia, 2008](#); [Arkolakis et al., 2023](#)) handles subset selection with complementarities, but assumes constant markups. The variable-markup literature ([Atkeson and Burstein, 2008](#); [Edmond et al., 2015](#)) endogenizes pricing, but takes the set of active firms as given. In many important settings—multinational production, retail chain location, platform entry—all three features interact simultaneously: firms choose *which subset* of markets to enter, entry changes market structure, and market structure feeds back into markups and entry incentives. Neither the entry game, the CDCP, nor the pricing game can be solved in isolation.

This paper proves a uniqueness theorem for threshold-strategy Bayesian Nash equilibria in combinatorial entry games with endogenous markups, and then verifies its hypotheses in two demand-and-pricing environments. The argument reduces equilibrium uniqueness to the contraction of a best-response map on the space of site-firm activation probabilities. Three economic forces make this map a contraction. *Density dampening*: diffuse private-cost beliefs smooth how one firm’s activation probability responds to shifts in rivals’ entry. *Marginal-value sensitivity*: bounded changes in profit per unit change in the rival pricing equilibrium limit how hard the circularity can push. *Threshold location*: when the game is either highly noisy or locally non-marginal, entry cutoffs sit far from the mode of the cost distribution and the map is locally flat. The resulting contraction strength has a distinctive *bathtub* shape in the noise scale: the diffuse-noise tail is automatic, while the near-complete-information tail requires localization to non-marginal regions; a single peak value below one then certifies uniqueness at every noise level. We verify these hypotheses in a Fréchet–CES environment with Atkeson–Burstein markups under Cournot pricing, which anchors the quantitative application, and in a nested-logit environment with Bertrand pricing, which confirms that the uniqueness engine is structural rather

than CES-specific. The peak value is numerically computable from primitives, providing a modular route to uniqueness for the broader class of combinatorial entry games whose pricing block admits a smooth selection.

We apply the framework to global cloud infrastructure entry, where five hyperscaler providers choose which of 42 countries to establish cloud regions in. The headline finding is distributional. Endogenous Atkeson–Burstein markups sustain 13 additional firm–market pairs relative to a constant-markup benchmark, concentrated among mid-ranked providers whose shares are highest under CES. Aggregate producer surplus falls by 31.8% when AB markups are replaced by the constant CES markup, with roughly one-third of producer surplus redistributed from leaders to mid-ranked rivals. A falsification test (Spec E) reassigns firm heterogeneity away from the productivity wedge and recovers the same redistribution sign with a 15% attenuation, isolating the markup mechanism from firm-heterogeneity parametrization. Two further exercises speak to baseline market structure rather than to pricing—they are robust across pricing regimes. EU compliance-cost shocks are regressive in market-structure terms. The binding constraint in digital-fragmentation counterfactuals is feasible-market access rather than trade costs: reducing China’s entry cost by 90% without feasible-set changes produces zero new entry. The game-theoretic results extend to other combinatorial entry settings—retail chains (Jia, 2008), platforms, or technology adoption—whenever the pricing block satisfies the hypotheses verified here.

Entry games and equilibrium selection. This paper contributes primarily to the literature on equilibrium uniqueness in entry games. The empirical entry literature, beginning with Bresnahan and Reiss (1991) and Berry (1992), has long confronted multiplicity as a barrier to estimation. Berry restricts equilibrium selection to obtain a tractable likelihood. Ciliberto and Tamer (2009) use partial identification to bound parameters without selecting an equilibrium. Seim (2006) introduces endogenous product-type choices, a structure related to combinatorial entry. Igami and Yang (2016) estimate a dynamic model with combinatorial product portfolios using simulation methods that must navigate multiplicity. Closest to our setting, Espin-Sanchez et al. (2023) prove BNE uniqueness in static entry games where each firm makes a binary enter-or-not decision, with one entry cutoff per firm. Many real-world entry decisions are combinatorial: firms choose which subset of markets, products, or technologies to serve, not simply whether to enter. The threshold

system then carries one cutoff per firm-site pair, and within-firm complementarities across actions become first-order. We extend their uniqueness result to this larger action space, in which within-firm actions interact through combinatorial-discrete-choice supermodularity and payoffs depend on an endogenous pricing equilibrium (Theorem 1, globalized across noise levels in Corollary 1 and extended to common cost shocks in Proposition 8). The two results are complementary. [Espin-Sanchez et al.](#) cover general payoffs with binary actions; we handle combinatorial actions under CES/AB payoffs; neither nests the other. Three results sharpen the distinction. First, a small-economy example (Section 4.4) constructs a configuration in which the binary threshold system does not apply, but the combinatorial structure still resolves uniqueness by internalizing within-firm complementarities. Second, the pricing feedback is quantitatively significant: its strength, measured by the spectral radius of the pricing-Jacobian, reaches 0.574 at stressed parameterizations (Appendix K). Third, endogenous variable markups *expand* the uniqueness region relative to constant markups (Appendix K); markup compression dampens the dominant-firm responses that drive multiplicity under constant markups.

A key negative result motivates the present approach. [Doraszelski and Satterthwaite \(2010\)](#) introduce private scrap values in dynamic Ericson–Pakes models. They obtain cutoff equilibria but explicitly demonstrate that multiplicity persists. Private information alone does not pin down the number of equilibria. The pricing contraction (Proposition 5) supplies the missing structure in our setting: it bounds how activation changes feed back through markups, converting the threshold system into a contraction. An alternative selection mechanism is the global-games approach of [Harrison and Jara-Moroni \(2021\)](#), who extend iterated-dominance arguments to binary-action games with strategic substitutes. Their approach requires binary action spaces and strict payoff asymmetry across player pairs. The present paper handles combinatorial actions and operates through pricing contraction rather than dominance propagation.

Quantitative entry with combinatorial or multi-location choice. A parallel quantitative literature estimates entry models where firms choose portfolios of products or locations. These studies typically restrict attention to equilibria that can be computed without resolving multiplicity. [Holmes \(2011\)](#) estimates a dynamic model of Wal-Mart’s spatial diffusion, using a perfect-foresight restriction to sidestep multiplicity when store-level economies

of density generate combinatorial interdependence. [Nishida \(2015\)](#) estimates a static multi-store network-choice model for convenience stores in Okinawa, selecting among equilibria via a maximum-profit rule. [Wollmann \(2018\)](#) studies equilibrium product-line choice in commercial trucks under oligopolistic competition; [Fan and Yang \(2020\)](#) does the same for smartphones. Both rely on equilibrium-selection assumptions to keep counterfactuals tractable. [Weintraub et al. \(2008\)](#) propose oblivious equilibrium for many-firm dynamic industries, approximating Markov-perfect behavior by treating the competitive environment as a long-run average. Relative to these, we provide primitive conditions on preferences and cost heterogeneity under which the static combinatorial entry game has a unique equilibrium. Counterfactuals in this class of applications can then be computed without invoking selection rules or many-firm approximations. The practical payoff is that comparative statics of the counterfactual activation profile inherit identification from model primitives rather than from the selection rule—an asset when the counterfactual perturbs the object (entry costs, markup structure) through which that rule selects.

Supermodular games. The lattice-theoretic approach connects to the literature on supermodular games ([Milgrom and Roberts 1990](#); [Vives 1990](#); [Milgrom and Shannon 1994](#); [Topkis 1998](#); [Amir 1996](#)). The challenge here is that the cross-player interaction exhibits strategic *substitutes* rather than complements: when one firm enters more markets, rivals’ incentives to enter weaken. We restore the Tarski fixed-point structure by composing the best-response map with itself; the squared map is monotone even when the single step is not. [Athey \(2001\)](#) establishes existence of pure-strategy equilibria under a single-crossing condition in own action and type. The within-player complementarity in our setting satisfies her conditions, but across-player substitutes violate single-crossing in rivals’ actions, requiring the bounding argument of Proposition 3.

Combinatorial discrete choice. [Jia \(2008\)](#) studies store location with strategic substitutes; [Arkolakis et al. \(2023\)](#) formalize the general combinatorial-discrete-choice framework. [Antràs et al. \(2017\)](#) prove supermodularity under constant CES markups when the elasticity of substitution exceeds the Fréchet shape parameter. Our extension (Proposition 2) shows that the same demand-curvature condition bounds the additional feedback introduced when markups respond to activation. Supermodularity holds analytically up

to a closed-form threshold on the largest firm’s market share. Above the threshold, a Jia–Tarski gap check (Remark 6) disciplines the upper region; the check passes at every iteration of the cloud calibration.

Variable markups and multinational production. The application draws on two literatures. The variable-markup literature (Atkeson and Burstein 2008; Edmond et al. 2015; Gaubert and Itskhoki 2021) endogenizes pricing but takes the set of active firms as given. The multinational-production literature (Tintelnot 2017; Antràs et al. 2026; Castro-Vincenzi 2024) endogenizes the active set but imposes constant CES markups. By closing the loop between activation and pricing, we make the three-way circularity explicit (Figure 2) and show that the pricing contraction bounds the resulting feedback, preserving both supermodularity and equilibrium uniqueness. Verification uses a spectral-radius condition on the pricing block (Proposition 5) and a direct enumeration certificate at the calibrated cost vector (Proposition 7); Theorem 2 consolidates them into the CES–AB uniqueness statement used in the application. The application itself follows the identification strategy of Arkolakis et al. (2023): demand and firm scale are fixed externally, while fixed costs are calibrated from entry moments across action spaces as large as a trillion candidate subsets per firm (algorithm comparison in Appendix J). The static game supplies the stage-game structure for dynamic oligopoly models such as Ericson and Pakes (1995) and Ifrach and Weintraub (2017). In the estimation of such dynamic games, Aguirregabiria and Mira (2007) develop sequential methods that require equilibrium selection at each iteration. Bajari et al. (2007) propose forward-simulation methods that bypass equilibrium computation altogether. The uniqueness conditions derived here offer a third path: they guarantee that the static stage game has a single equilibrium to estimate. The nested-logit verification in Section 4.2 additionally draws on single-stage Bertrand-pricing uniqueness under (nested) logit as a black-box input (Anderson et al., 1992; Mizuno, 2003; Konovalov and Sándor, 2010; Nocke and Schutz, 2018).

2. ENVIRONMENT

This section states the entry game at the level of abstraction used by the uniqueness theorem of Section 4. The abstract environment is instantiated in two specific demand-

and-pricing environments: Fréchet–CES–Atkeson–Burstein with Cournot pricing (Environment A, Section 4.1), which supplies the quantitative application of Section 6; and nested logit with Bertrand pricing (Environment B, Section 4.2), which shows that the uniqueness engine is a property of the threshold system rather than of the CES specification.

2.1 Players, actions, and private fixed-cost shocks

There are N markets indexed by $n \in \{1, \dots, N\}$ and F firms indexed by $f \in \mathcal{F} = \{1, \dots, F\}$. Each firm f is endowed with a menu of *available sites* $\mathcal{A}_f \subseteq \{1, \dots, N\}$. Firm f 's action is an *activation set* $\mathcal{I}_f \subseteq \mathcal{A}_f$, a subset of its menu; its action space has cardinality $2^{|\mathcal{A}_f|}$. Let $M \equiv \sum_f |\mathcal{A}_f|$ denote the total number of site-firm pairs, indexed by $m = (i, f)$.

For each $m = (i, f)$, firm f privately observes a fixed-cost shock ε_{if} drawn i.i.d. across pairs from a CDF $G_\sigma(\varepsilon) = G_0(\varepsilon/\sigma)$, where G_0 is a standardized continuously-differentiable CDF on $\mathbb{R}$ with density g_0 and scale parameter $\sigma > 0$. The total fixed cost of activating site i for firm f is $(\phi_i + \varepsilon_{if})w_i$, where ϕ_i is a common site- i shifter and $w_i > 0$ is the site- i wage. Firm f observes its own shock profile but not rivals'.

2.2 Threshold strategies and activation probabilities

Attention is restricted to *site-independent threshold strategies*: for firm f , there exist cutoffs $(\bar{\varepsilon}_{if})_{i \in \mathcal{A}_f}$ such that site i is activated if and only if $\varepsilon_{if} < \bar{\varepsilon}_{if}$, independently across sites within the firm. The cutoffs depend on rival activation probabilities and on own expected within-firm complementarities, but not on the realized own-site shocks at other sites. Remark 8 (Section 4.1.9) discusses why this restriction is substantive and gives its economic interpretation.

Given cutoffs $\bar{\varepsilon}_{if}$, the *activation probability* of site-firm pair (i, f) is $v_{if} = G_\sigma(\bar{\varepsilon}_{if}) \in [0, 1]$. Stacking across pairs yields a vector $\mathbf{v} \in [0, 1]^M$; we use the single-index notation v_m interchangeably with v_{if} .

2.3 Payoffs and the static Bayesian Nash equilibrium

Firm f 's variable profit depends on its own activation set and on a *payoff-relevant sufficient statistic* D_{-f} that summarizes rivals' behavior through the environment's pricing block.

In Environment A, D_{-f} is the vector of rival demand indices (4.11); in Environment B, it is the nested-logit aggregator of rival quality–price indices. Given D_{-f} , the environment supplies a unique pricing equilibrium that maps $(\mathcal{I}_f, D_{-f})$ to variable profit $V_f(\mathcal{I}_f, D_{-f})$; total profit is

$$\pi_f(\mathcal{I}_f; D_{-f}, \varepsilon_f) = V_f(\mathcal{I}_f, D_{-f}) - \sum_{i \in \mathcal{I}_f} (\phi_i + \varepsilon_{if}) w_i. \quad (2.1)$$

Under site-independent threshold strategies and i.i.d. shocks, firm f 's optimal cutoff for site i is the expected marginal value of that site, integrated over own-firm sibling shocks and evaluated at rivals' activation probabilities. Denote this scalar threshold index by $v_m(v)$ for $m = (i, f)$; the formal construction appears in Section 3.1. The best-response threshold satisfies $\bar{\varepsilon}_{if} = v_m(v)$, so the threshold system collapses to

$$v_m = G_0\left(\frac{v_m(v)}{\sigma}\right), \quad m = 1, \dots, M. \quad (2.2)$$

A *site-independent threshold-strategy Bayesian Nash equilibrium* is a fixed point $v^* \in [0, 1]^M$ of (2.2).

2.4 Three-way circularity and roadmap

The game exhibits a three-way circularity: activation decisions determine the payoff-relevant statistic D_{-f} that each firm faces; D_{-f} determines pricing and market structure; and market structure determines the profitability of each activation decision. The environment-specific middle link is what prior literature has typically foreclosed by assumption. Antràs et al. (2017) and Jia (2008) close it by imposing constant markups; Atkeson and Burstein (2008) close it by taking the active set as given; Espin-Sanchez et al. (2023) close it by restricting actions to binary entry. Figure 2 illustrates the loop as instantiated in Environment A.

The remainder of the paper proceeds in four steps. Section 4.1 instantiates the abstract primitives in Environment A (Fréchet–CES–AB with Cournot pricing), specifying demand, production, and pricing; Environment B (nested logit with Bertrand pricing) is developed in Section 4.2. Section 4 states the abstract uniqueness theorem: the Jacobian of the threshold map factors as $\text{diag}(g_0(v/\sigma)/\sigma) J_v(v)$, and when the contraction modulus $\gamma(\sigma) \equiv \sup_v \|J_{T_v}(v)\|_\infty$ stays below one the fixed point of (2.2) is unique. Verifying the theorem

requires a C^1 threshold index v and a uniform bound on J_v ; both are supplied explicitly in Environments A and B. Section 5 presents a three-layer computational algorithm used in the quantitative application, and Section 6 calibrates Environment A to global cloud infrastructure entry.

3. ABSTRACT UNIQUENESS THEOREM

This section states the paper’s main result: a threshold-strategy uniqueness theorem that applies to any combinatorial entry game whose marginal-value map is C^1 , whose operator-norm Jacobian is uniformly bounded, and whose complete-information game has a strict profit gap. Section 3.1 records the formal statement and proof; Section 3.2 develops the bathtub geometry of the contraction modulus that makes the economic mechanism explicit. Section 4 then verifies these structural hypotheses in two canonical environments—Fréchet–CES with Atkeson–Burstein markups (Section 4.1, where the abstract v is built from the nested-CES pricing equilibrium) and nested logit with Bertrand pricing (Section 4.2)—and discusses further environments (Section 4.3).

3.1 Abstract Threshold-Uniqueness Framework

The uniqueness argument below operates at a level of abstraction broader than the Fréchet–CES–AB environment of Section 4.1. What the nested-CES demand structure supplies is a *tractable* verification environment in which the smoothness of the marginal-value map, the contraction of the pricing block, and the complete-information gap can all be checked in closed form. None of those ingredients is specific to CES: any Bayesian combinatorial entry game whose threshold map is smooth and whose contraction modulus is below one admits a unique threshold-strategy BNE. This subsection records the abstract result; Section 4.1.9 supplies the CES–AB verification; Section 4.2 supplies a nested-logit verification on the consumer side; Section 4.3 discusses further environments.

Abstract setup. Fix a combinatorial entry game with F firms and site sets $(\mathcal{A}_f)_{f=1}^F$; let $M = \sum_f |\mathcal{A}_f|$ be the total number of site-firm pairs. Each firm f privately observes site-specific shocks $\{\varepsilon_{if}\}_{i \in \mathcal{A}_f}$ i.i.d. across (i, f) with CDF $G_\sigma(\varepsilon) = G_0(\varepsilon/\sigma)$, where G_0 is a standardized continuously-differentiable CDF on $\mathbb{R}$ with density g_0 and scale parameter

$\sigma > 0$. The symbol σ in the abstract framework plays the role of σ_ε in Assumption 2; we retain the shorter notation here because the abstract theorem never interacts with the CES elasticity σ of Section 4.1, which appears only in the concrete verification of Section 4.1.9. Attention is restricted to the class of site-independent threshold strategies defined in the CES–AB setting by Assumption 2 and its nested-logit analog in Assumption 3 below. Let $v \in [0, 1]^M$ collect the site-firm activation probabilities induced by a profile of thresholds, and let $v : \mathcal{N} \rightarrow \mathbb{R}^M$ be a C^1 map on an open neighborhood $\mathcal{N}$ of $[0, 1]^M$ returning the scaled expected per-site marginal value at an activation-probability profile: $v_m(v)$ aggregates the firm’s own expected within-firm complementarities over sibling shocks and the rival pricing equilibrium induced by v into a single scalar threshold index. The threshold map is

$$T_\sigma : [0, 1]^M \rightarrow [0, 1]^M, \quad T_{\sigma,m}(v) = G_0\left(\frac{v_m(v)}{\sigma}\right), \quad m = 1, \dots, M, \quad (3.1)$$

and the contraction modulus is

$$\gamma(\sigma) \equiv \sup_{v \in [0, 1]^M} \|J_{T_\sigma}(v)\|_\infty. \quad (3.2)$$

Every site-independent threshold-strategy BNE corresponds to a fixed point of T_σ ; the verification environments of Sections 4.1.9 and 4.2 make this correspondence explicit. Proofs of Theorem 1, Lemmas 1–2, and Proposition 1 are collected in Appendix A.

Theorem 1 (Abstract Threshold-Strategy Uniqueness at Fixed Noise). *Let G_0 be a continuously differentiable CDF on $\mathbb{R}$ and let v be C^1 on an open neighborhood of $[0, 1]^M$. Then*

- (i) *the threshold map T_σ defined in (3.1) has at least one fixed point on $[0, 1]^M$;*
- (ii) *its Jacobian admits the factorization*

$$J_{T_\sigma}(v) = \text{diag}\left(\frac{g_0(v_m(v)/\sigma)}{\sigma}\right)_{m=1}^M J_v(v); \quad (3.3)$$

- (iii) *if $\gamma(\sigma) < 1$, the fixed point is unique.*

Corollary 1 (Global-in- σ Uniqueness). *Under the hypotheses of Theorem 1, if $\gamma_{\max} \equiv \sup_{\sigma > 0} \gamma(\sigma) < 1$, then the threshold-strategy BNE is unique for every $\sigma > 0$.*

The proof is immediate: any $\sigma > 0$ satisfies $\gamma(\sigma) \leq \gamma_{\max} < 1$, so Theorem 1(iii) applies at that σ . The finite peak $\gamma_{\max}$ is the single primitive-level number whose control certifies uniqueness uniformly across σ ; the verification sections below produce explicit decompositions of $\gamma(\sigma)$ into density dampening and marginal-value sensitivity that render $\gamma_{\max} < 1$ verifiable by evaluation on a grid of σ .

Lemma 1 (Large-Noise Tail). *Assume additionally that $\|g_0\|_\infty < \infty$ and $K_v \equiv \sup_{v \in [0,1]^M} \|J_v(v)\|_\infty < \infty$. Then $\gamma(\sigma) \leq \|g_0\|_\infty K_v / \sigma$, and in particular $\gamma(\sigma) \rightarrow 0$ as $\sigma \rightarrow \infty$.*

The analog of Lemma 1 as $\sigma \rightarrow 0$ is subtler. The modulus $\gamma(\sigma)$ is a global supremum over all off-equilibrium profiles $v \in [0,1]^M$, so a strictly positive complete-information gap at the equilibrium profile does not automatically push the density factor to zero at every off-equilibrium v : some v may keep $|v_m(v)|$ small even when the equilibrium profile has strictly separated thresholds. The following lemma closes the gap by *localizing* the contraction check to an invariant set on which all threshold indices are uniformly bounded away from zero.

Lemma 2 (Small-Noise Localization). *Suppose there exists a family $\{S_\sigma\}_{0 < \sigma < \sigma_0} \subseteq [0,1]^M$ of nonempty, closed, convex sets such that:*

- (i) $T_\sigma(S_\sigma) \subseteq S_\sigma$ for every $\sigma \in (0, \sigma_0)$ (invariance);
- (ii) every fixed point of T_σ in $[0,1]^M$ lies in S_σ (fixed-point inclusion);
- (iii) there exists $b > 0$ with $\inf_{v \in S_\sigma} \min_m |v_m(v)| \geq b$ for every $\sigma \in (0, \sigma_0)$ (uniform non-marginality);
- (iv) $|z|g_0(z) \rightarrow 0$ as $|z| \rightarrow \infty$ (tail regularity on the shock density);
- (v) $K_S \equiv \sup_{0 < \sigma < \sigma_0} \sup_{v \in S_\sigma} \|J_v(v)\|_\infty < \infty$ (uniform J_v bound on S_σ).

Then $\sup_{v \in S_\sigma} \|J_{T_\sigma}(v)\|_\infty \rightarrow 0$ as $\sigma \rightarrow 0$. In particular, for all σ sufficiently small, T_σ is a contraction on $(S_\sigma, \|\cdot\|_\infty)$, and by (ii) its fixed point is globally unique.

Proposition 1 (Lipschitz Bound on the Contraction Modulus). *Assume g_0 is a standard normal density, $K_v \equiv \sup_v \|J_v(v)\|_\infty < \infty$, and $v(v)$ is σ -independent. Then γ is continuously differentiable on $(0, \infty)$, and for every $\sigma > 0$,*

$$|\gamma'(\sigma)| \leq \frac{c_g K_v}{\sigma^2}, \quad c_g \equiv \sup_z |\phi(z)(1 - z^2)| = \phi(0) = \frac{1}{\sqrt{2\pi}} \approx 0.3989. \quad (3.4)$$

Consequently, for any $0 < \sigma_\ell < \sigma_u < \infty$,

$$\sup_{\sigma \in [\sigma_\ell, \sigma_u]} \gamma(\sigma) \leq \max\{\gamma(\sigma_\ell), \gamma(\sigma_u)\} + \frac{c_g K_v}{2} \left(\frac{1}{\sigma_\ell} - \frac{1}{\sigma_u} \right). \quad (3.5)$$

Remark 1 (Numerical grid certificate). Proposition 1 promotes any finite grid evaluation $\{\gamma(\sigma_i)\}_{i=1}^K$ into a certified upper bound on $\sup_\sigma \gamma(\sigma)$ via (3.5) applied to each consecutive interval $[\sigma_i, \sigma_{i+1}]$, plus the tail bounds $\gamma(\sigma) \leq \|g_0\|_\infty K_v / \sigma$ (Lemma 1) on $[\sigma_K, \infty)$ and a symmetric Lemma 2-type bound on $(0, \sigma_1]$. The certificate requires only (a) evaluating γ at K grid points and (b) bounding $K_v = \sup_v \|J_v(v)\|_\infty$ from primitives. In practice K_v is tight once σ is large enough that $h_{m^*} \approx c_g / \sigma$, so $K_v \approx \sigma_{\max} \gamma_{\max} / c_g$ is a self-consistent estimate that $K = 5$ – 10 grid points certify to arbitrary tolerance.

Remark 2 (Verification strategy across environments). Theorem 1, Corollary 1, and Lemmas 1–2 reduce the uniqueness problem to three structural checks on the demand and payoff environment:

- (a) *Smooth threshold index.* The marginal-value map $v(v)$ must be C^1 . Under Fréchet–CES–AB, this follows from Proposition 5: the pricing equilibrium selection $v \mapsto (p, s)$ is smooth whenever $\rho(\Lambda) < 1$, and expected profits integrate smoothly over the IID sibling shocks. Under nested logit (Section 4.2), v inherits smoothness from the Bertrand best-response map under log-concavity of the logit shocks.
- (b) *Bounded J_v .* The operator norm $\|J_v\|_\infty$ must be uniformly finite. Both CES–AB (Appendix G, Proposition 10) and nested logit deliver explicit Lipschitz bounds involving the Perron–Frobenius amplification $1/(1 - \rho(\Lambda))$ of the pricing block.
- (c) *Small- σ localization.* A positive complete-information gap is useful for constructing a candidate non-marginal region, but it does not by itself verify the invariance and fixed-point-inclusion requirements of Lemma 2. The environment-specific verification must therefore construct or numerically check a family S_σ on which $\min_m |v_m(v)|$ is uniformly bounded away from zero and all fixed points of T_σ are contained.

The abstract theorem is the uniqueness engine; each verification environment does the application-specific work of constructing a smooth v , bounding J_v , and supplying the non-marginal localization.

3.2 The Bathtub Modulus: Private-Cost Noise as Uniqueness Discipline

The factorization (3.3) yields the paper’s central economic insight, which we separate here because it is conceptually distinct from the mechanical uniqueness proof. The contraction modulus $\gamma(\sigma)$ has a *bathtub* shape in the noise scale σ : uniqueness tightens at both ends of the private-information spectrum, and loosens only at intermediate scales where entry decisions are most coupled.

Two vanishing tails. The two tails dissipate multiplicity through different channels:

- *Large σ (diffuse beliefs).* The density prefactor $g_0(\cdot)/\sigma$ vanishes (Lemma 1): each marginal shift in rivals’ activation probabilities is smoothed across a wide distribution of thresholds, so the best-response map becomes nearly flat. Noise itself is the discipline.
- *Small σ (near-complete information).* When the complete-information equilibrium is unique with strictly positive marginal-value gap δ_v , that gap helps construct a non-marginal localized region. On such a region, Lemma 2 pushes the relevant thresholds into the tails of g_0 , so activation probabilities become locally insensitive to perturbations of rivals’ strategies.

Multiplicity therefore concentrates at intermediate σ where some thresholds sit near the density mode. The peak $\gamma_{\max} \equiv \sup_{\sigma>0} \gamma(\sigma)$ is the single number that witnesses global uniqueness: if $\gamma_{\max} < 1$, the BNE is unique for every $\sigma > 0$ (Corollary 1).

Verification strategy. The condition $\gamma_{\max} < 1$ is verifiable from primitives without enumerating equilibria: evaluate $\gamma(\sigma)$ on a log-spaced grid covering $(0, \infty)$ and take the maximum. Because γ is continuously differentiable in σ (Proposition 1) with derivative bounded by $c_g K_v / \sigma^2$, a finite grid upgrades to a *certified* upper bound on $\sup_{\sigma} \gamma(\sigma)$ via (3.5) on each interval plus the closed-form tail bounds of Lemmas 1–2. Figure 1 illustrates the bathtub in a small economy ($N = 3, F = 4$) where the Jacobian can be enumerated exactly; the peak $\gamma_{\max} = 0.951 < 1$ lies at $\sigma/\bar{\phi} \approx 0.3$. The same numerical procedure applied to the calibrated cloud economy ($N = 42, F = 5$; Section 6) yields $\gamma_{\max} = 0.842 < 1$ at $\sigma_{\varepsilon}/\bar{\phi} = 1.4$ (Table 3; margin 0.158 below one). The grid is 10 points on $\sigma_{\varepsilon}/\bar{\phi} \in \{0.100, 0.316, 0.500, 0.700, 1.000, 1.400, 2.000, 2.800, 3.162, 10.000\}$, built by re-

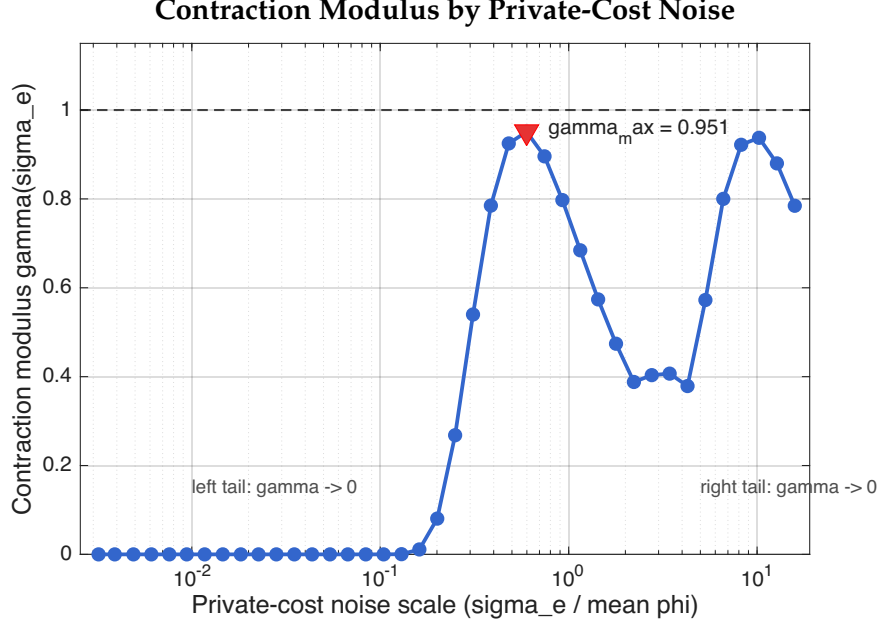


Figure 1: Contraction modulus $\gamma(\sigma)$ for a small economy ($N = 3, F = 4$). The plotted $\gamma(\sigma)$ is the global supremum over $\nu \in [0, 1]^M$, not the restricted supremum on the invariant set of Lemma 2; the two agree whenever the complete-information gap is strict, which holds here. The bathtub shape confirms Corollary 1: uniqueness tightens at both extremes of the noise scale, and the peak $\gamma_{\max} = 0.951$ (marked) is below the $\gamma = 1$ threshold. See Appendix K for scaling across economy sizes.

fining the five log-spaced anchors $\{0.1, 0.316, 1, 3.162, 10\}$ with five additional points that bracket the peak region where the coarse grid showed the steepest rise.¹ Combined with Proposition 1 and the implied $K_v \approx 0.283$, the grid certifies BNE uniqueness at every σ_ε in a neighborhood of the calibrated scale.

4. VERIFICATION ENVIRONMENTS

The abstract uniqueness theorem (Theorem 1, Corollary 1) reduces uniqueness of threshold-strategy BNE to three structural checks on the demand and payoff environment: a smooth marginal-value map $v(\nu)$, a uniformly bounded J_v , and a non-marginality family $\{S_\sigma\}$ on which the complete-information gap is uniformly positive. This section verifies these hypotheses in two canonical environments—Fréchet-CES with Atkeson–Burstein markups

¹An earlier coarse-only evaluation located the peak at $\sigma_\varepsilon/\bar{\phi} = 1.0$ with $\gamma = 0.821$; the refinement raised the located peak to $\sigma_\varepsilon/\bar{\phi} = 1.4$ with $\gamma = 0.842$. The 0.021 revision is well within the Lipschitz-derivative bound from Proposition 1 across the refined $[\sigma_\varepsilon/\bar{\phi} \in (1.0, 1.4)]$ interval: $(c_g K_v/2)(1/\sigma_\ell - 1/\sigma_u) \approx 0.168$ with $c_g = 1/\sqrt{2\pi} \approx 0.399$ and $K_v \approx 0.283$.

under Cournot pricing (Environment A, Section 4.1) and nested logit with Bertrand pricing (Environment B, Section 4.2)—and outlines further environments to which the same template applies (Section 4.3). Section 4.4 illustrates the bathtub geometry on a small economy where the Jacobian can be enumerated in closed form.

Environment A is the running example: it anchors the quantitative application of Section 6 and, through its Fréchet cost structure and AB perceived elasticities, exhibits all three channels—density dampening, marginal-value sensitivity, and threshold location—in analytically tractable form. Environment B shows that the same uniqueness verification goes through with a different micro-foundation for demand and a different pricing concept, isolating the abstract theorem from CES-specific features.

4.1 Environment A: Fréchet–CES–AB with Cournot Pricing

Environment A instantiates the abstract environment of Section 2 under Fréchet productivity draws, nested CES demand, and Atkeson–Burstein Cournot pricing. This environment supplies the quantitative application of Section 6 and one of the two verifications of the abstract uniqueness theorem (Section 4.1.9). Figure 2 shows the three-way circularity instantiated in this environment.

4.1.1 Setting

There are N countries, indexed by $n \in \{1, \dots, N\}$. Each country serves as both a potential production site and a destination market. Wages w_n are exogenous.

There are F firms, indexed by $f \in \mathcal{F} = \{1, \dots, F\}$, that produce differentiated varieties in a single sector. Each firm f is characterized by two primitives: (i) a *total factor productivity* (TFP) $a_f > 0$, and (ii) a *set of available production sites* $\mathcal{A}_f \subseteq \{1, \dots, N\}$; the firm’s strategic decision is which sites in $\mathcal{A}_f$ to activate for production. The private fixed-cost shocks $\{\varepsilon_{if}\}$ and the threshold-strategy restriction follow Section 2; the formal Bayesian-game statement used in the uniqueness argument appears in Section 4.1.9.

4.1.2 Demand

A representative household in each country has income $E_n > 0$ and nested CES preferences. Following Atkeson and Burstein (2008), there is a continuum of sectors indexed by

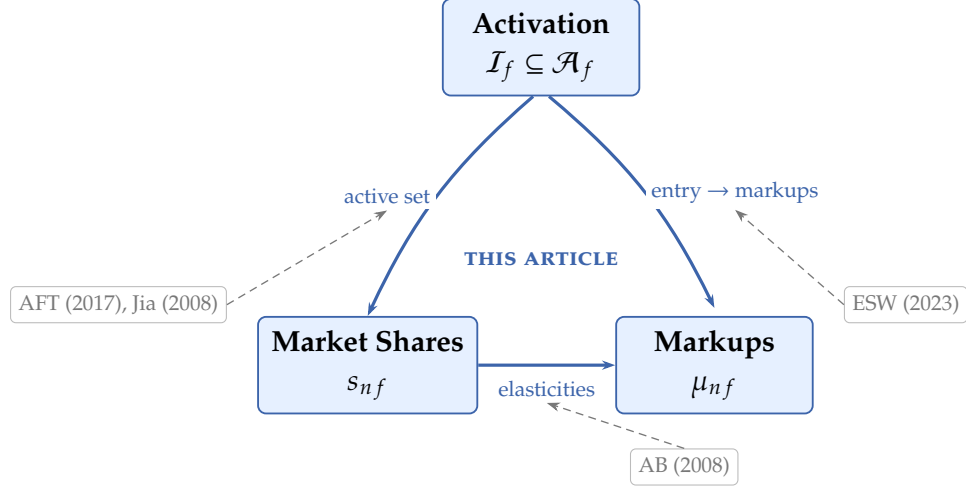


Figure 2: Three-way circularity and the literature. Solid arrows show the full equilibrium loop closed in this article. Prior work addresses subsets: AB (2008), i.e., [Atkeson and Burstein \(2008\)](#), endogenize markups given active firms (bottom edge); AFT (2017), i.e., [Antràs et al. \(2017\)](#), and [Jia \(2008\)](#) solve activation under constant markups (left edge); ESW (2023), i.e., [Espin-Sanchez et al. \(2023\)](#), prove uniqueness for binary entry decisions given a fixed payoff function (right edge; their result covers the activation channel but does not close the pricing loop).

$k \in [0, 1]$, aggregated with elasticity of substitution $\eta > 1$:

$$C_n = \left[\int_0^1 C_{nk}^{\frac{\eta-1}{\eta}} dk \right]^{\frac{\eta}{\eta-1}}. \quad (4.1)$$

The F oligopolistic firms compete in a single differentiated-good sector (say $k = 0$). Because this sector has measure zero in the continuum, its price index P_n does not affect the aggregate price index $\tilde{P}_n$, so the demand shifter Ω_n defined below is truly exogenous to the firms' decisions. Within the differentiated sector, varieties are aggregated through a CES technology with elasticity $\sigma > \eta$:

$$Q_n = \left[\sum_{f \in \mathcal{F}_n} q_{nf}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad (4.2)$$

where $\mathcal{F}_n$ is the set of firms actively supplying market n . Since all production sites can serve all markets via iceberg trade costs, $f \in \mathcal{F}_n$ if and only if $I_f \neq \emptyset$: a firm is present in

every market once it activates at least one site. The within-sector price index is:

$$P_n = \left[\sum_{f \in \mathcal{F}_n} p_{nf}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (4.3)$$

Two-stage budgeting implies that total expenditure on the differentiated sector in market n is:

$$R_n = \Omega_n P_n^{1-\eta}, \quad (4.4)$$

where $\Omega_n \equiv E_n / \tilde{P}_n^{1-\eta} > 0$ collects the household's income and the aggregate price index $\tilde{P}_n$. Because the differentiated sector is measure zero in the continuum, $\tilde{P}_n$ is determined by the remaining sectors and is therefore exogenous to the oligopolistic firms. Firm f 's demand in market n is:

$$q_{nf} = p_{nf}^{-\sigma} P_n^{\sigma-\eta} \Omega_n. \quad (4.5)$$

Assumption 1 (Fixed sectoral expenditure). Sectoral expenditure $R_n = \Omega_n P_n^{1-\eta}$ is treated as exogenous: we fix R_n at its initial value and suppress the price-index feedback $P_n \rightarrow R_n$.

This assumption is maintained throughout the theoretical analysis and the quantitative illustration. The omitted channel—the price-index feedback $P_n \rightarrow R_n = \Omega_n P_n^{1-\eta}$ —has an ambiguous effect on uniqueness. For $\eta > 1$, a lower P_n raises R_n , which increases demand and makes entry more profitable: this positive feedback could either tighten or loosen the contraction condition depending on the relative magnitude of the demand amplification and the pricing dampening. A formal resolution would require bounding the spectral radius of the augmented Jacobian that includes the $P_n \rightarrow R_n$ loop jointly with the pricing contraction; we leave this extension to future work. The Jia–Tarski gap of zero at all tested parameterizations is consistent with the lattice structure used by the fixed- R_n solver, but it does not replace an analytical proof of aggregate supermodularity.

4.1.3 Production Technology and Costs

Firms produce using labor with heterogeneous productivity across production sites. Firm f with TFP a_f delivering one unit of output from production site i to market n faces unit cost:

$$c_{fin}(\omega) = \frac{w_i \tau_{in}}{a_f z(\omega)}, \quad (4.6)$$

where w_i is the wage at site i , $\tau_{in} \geq 1$ is the iceberg trade cost from site i to market n , and $z(\omega)$ is an i.i.d. Fréchet productivity draw with shape parameter $\theta > 0$ across varieties $\omega \in [0, 1]$.

A firm producing at multiple sites sources each variety from the lowest-cost site. As shown by [Eaton and Kortum \(2002\)](#)—and applied at the firm level by [Tintelnot \(2017\)](#) and [Antràs et al. \(2017\)](#)—the Fréchet assumption implies that the firm-level cost index aggregates across active production sites:

$$c_{nf} = \frac{\Gamma_\theta}{a_f} \left[\sum_{i \in \mathcal{I}_f} (w_i \tau_{in})^{-\theta} \right]^{-\frac{1}{\theta}}, \quad (4.7)$$

where $\mathcal{I}_f \subseteq \mathcal{A}_f$ is the set of production sites that firm f activates and $\Gamma_\theta = \Gamma(1 + \frac{1-\sigma}{\theta})^{\frac{1}{1-\sigma}}$ with $\Gamma(\cdot)$ the Gamma function. Higher TFP a_f lowers the cost index uniformly across all markets. Because Γ_θ is common across all firms, we normalize $\Gamma_\theta = 1$ without loss of generality; it can be absorbed into the TFP parameters $\{a_f\}$.

Remark 3 (Cost Complementarity). Equation (4.7) embodies a key complementarity: activating an additional production site i' adds a term $(w_{i'} \tau_{i'n})^{-\theta}$ to the sum inside the brackets, *reducing* the cost index c_{nf} in every market n . This cost reduction is larger when the firm already has fewer active sites—a form of increasing differences in the activation indicators. [Antràs et al. \(2017\)](#) establish that this cost complementarity generates supermodularity in the global sourcing problem under constant CES markups when $(\sigma - 1)/\theta > 1$.

4.1.4 Pricing and Markups

Firms compete à la Cournot: each firm chooses the quantity q_{nf} of its variety to supply to market n , taking rivals' quantities as given. The nested CES demand system with a finite number of firms generates endogenous variable markups through a mechanism first analyzed by [Atkeson and Burstein \(2008\)](#).

Firm f 's revenue share in market n is:

$$s_{nf} \equiv \frac{p_{nf}^{1-\sigma}}{p_n^{1-\sigma}} = \frac{p_{nf} q_{nf}}{\sum_{f'} p_{nf'} q_{nf'}}. \quad (4.8)$$

Under Cournot competition, the firm internalizes its effect on the sectoral aggregate Q_n , yielding a *perceived inverse elasticity* of demand (see [Atkeson and Burstein, 2008](#), Section II):

$$\frac{1}{\epsilon_{nf}} = \frac{1 - s_{nf}}{\sigma} + \frac{s_{nf}}{\eta}. \quad (4.9)$$

The first-order condition $p_{nf}(1 - 1/\epsilon_{nf}) = c_{nf}$ yields the optimal price as a markup over marginal cost:

$$p_{nf} = \mu_{nf} \cdot c_{nf}, \quad \mu_{nf} \equiv \frac{\epsilon_{nf}}{\epsilon_{nf} - 1}. \quad (4.10)$$

Remark 4 (Bounded Markups and the Cannibalization Channel). Equation (4.9) implies that the perceived inverse elasticity is a harmonic-mean weight between the two CES parameters σ and η :

- (i) **Small firm** ($s_{nf} \rightarrow 0$): $\epsilon_{nf} \rightarrow \sigma$, $\mu_{nf} \rightarrow \sigma/(\sigma - 1)$. The firm faces the within-sector CES markup.
- (ii) **Dominant firm** ($s_{nf} \rightarrow 1$): $\epsilon_{nf} \rightarrow \eta$, $\mu_{nf} \rightarrow \eta/(\eta - 1)$. The firm's pricing power is bounded by across-sector substitution.

Markups are bounded: $\mu_{nf} \in [\sigma/(\sigma - 1), \eta/(\eta - 1)]$. Activating additional production sites lowers marginal cost (4.7), which raises output and market share, thereby *compressing* the perceived elasticity and *raising* the markup. But the markup is bounded above by $\eta/(\eta - 1)$, so the self-cannibalization force is finite. This *bounded* cannibalization—a direct consequence of the nested CES structure ([Atkeson and Burstein, 2008](#))—competes against the cost complementarity of Remark 3. Whether supermodularity of the activation problem survives this tension is the central question of Section 4. A second, cross-firm channel arises because activation changes the rival demand index (4.11), feeding back into all firms' markups—a pricing interaction absent under constant markups that creates a new obstacle to uniqueness (Proposition 5).

Remark 5 (Cournot vs. Bertrand). Under Cournot, the perceived inverse elasticity is $1/\epsilon^C = (1 - s)/\sigma + s/\eta$ (equation 4.9). Under Bertrand with the same nested CES demand, the perceived elasticity is $\epsilon^B = \sigma(1 - s) + \eta s$, which differs for interior market shares: the two coincide only at $s = 0$ (both yield σ) and $s = 1$ (both yield η). The theoretical framework—pricing contraction via the interaction matrix, supermodularity

via cost-aggregator convexity—applies to both cases, but the quantitative conditions (the feedback coefficient κ , the per-market threshold, and the contraction modulus $\gamma_{\max}$) must be re-derived using the Bertrand markup function $\mu^B(s) = \epsilon^B/(\epsilon^B - 1)$. Since Bertrand markups are lower than Cournot markups at interior shares, the pricing contraction is typically weaker, but this must be verified case by case. We adopt the Cournot assumption throughout, which is natural for the multinational production application where firms choose production scale and capacity.

Rival demand index. The competitive environment facing firm f in market n is summarized by a *rival demand index*:

$$D_{-f}(n) = \sum_{f' \neq f, f' \in \mathcal{F}_n} p_{nf'}^{1-\sigma}, \quad (4.11)$$

which is a sufficient statistic for competition: firm f needs only $D_{-f}(n)$ to compute its optimal price and variable profit in market n . The firm's market share can be written as:

$$s_{nf} = \frac{p_{nf}^{1-\sigma}}{p_{nf}^{1-\sigma} + D_{-f}(n)}, \quad (4.12)$$

making explicit the dependence of the markup (4.9) on the rival demand index.

4.1.5 The Activation Problem

Each firm f chooses which production sites to activate. Let $I_{fi} \in \{0, 1\}$ be an indicator equal to 1 if firm f activates production site $i \in \mathcal{A}_f$, and 0 otherwise. The firm's *activation set* is:

$$\mathcal{I}_f = \{i \in \mathcal{A}_f : I_{fi} = 1\}. \quad (4.13)$$

Activating production site i incurs a fixed operating cost $\phi_i \cdot w_i$ in units of local labor, where ϕ_i may vary across countries (reflecting, e.g., differences in regulatory or infrastructure costs).

Firm f 's total static profit, given its own activation set $\mathcal{I}_f$ and the equilibrium activation

sets of all rivals $\mathcal{I}_{-f}^*$, is:

$$\pi_f(\mathcal{I}_f; \mathcal{I}_{-f}^*) = \sum_{n=1}^N \left[\text{rev}_{nf}(\mathcal{I}_f, \mathcal{I}_{-f}^*) \cdot \frac{\mu_{nf} - 1}{\mu_{nf}} \right] - \sum_{i \in \mathcal{I}_f} \phi_i w_i, \quad (4.14)$$

where $\text{rev}_{nf} = p_{nf} \cdot q_{nf}$ is firm f 's revenue in market n and $(\mu_{nf} - 1)/\mu_{nf}$ is the variable profit margin. Revenue and markups depend on the full industry activation profile: own activation enters through the cost index (4.7), and rivals' activation sets enter through the rival demand index $D_{-f}(n)$ defined in (4.11).

4.1.6 Static Equilibrium

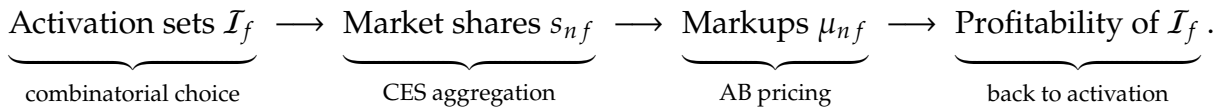
Definition 1 (Static Equilibrium). A *static equilibrium* is a profile $(\{\mathcal{I}_f^*\}_{f \in \mathcal{F}}, \{\mu_{nf}^*\}, \{s_{nf}^*\})$ of activation sets, markups, and market shares such that:

E1. Profit maximization: For each firm f , the activation set $\mathcal{I}_f^*$ solves

$$\mathcal{I}_f^* \in \arg \max_{\mathcal{I}_f \subseteq \mathcal{A}_f} \pi_f(\mathcal{I}_f; \mathcal{I}_{-f}^*). \quad (4.15)$$

E2. Markup consistency: For each (n, f) with $\mathcal{I}_f^* \neq \emptyset$, the markup μ_{nf}^* satisfies the pricing equation (4.10) with perceived elasticity (4.9), and market shares s_{nf}^* are consistent with equilibrium prices via (4.12).

The static equilibrium involves a three-way circularity:



With constant markups (as in Antràs et al., 2017), the last link is severed: activation does not feed back through markups. The theoretical contribution of this article is to show that the CDCP structure—and in particular, supermodularity—survives this feedback (Section 4). A complementary computational contribution compares alternative algorithms for resolving the three-way fixed point (Section 5).

4.1.7 Lattice Structure and Supermodularity

The equilibrium of Definition 1 is a joint fixed point over activation sets, markups, and shares. Before turning to uniqueness (Section 4.1.8 for the intensive margin, Section 4.1.9 for the extensive margin), we establish that the single-firm activation problem retains lattice structure under AB endogenous markups: Proposition 2 gives supermodularity, Proposition 3 bounds the Nash set between extremal fixed points of the squared best-response map, and Proposition 4 establishes convergence of Jacobi iteration. These results play a dual role: inside Environment A they certify that Jia–Tarski-style solvers apply; viewed from the abstract framework of Section 3, they are the lattice scaffolding the abstract hypotheses presume, and Environment B will reproduce them under nested logit demand (Section 4.2).

Since the rival demand index $D_{-f}(n)$ is a sufficient statistic for competition (Section 4.1.4), we write $\pi_f(\mathcal{I}_f; D_{-f})$ for firm f 's profit (4.14) evaluated at the unique pricing equilibrium given D_{-f} .

Proposition 2 (Single-Agent Supermodularity). *Fix the rival demand index $D_{-f}(n)$ for all markets n and suppose $(\sigma - 1)/\theta > 1$.*

- (i) *Per-market sufficient condition (CES benchmark). If the equilibrium market shares satisfy $\max_{n,f} s_{nf} < \bar{s}^*$, where $\bar{s}^* \equiv (\alpha - 1)/(2\alpha)$ with $\alpha = (\sigma - 1)/\theta$ is a tractable conservative threshold derived under constant CES markups (Appendix E, Step 4), then each market's contribution to variable profit is convex in the cost aggregator, which implies that firm f 's activation profit $\pi_f(\mathcal{I}_f; D_{-f})$ exhibits increasing differences in the activation indicators $\{I_{fi}\}_{i \in \mathcal{A}_f}$. Under AB endogenous markups, the exact per-market condition is given implicitly by $H_{AB}(s) \geq 0$ (Appendix E, Step 3'):*

$$\pi_f(\mathcal{I}_f \cup \{i, i'\}; D_{-f}) - \pi_f(\mathcal{I}_f \cup \{i\}; D_{-f}) \geq \pi_f(\mathcal{I}_f \cup \{i'\}; D_{-f}) - \pi_f(\mathcal{I}_f; D_{-f}). \quad (4.16)$$

- (ii) *Computational diagnostic. When some market shares exceed $\bar{s}^*$, the per-market condition is not met, but aggregate supermodularity across all markets can still hold because low-share markets (positive curvature) dominate high-share markets (negative curvature). A zero Jia–Tarski gap, together with the interval-audited sign checks reported below, is treated as a computational diagnostic for the solved single-agent subproblems. This diagnostic is*

consistent with aggregate supermodularity at the tested parameterization, but it is not an analytical proof, and we do not currently have a closed-form sufficient condition for aggregate supermodularity under heterogeneous market shares.

Under (i), or under a maintained computational certificate that the solved single-agent problem has the requisite lattice structure at the tested instance, the set of optimal solutions to $\max_{\mathcal{I}_f \subseteq \mathcal{A}_f} \pi_f(\mathcal{I}_f; D_{-f})$ has a greatest element and a least element, each attainable by the Jia–Tarski algorithm.

Proof sketch. The marginal return to activating site i' can be decomposed into two effects. The *cost complementarity effect*: by equation (4.7), adding site i' contributes a term $(w_{i'} \tau_{i'n})^{-\theta}$ to the cost index in every market n . The proportional cost reduction is larger when the existing cost index is higher—i.e., when fewer other sites are already active. Under the condition $(\sigma - 1)/\theta > 1$, variable profit is sufficiently convex in the cost index that the scale expansion from lower costs generates increasing differences.

The *self-cannibalization effect*: activating i' lowers cost, which lowers price, which raises market share s_{nf} , which compresses the perceived elasticity ϵ_{nf} via (4.9) and raises the markup μ_{nf} . This partially offsets the quantity gain. Under the CES benchmark (constant markups), per-market convexity of variable profit in the cost aggregator $\Sigma_{fn} \equiv \sum_{i \in \mathcal{I}_f} (w_i \tau_{in})^{-\theta}$ from (4.7) requires $s < \bar{s}^*$. Under AB endogenous markups, the exact per-market condition is $H_{AB}(s) \geq 0$, where H_{AB} is an explicit function of $(s, \sigma, \eta, \theta)$ derived in Appendix E, Step 3'; the AB threshold is parameter-dependent and need not exceed the CES threshold. When some shares exceed both thresholds, part (ii) supplies only a computational diagnostic, not an analytical supermodularity proof.

The condition $(\sigma - 1)/\theta > 1$ is the same as in Antràs et al. (2017), Theorem 1, who establish supermodularity under constant markups. The extension here shows that the additional markup feedback can be incorporated analytically at the per-market level and does not overturn supermodularity in the tested parameterizations. Full proof in Appendix E. $\square$

Under Atkeson–Burstein endogenous markups, the exact per-market convexity condition $H_{AB}(s) \geq 0$ defines a *unique* implicit threshold $\bar{s}_{AB}^*$ (Appendix E, Proposition 9): H_{AB} is strictly concave, so the threshold is the unique zero of a concave function that is positive at $s = 0$ and negative at $s = 1$. The AB threshold is parameter-dependent and

need not uniformly exceed the CES benchmark $\bar{s}^*$. In the cloud application (Section 6), $\max_{n,f} s_{nf} = 0.438$ exceeds both thresholds, so part (ii) is used only as a computational diagnostic for the solved single-agent subproblems (see Remark 6).

Remark 6 (Logical status of supermodularity). Part (i) of Proposition 2 provides an *analytical sufficient condition*: if all market shares lie below $\bar{s}^*$, supermodularity is proven. Part (ii) provides a *computational diagnostic*: when some shares exceed $\bar{s}^*$ (as in the cloud application, where $\max s_{nf} = 0.438 > \bar{s}^*$), the Jia–Tarski gap = 0 certifies that the algorithm found matching upper and lower bounds on the optimal activation set at the solved rival-demand vectors, and the interval audit checks the decisive marginal-value signs on those solved instances. This is verification of a consequence (lattice optimum coincidence at the solved instance) rather than a direct proof of the increasing-differences property on the full activation lattice. Downstream propositions therefore rely on analytical supermodularity when part (i) applies and otherwise on a maintained computational certification of the single-agent lattice structure at the tested parameterization. For tolerance hygiene on the computational pathway, the Jia–Tarski MV-sign test uses a threshold of 10^{-12} ; a diagnostic counter (`jia_tarski.m`, `output_near_zero_count`) records the number of evaluations with $|\text{MV}| \in [10^{-14}, 10^{-10}]$, i.e., inside the tolerance band or within two orders of magnitude of it. Across every iteration of the reported calibration and counterfactual runs this counter equals zero: the gap-zero diagnostic is not driven by near-zero sign ambiguity.

Existence. A Nash equilibrium of the complete-information game exists by Nash’s theorem, since the activation lattice $L = \prod_f 2^{\mathcal{A}_f}$ is finite. The following two propositions characterize the equilibrium set and establish convergence of iterative algorithms.

Proposition 3 (Equilibrium Bounds). *Suppose each firm’s activation profit is supermodular (Proposition 2).*

- (i) *The joint best-response map Φ is anti-monotone (order-reversing) on the joint activation lattice L , so the squared map $\Phi^2 = \Phi \circ \Phi$ is monotone. By Tarski’s (1955) fixed-point theorem, Φ^2 has a greatest fixed point $\bar{\mathbf{I}}$ and a least fixed point $\underline{\mathbf{I}}$ on L .*
- (ii) *Any Nash equilibrium $\mathbf{I}^*$ satisfies $\underline{\mathbf{I}} \leq \mathbf{I}^* \leq \bar{\mathbf{I}}$: the extremal fixed points of Φ^2 bound the entire Nash equilibrium set.*

(iii) If $\bar{\mathbf{I}}$ is itself a Nash equilibrium (i.e., $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$, not merely a two-cycle of Φ), it is the greatest Nash equilibrium.

Proof. Since activation choices are *strategic substitutes*—when rival firms activate more sites, the rival demand index D_{-f} increases, reducing the profitability of firm f 's sites—the joint best-response map $\Phi : (I_1, \dots, I_F) \mapsto (I_1^*, \dots, I_F^*)$ is *anti-monotone*: $\mathbf{I} \leq \mathbf{I}'$ implies $\Phi(\mathbf{I}) \geq \Phi(\mathbf{I}')$ (cf. Amir, 1996), where $\leq$ is the component-wise set-inclusion order on $L \equiv \prod_f 2^{\mathcal{A}_f}$.

Part (i): Φ^2 is monotone because anti-monotonicity composes to give $\mathbf{I} \leq \mathbf{I}' \Rightarrow \Phi^2(\mathbf{I}) \leq \Phi^2(\mathbf{I}')$. Tarski's theorem on the finite complete lattice L yields the extremal fixed points.

Part (ii): any Nash equilibrium satisfies $\Phi(\mathbf{I}^*) = \mathbf{I}^*$, hence $\Phi^2(\mathbf{I}^*) = \mathbf{I}^*$. By Tarski's theorem, $\underline{\mathbf{I}} \leq \mathbf{I}^* \leq \bar{\mathbf{I}}$.

Part (iii): if $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$, then $\bar{\mathbf{I}}$ is a Nash equilibrium; any other Nash equilibrium $\mathbf{I}^*$ satisfies $\mathbf{I}^* \leq \bar{\mathbf{I}}$ by (ii).

In general, $\bar{\mathbf{I}}$ may be a two-cycle rather than a Nash equilibrium: it is possible that $\Phi(\bar{\mathbf{I}}) \neq \bar{\mathbf{I}}$ while $\Phi^2(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$. However, at all parameterizations tested in Section 6 and Appendix K, $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$ and the equilibrium is unique. Full proof in Appendix F. $\square$

Proposition 4 (Convergence of Jacobi Iteration). *Suppose each firm's activation profit is supermodular. Starting from the supremum $\hat{\mathbf{I}} = (\mathcal{A}_1, \dots, \mathcal{A}_F)$, the even subsequence $\{\Phi^{2k}(\hat{\mathbf{I}})\}$ is decreasing and converges to $\bar{\mathbf{I}}$ (the greatest fixed point of Φ^2). The odd subsequence $\{\Phi^{2k+1}(\hat{\mathbf{I}})\}$ is increasing and converges to $\Phi(\bar{\mathbf{I}})$. If $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$, the full sequence converges to the greatest Nash equilibrium. Analogously from the infimum.*

Proof. Since Φ is anti-monotone and $\Phi(\hat{\mathbf{I}}) \leq \hat{\mathbf{I}}$, the even subsequence $\{\Phi^{2k}(\hat{\mathbf{I}})\}$ is decreasing (by monotonicity of Φ^2) and the odd subsequence $\{\Phi^{2k+1}(\hat{\mathbf{I}})\}$ is increasing, with $\Phi^{2k+1}(\hat{\mathbf{I}}) \leq \Phi^{2k}(\hat{\mathbf{I}})$ for all k . On the finite lattice L , both subsequences stabilize in finitely many steps. The even limit is $\bar{\mathbf{I}}$ and the odd limit is $\Phi(\bar{\mathbf{I}})$, which satisfy $\Phi^2(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$ and $\Phi(\bar{\mathbf{I}}) \leq \bar{\mathbf{I}}$. Full proof in Appendix F. $\square$

Remark 7 (Gauss–Seidel convergence). The sequential update map Ψ that updates firms in order $(f_1, \dots, f_F)$ using the latest available rivals' strategies converges at all tested parameterizations (Section 6 and Appendix K). Formal convergence of Ψ under anti-monotone best responses is not guaranteed in general—sequential updating breaks the even/odd

subsequence structure that yields monotonicity of Φ^2 —but empirical convergence is robust across all parameterizations tested.

At all parameterizations tested in Section 6 and Appendix K, $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$: both Jacobi and Gauss–Seidel converge to the same Nash equilibrium, and no two-cycles are observed. Gauss–Seidel uses updated rival information within each round and typically converges in fewer rounds, whereas Jacobi is naturally parallelizable. Section 5 presents the full comparison.

4.1.8 Pricing Uniqueness and the Intensive Margin

The uniqueness question has two distinct margins. On the *intensive margin* (pricing), uniqueness requires that the AB markup fixed point is contractive for any given activation profile. On the *extensive margin* (site activation), uniqueness requires that no “marginal sites”—sites whose profitability depends on rival behavior—switch activation status across equilibria. This subsection resolves the intensive margin via a contraction condition on the feedback coefficient (Proposition 5); Section 4.1.9 resolves the extensive margin by introducing private information over fixed costs.

Feedback coefficient. The strength of the strategic interaction through the AB markup channel is captured by a *feedback coefficient*. For firm f in market n with share s_{nf} and markup μ_{nf} , define:

$$\kappa_{nf} = \frac{s_{nf}^2 \beta_{nf}}{1 + s_{nf}(1 - s_{nf}) \beta_{nf}}, \quad \beta_{nf} = \frac{(\sigma - 1) \mu_{nf} (\sigma - \eta)}{\sigma \eta}. \quad (4.17)$$

The coefficient $\kappa_{nf} = \partial p_{nf}^{1-\sigma} / \partial D_{-f}(n)$ measures how firm f ’s pricing power responds to a change in the rival demand index, holding the activation set fixed. When $s_{nf} = 0$ (negligible firm), $\kappa_{nf} = 0$; as s_{nf} rises, the markup channel amplifies the pricing response, so κ_{nf} increases.

Proposition 5 (Pricing Uniqueness). *For any fixed activation profile $(\mathcal{I}_1, \dots, \mathcal{I}_F)$, define the markup interaction matrix Λ with entries*

$$\Lambda_{(n,f),(n,g)} = \kappa_{nf} \cdot \frac{p_{ng}^{1-\sigma}}{D_{-f}(n)}, \quad g \neq f, \quad (4.18)$$

with all other entries zero. If the spectral radius satisfies

$$\rho(\Lambda) < 1, \quad (4.19)$$

then the AB pricing equilibrium is unique.

Proof. For a fixed activation profile, the AB pricing system defines a map $\Psi : D \mapsto D'$ on the space of rival demand indices. The Jacobian of Ψ is bounded entrywise by Λ , which is nonnegative. When $\rho(\Lambda) < 1$, the Perron–Frobenius theorem guarantees a positive vector $w \gg 0$ and a scalar $\lambda < 1$ such that $\Lambda w \leq \lambda w$. Define the weighted sup-norm $\|x\|_w = \max_i |x_i|/w_i$. By the mean value theorem, for any two demand vectors D, D' :

$$\|\Psi(D) - \Psi(D')\|_w \leq \lambda \|D - D'\|_w,$$

so Ψ is a contraction in $\|\cdot\|_w$. Since the domain is a compact convex subset of $\mathbb{R}_+^{NF}$ (bounded markups imply bounded demand indices), Banach’s fixed-point theorem yields a unique pricing equilibrium. $\square$

The matrix Λ is block-diagonal across markets (N blocks of size $F \times F$), so $\rho(\Lambda) = \max_n \rho(\Lambda_n)$ is cheap to compute. A conservative closed-form sufficient condition uses the infinity norm:

Corollary 2. *If $\max_{n,f} \kappa_{nf} < 1/(F - 1)$, then $\rho(\Lambda) < 1$ and the pricing equilibrium is unique.*

At the cloud application’s equilibrium profile ($F = 5$, Section 6), both conditions are satisfied: $\max \kappa = 0.239 < 1/(F - 1) = 0.25$ and $\rho(\Lambda) = 0.109$. *Off-equilibrium, the infinity-norm condition is tight.* Across the 1,302 per-market subgames enumerated in Proposition 7 (Table 2), $\|\Lambda_n\|_\infty$ exceeds $1/(F - 1)$ in a nontrivial fraction of subgames, yet $\rho(\Lambda_n) \leq 0.314$ throughout. This gap between the row-sum bound and the spectral radius motivates the spectral refinement of the downstream Jacobian bound in Proposition 10: the Neumann-series amplification $1/(1 - \|\Lambda\|_\infty)$ is not available uniformly, but the Perron–Frobenius amplification $1/(1 - \rho(\Lambda))$ is. For applied work, the spectral radius (4.19) is the recommended diagnostic. At the synthetic economy ($F = 8$, Appendix K), the same pattern obtains: $\max \kappa = 0.193 > 1/(F - 1) = 0.143$ but $\rho(\Lambda) = 0.096$.

Proposition 5 resolves the intensive margin but does *not* guarantee uniqueness of the full game, because different activation profiles lead to different demand indices. Define

the composite map $\tilde{\Phi} : D \mapsto D'$ that maps rival demands to best-response activation sets, then evaluates the unique pricing equilibrium to produce new demands. Multiple equilibria arise when $\tilde{\Phi}$ has multiple fixed points—equivalently, when there exist marginal sites whose activation profit $\Delta\pi_{if}$ changes sign across equilibrium demands. The number of marginal sites is the key diagnostic for multiplicity; the cloud application (Section 6.5) and synthetic stress tests (Appendix K) show it is zero at all tested parameterizations. To resolve the extensive margin analytically, we introduce private information over fixed costs, converting the discrete activation problem into a continuous threshold system on which contraction arguments apply. Section 3.1 states the uniqueness result at the abstract level of a generic threshold map; Section 4.1.9 then verifies its hypotheses under the Fréchet–CES–AB environment of Section 4.1; Section 4.2 verifies them under nested logit on the consumer side; and Section 4.3 discusses further environments.

4.1.9 Bayesian Perturbation and the Threshold System

We now verify the hypotheses of Theorem 1 and Corollary 1 for the Fréchet–CES–AB environment of Section 4.1. Firms' fixed costs are not publicly observed in practice; we introduce private information over fixed costs, which serves two purposes: it captures the empirical reality that site-specific deployment costs are unobserved, and it converts the discrete activation problem into the continuous threshold system (3.1) on which the abstract contraction argument applies.

Assumption 2 (Private cost shocks and threshold strategies). (*Information.*) Each firm f privately observes fixed-cost shocks $\{\varepsilon_{if}\}_{i \in \mathcal{A}_f}$ that are i.i.d. across sites and across firms, drawn from a distribution G with full-support density g , mean zero, and scale parameter $\sigma_\varepsilon > 0$. The total fixed cost of activating site i is $(\phi_i + \varepsilon_{if})w_i$. Firm f observes its own draws but not rivals'. (*Strategy space.*) Firm f 's pure strategy is a measurable decision rule $\sigma_f : \mathbb{R}^{|\mathcal{A}_f|} \rightarrow 2^{\mathcal{A}_f}$ of the site-independent threshold form: there exist cutoffs $(\bar{\varepsilon}_{if})_{i \in \mathcal{A}_f}$ such that site i is activated if and only if $\varepsilon_{if} < \bar{\varepsilon}_{if}$, independently across sites within the firm. The cutoffs $\bar{\varepsilon}_{if}$ depend on rival activation probabilities and on own expected complementarities, but not on the realized own-site shocks ε_{jf} for $j \neq i$.

Remark 8 (Why the threshold-strategy restriction). Two comments on the strategy-space restriction in Assumption 2. First, restricting to site-independent threshold strategies is

not a free consequence of supermodularity plus i.i.d. shocks: under full joint observation of ε_f and supermodular within-firm payoffs, a firm could in principle condition its site- i decision on realized ε_{jf} through the complementarity channel, producing an activation region with step (not product-form) structure. We rule this out by restriction rather than by proof. Second, the restriction has a natural economic interpretation: in decentralized multinational or platform-entry organizations, each country's entry is decided by a local team on the basis of local cost conditions, without conditioning in real time on the realized idiosyncratic draws at other sites. The maintained restriction is what permits the threshold-system representation on $[0, 1]^M$ used in Proposition 6 and Theorem 2. All uniqueness claims below are statements about the unique BNE *within the class of site-independent threshold strategies*; when the complete-information game has a unique pure-strategy Nash equilibrium and the bathtub peak $\gamma_{\max}$ is below one, no other threshold-strategy BNE exists.

The i.i.d. assumption is substantive: combined with the site-independent threshold-strategy restriction, it keeps each site's activation decision one-dimensional (Proposition 6(i)), enabling the Jacobian factorization (4.21). With correlated shocks—for example, a hierarchical structure $\varepsilon_{if} = \eta_f + \tilde{\varepsilon}_{if}$ with a common firm-level component η_f —the activation decision for site i would depend on η_f through the conditional expected marginal value, yielding a threshold that varies with the firm-level draw rather than an unconditional scalar. The Jacobian factorization breaks because the density factor becomes conditional on η_f . A partial extension is possible: conditional on η_f , the site-level shocks $\tilde{\varepsilon}_{if}$ remain i.i.d., so a conditional version of the contraction argument applies at each realization, but the fixed point is then over a higher-dimensional system (one threshold per site per firm-level type) and the contraction condition is harder to verify. In practice, the i.i.d. assumption is standard in the CDCP literature (Jia, 2008; Arkolakis et al., 2023); the restriction here is no more binding than in those frameworks.

Economically, the i.i.d. assumption rules out two forms of dependence that arise naturally in multinational production. First, *within-firm correlation*: a firm-level “headquarters quality” draw that shifts fixed costs at all sites simultaneously—for example, a management practice that raises or lowers setup costs uniformly across countries. Second, *spatial correlation*: regulatory or institutional costs that co-move across neighboring countries sharing a border or regulatory regime. Either form of dependence would break the

one-dimensional threshold structure exploited in Proposition 6(i).

Two distinct roles of the i.i.d. assumption deserve emphasis. Independence across sites *within* a given firm combined with the threshold-strategy restriction in Assumption 2 ensures that each site’s activation reduces to a scalar threshold comparison, enabling the Jia–Tarski algorithm and the Jacobian factorization (4.21). Independence *across firms* makes each firm’s threshold $\bar{\varepsilon}_{if}$ a deterministic function of rival activation probabilities rather than of rival cost realizations, so the cross-player Jacobian J_T has the product structure that supports the contraction argument. With correlated shocks across firms—for example, a common macroeconomic shock—the thresholds would be conditional on the common state, and the contraction condition would need to hold state-by-state (Proposition 8).

Extension: across-firm common shocks. Appendix D develops a public-state extension $\varepsilon_{if} = \zeta + \tilde{\varepsilon}_{if}$ in which ζ is observed by all firms and $\tilde{\varepsilon}_{if}$ is i.i.d. (Proposition 8): uniqueness holds state-by-state whenever $\sup_{\zeta} \gamma_{\max}(\zeta) < 1$, a condition we verify in Table 12 on a $\pm 40\%$ grid of $\zeta/\bar{\phi}$ in the small economy of Section 4.4. The remaining case—*within-firm* common shocks $\varepsilon_{if} = \eta_f + \tilde{\varepsilon}_{if}$ with η_f privately observed by firm f —is genuinely outside the paper’s framework: it breaks the sibling independence used to reduce each firm’s CDCP to $|\mathcal{A}_f|$ scalar threshold comparisons. A formal treatment would condition the threshold system on firm-level types and replace Banach contraction with a family of contractions indexed by η_f ; we leave this extension to future work. Within-firm correlation is arguably the less empirically relevant case in our application, because calibrated site-specific fixed costs ϕ_i already absorb the firm-invariant component of country-site heterogeneity.

The solution concept is a Bayesian Nash equilibrium (BNE) *within the class of site-independent threshold strategies* defined in Assumption 2. Existence is immediate: the threshold map $T : [0, 1]^M \rightarrow [0, 1]^M$ (defined below) is continuous, so Brouwer’s fixed-point theorem guarantees a threshold-strategy BNE. All uniqueness claims in what follows are statements about this restricted equilibrium concept.

Threshold strategies. Given rivals’ threshold strategies, the single-agent activation problem is supermodular in $(\mathcal{I}_f, -\varepsilon_f)$: higher fixed-cost shocks make activation less attractive, and the within-firm interaction is complementary by Proposition 2. The monotone comparative statics of Athey (2001) imply that *if* the firm restricts attention to site-independent

threshold strategies—the restriction maintained in Assumption 2—then firm f 's best response activates site i if and only if $\varepsilon_{if} < \bar{\varepsilon}_{if}$, where $\bar{\varepsilon}_{if}$ is an endogenous threshold pinned down by rival activation probabilities and own expected complementarities. Given rivals' thresholds, each site i of rival g is active independently with probability $v_{ig} = G(\bar{\varepsilon}_{ig})$.

Threshold system. The threshold $\bar{\varepsilon}_{if}$ is pinned down by the zero-marginal-value condition: at $\varepsilon_{if} = \bar{\varepsilon}_{if}$, the expected marginal value of activating site i , integrating over rivals' random activation sets, equals zero. Define the best-response map T that maps rival activation probabilities $\{v_{ig}\}_{g \neq f}$ to own activation probabilities $\{v_{if}\}$ via:

$$\{v_{ig}\}_{g \neq f} \xrightarrow{\text{expected demand}} \mathbb{E}[D_{-f}] \xrightarrow{\rho < 1} \text{pricing eq.} \xrightarrow{\text{bounded markups}} \bar{\varepsilon}_{if} \xrightarrow{G} v_{if}. \quad (4.20)$$

The nested CES demand structure provides three ingredients for this map to be contractive:

1. *Bounded markups* $\mu \in [\sigma/(\sigma-1), \eta/(\eta-1)]$: the marginal value of each site is Lipschitz-continuous in the rival demand index, so small changes in the pricing equilibrium produce bounded changes in the threshold $\bar{\varepsilon}_{if}$, with Lipschitz constant depending on (σ, η) .
2. *Pricing contraction* $\rho(\Lambda) < 1$: the map from expected rival demands to the pricing equilibrium is contractive (Proposition 5), ensuring that a small shift in activation probabilities feeds back into prices with bounded magnification.
3. *Smoothing via G* : the map from thresholds to probabilities (via the CDF) is smooth and bounded, converting the discrete extensive margin into a continuous probability.

Jacobian decomposition. Before stating the uniqueness result, we develop its key tool: a quantitative bound on the contraction modulus of the threshold map T defined in (4.20). Write $G(x) = G_0(x/\sigma_\varepsilon)$ with standardized CDF G_0 and density $g_0 = G'_0$, so that $g(x) = g_0(x/\sigma_\varepsilon)/\sigma_\varepsilon$. The Jacobian of T factors as

$$J_T(v) = \underbrace{\text{diag}\left(\frac{g_0(\bar{\varepsilon}_{if}/\sigma_\varepsilon)}{\sigma_\varepsilon}\right)}_{\text{density dampening}} \cdot \underbrace{\text{diag}\left(\frac{1}{w_i}\right)}_{\text{scaling}} \cdot \underbrace{\left[\frac{\partial \mathbb{E}[\Delta\pi_{if}]}{\partial v_{jg}}\right]}_{\equiv J_\Delta: \text{marginal-value sensitivity}}, \quad (4.21)$$

where $\bar{\varepsilon}_{if} = \mathbb{E}[\Delta\pi_{if}]/w_i$ is the threshold of site (i, f) . Define $J_v \equiv \text{diag}(1/w_i) \cdot J_\Delta$, which absorbs the wage scaling into the sensitivity matrix. The three ingredients above appear as three factors governing the contraction modulus $\gamma(\sigma_\varepsilon) \equiv \sup_v \|J_T(v)\|_\infty$:

- (a) *Density dampening*: the prefactor $g_0(\cdot)/\sigma_\varepsilon$ vanishes as $\sigma_\varepsilon \rightarrow \infty$.
- (b) *Marginal-value sensitivity*: the matrix J_Δ inherits bounded entries from the CES demand structure. The across-firm entries are non-positive (strategic substitutes through $D_{-f}(n)$), while within-firm entries are non-negative (complementarity from Proposition 2). Both are bounded by the markup range $[\sigma/(\sigma-1), \eta/(\eta-1)]$, and the pricing contraction amplifies demand perturbations by at most $1/(1-\rho(\Lambda))$ in the Perron–Frobenius weighted sup-norm (Proposition 10; Appendix G).
- (c) *Threshold location*: through $g_0(\bar{\varepsilon}_{if}/\sigma_\varepsilon)$, the density is small when the threshold is far from the mode—i.e., when site i is far from marginal.

Proposition 6 (Threshold-Strategy BNE Uniqueness under Fréchet–CES–AB). *Proposition 6 verifies the hypotheses of Theorem 1 and Corollary 1 for the Fréchet–CES–AB environment. Suppose each firm’s activation profit is supermodular and the pricing equilibrium is unique ($\rho(\Lambda) < 1$, Proposition 5; a conservative sufficient condition is $\max \kappa < 1/(F-1)$, Corollary 2).*

- (i) *Within the maintained threshold-strategy class of Assumption 2, firm f ’s best-response activation rule at site i takes the cutoff form $I_{fi} = \mathbf{1}\{\varepsilon_{if} < \bar{\varepsilon}_{if}\}$.*
- (ii) *Define the contraction modulus*

$$\gamma(\sigma_\varepsilon) \equiv \sup_{v \in [0,1]^M} \|J_T(v)\|_\infty \leq \frac{\|g_0\|_\infty}{\sigma_\varepsilon} \cdot \|J_v\|_\infty, \quad (4.22)$$

where $M = \sum_f |\mathcal{A}_f|$, $\|g_0\|_\infty = \sup_z g_0(z)$ is the peak density of the standardized distribution, and $\|J_v\|_\infty = \sup_v \max_{(i,f)} (1/w_i) \sum_{(j,g)} |\partial \mathbb{E}[\Delta\pi_{if}]/\partial v_{jg}|$ is the weighted row-sum norm of the marginal-value sensitivity. If $\gamma(\sigma_\varepsilon) < 1$, the threshold map T is a contraction and the BNE is unique.

- (iii) *As $\sigma_\varepsilon \rightarrow 0$, the BNE activation probabilities converge to $\{0, 1\}$, recovering a Nash equilibrium of the complete-information game.*

(iv) Assume additionally that G_0 has log-concave tails (e.g., normal or logistic). The large-noise tail is automatic: $\gamma(\sigma_\varepsilon) \rightarrow 0$ as $\sigma_\varepsilon \rightarrow \infty$ by density dampening. The small-noise tail requires the localized non-marginality conditions of Lemma 2. If the complete-information game has a unique Nash equilibrium with gap $\delta_v = \min_{i,f} |\Delta\pi_{if}^*|/w_i > 0$, and the environment-specific verification constructs or numerically checks a family S_σ satisfying Lemma 2, and $\gamma_{\max} \equiv \sup_{\sigma_\varepsilon > 0} \gamma(\sigma_\varepsilon) < 1$, then the BNE is unique for all $\sigma_\varepsilon > 0$. The condition $\gamma_{\max} < 1$ is verifiable numerically by evaluating γ on a grid of σ_ε values.

Proof sketch. Part (i): within the threshold-strategy class imposed by Assumption 2, firm f 's cutoffs $(\bar{\varepsilon}_{if})_{i \in \mathcal{A}_f}$ depend on rival activation probabilities $\{v_{jg}\}_{g \neq f}$ and on own expected complementarities integrated over the i.i.d. own-site shocks, but are deterministic functions of v rather than of the realized own-site shocks ε_{jf} ($j \neq i$). Given rivals' threshold strategies (which determine $\{v_{jg}\}_{g \neq f}$), each cutoff $\bar{\varepsilon}_{if}$ is pinned down by the zero-marginal-value condition $\mathbb{E}[\Delta\pi_{if}(v) \mid \varepsilon_{if} = \bar{\varepsilon}_{if}] = 0$, where the expectation integrates over both rivals' realizations and firm f 's own other-site draws. The resulting activation rule $I_{fi} = \mathbf{1}\{\varepsilon_{if} < \bar{\varepsilon}_{if}\}$ is best-response within the threshold-strategy class, by Proposition 2 and the monotone comparative statics of Topkis (1998), Theorem 4.2.2. (We do not claim that every BNE is in threshold strategies; Assumption 2 is a maintained restriction on the strategy space.)

For (ii), the factorization (4.21) decomposes each row of $|J_T|$ into density dampening ($\leq \|g_0\|_\infty/\sigma_\varepsilon$) times the weighted marginal-value sensitivity J_v . Since the expected marginal value is a polynomial in $\{v_{ig}\}$, it is smooth; its row sums are bounded by composing Lipschitz constants along the chain $v \rightarrow \mathbb{E}[D_{-f}] \rightarrow \text{pricing} \rightarrow \Delta\pi_{if}$, with the pricing contraction amplifying perturbations by at most $1/(1 - \rho(\Lambda))$ in the Perron–Frobenius weighted sup-norm (Proposition 10, Appendix G). This yields $\|J_v\|_\infty \leq L_D \cdot L_\mu/(1 - \rho(\Lambda)) + L_C < \infty$, so $\gamma \leq (\|g_0\|_\infty/\sigma_\varepsilon) \cdot \|J_v\|_\infty < 1$ for σ_ε large enough; Banach's theorem yields uniqueness.

Part (iii) follows from upper hemicontinuity of the equilibrium correspondence: as $\sigma_\varepsilon \rightarrow 0$, the perturbed game converges to the complete-information game, and any sequence of BNE has a limit point that is a Nash equilibrium of the complete-information game (cf. Harsanyi 1973; for entry games with private information, see Espin-Sanchez et al. 2023, Proposition 2).

For (iv), $\gamma \rightarrow 0$ as $\sigma_\varepsilon \rightarrow \infty$ is immediate from (4.22). For $\sigma_\varepsilon \rightarrow 0$, the relevant statement

is Lemma 2: once the environment-specific verification supplies a localized family S_σ with invariance, fixed-point inclusion, and uniform non-marginality, the contraction modulus on that region decays to zero. At any σ_ε where $\gamma < 1$, the implicit function theorem ensures that the BNE extends as a unique C^1 branch. Since $\gamma \rightarrow 0$ at both extremes and $\gamma_{\max} < 1$ implies $\gamma < 1$ everywhere, uniqueness holds for all $\sigma_\varepsilon > 0$. $\square$

The interaction matrix Λ controls *both* margins through a single channel. On the intensive margin, $\rho(\Lambda) < 1$ guarantees pricing uniqueness (Proposition 5). On the extensive margin, $\rho(\Lambda)$ enters the contraction modulus (4.22) through the amplification factor $1/(1-\rho(\Lambda))$: the sensitivity of the pricing equilibrium to a demand perturbation is bounded by $\|(I - \Lambda)^{-1}\| \leq 1/(1 - \rho(\Lambda))$. The infinity-norm condition $\|\Lambda\|_\infty < 1$ of Corollary 2 is a conservative sufficient condition; the spectral condition $\rho(\Lambda) < 1$ (Proposition 5) is tighter and is used in all numerical diagnostics. At the baseline parameterization of Section 6 and Appendix K, $\rho(\Lambda) = 0.096$ (amplification factor 1.11): the outer activation problem is *nearly* a constant-markup entry game. Across the robustness experiments (Appendix K), $\rho(\Lambda)$ ranges from 0.025 (low σ , high η) to 0.574 (high σ , low η , concentrated market), remaining well below one throughout. When $\rho(\Lambda) \rightarrow 1$, the amplification diverges and larger σ_ε is required to restore the contraction.

Remark 9 (Structural vs. CES-specific elements). The Jacobian decomposition (4.21) and the bathtub shape of $\gamma(\sigma_\varepsilon)$ are *structural*: they hold for any Bayesian game with private cost shocks, a contractive pricing map, and Lipschitz-continuous marginal values—regardless of the demand system. What the CES structure provides is closed-form expressions for the Lipschitz constants (L_D, L_μ, L_C) , the feedback coefficient κ_{nf} , and the share thresholds $\bar{s}^*$ and $\bar{s}$. Under more general demand systems (e.g., translog), markups need not be bounded, so the Lipschitz constants may diverge and the contraction modulus may exceed one. The qualitative insight—that uniqueness is governed by a bathtub-shaped modulus driven by three interpretable forces—extends beyond CES, but the quantitative verification $\gamma_{\max} < 1$ would require numerical Jacobian bounds tailored to the specific demand system.

Remark 10 (Role of private information). Proposition 6 establishes uniqueness of the Bayesian game for a given noise distribution G and scale σ_ε . The BNE is a well-defined equilibrium concept in its own right—not merely a device for selecting among complete-information equilibria. Non-marginal sites ($|\Delta\pi_{if}|$ bounded away from zero) activate

with probability near one or zero; only marginal sites ($|\Delta\pi_{if}| \approx 0$) have interior activation probabilities that depend on G . When the complete-information game has a unique Nash equilibrium (as at all tested parameterizations), the BNE converges to it as $\sigma_\varepsilon \rightarrow 0$ regardless of G .

The following theorem consolidates all assumptions for the main result in one place, referencing Propositions 2–6 as building blocks. Conditions (a)–(b) are structural properties of the model; condition (c) is a distributional regularity; conditions (d)–(e) are established by environment-specific verification and numerical diagnostics at the parameterization of interest—condition (d) through an explicit localization check satisfying Lemma 2, and condition (e) by evaluating γ on a grid of σ_ε values.

Theorem 2 (Main Result: Uniqueness of the Threshold-Strategy BNE). *Impose Assumption 2 (i.i.d. private cost shocks; site-independent threshold strategies). Suppose further:*

- (a) $(\sigma - 1)/\theta > 1$ and each firm’s activation problem has the lattice structure required by Proposition 2 (analytically when $\max_{n,f} s_{nf} \leq \bar{s}^*$ or, at the cloud calibration where the per-market threshold is violated, under the maintained computational certification of Remark 6);
- (b) $\rho(\Lambda(v)) < 1$ uniformly over every activation profile $v \in [0, 1]^M$ that the threshold map T may visit, where $\Lambda(v)$ is the markup interaction matrix at activation profile v (Proposition 5). This hypothesis admits three operational tests: (i) Corollary 3 gives a closed-form sufficient condition $\max_{n,f} s_{nf} < \bar{s}$; (ii) because Λ is block-diagonal across markets, $\rho(\Lambda) = \max_n \rho(\Lambda_n)$, so condition (b) reduces to per-market pricing uniformity, which admits the direct enumeration certificate in Proposition 7 below at the calibrated marginal-cost vector (see Remark 13); (iii) when neither (i) nor (ii) is available, the continuity of $\rho(\Lambda(v))$ on $[0, 1]^M$ permits a finite-grid evaluation with sufficient resolution to bound the variation of ρ between grid points.²

- (c) the standardized distribution G_0 has log-concave tails (e.g., normal or logistic);

²The infinity-norm sufficient condition $\max \kappa < 1/(F - 1)$ (Corollary 2) is uniform by construction but conservative. In the cloud calibration, $\|\Lambda\|_\infty \geq 1/(F - 1)$ on roughly 12% of enumerated per-market subgames, so the Neumann-series amplification $1/(1 - \|\Lambda\|_\infty)$ used at step 3 of Appendix G is not available throughout. Proposition 10 below refines the downstream Jacobian bound to the Perron–Frobenius amplification $1/(1 - \rho(\Lambda))$, so the spectral condition $\rho(\Lambda) < 1$ alone suffices; this is the condition certified by Proposition 7 at $\bar{\rho} \leq 0.314$ across the 1,302 enumerated subgames (Table 2).

- (d) *there exists a family $\{S_\sigma\}_{0 < \sigma < \sigma_0}$ satisfying the localization conditions of Lemma 2; in the calibrated application, the complete-information gap and the supremum/infimum diagnostics are used only to construct and numerically check this localization;*
- (e) $\gamma_{\max} \equiv \sup_{\sigma_\varepsilon > 0} \gamma(\sigma_\varepsilon) < 1$ *(verifiable by evaluating γ on a log-spaced grid of σ_ε values that brackets the bathtub peak).*

Then the BNE in site-independent threshold strategies is unique for all $\sigma_\varepsilon > 0$, and the uniqueness is conditional on hypotheses (a)–(e): it is a statement of the form “if (a)–(e) hold at the parameters of interest, then the threshold-strategy BNE is unique,” not an unconditional uniqueness claim.

Proof. Conditions (a)–(b) ensure that the single-agent problem has the maintained lattice structure and that the pricing equilibrium is unique for every activation profile, so the threshold map T in (4.20) is well-defined on $[0, 1]^M$. The limit $\gamma(\sigma_\varepsilon) \rightarrow 0$ as $\sigma_\varepsilon \rightarrow \infty$ follows from $\|g_0\|_\infty/\sigma_\varepsilon \rightarrow 0$ (Lemma 1). Conditions (c)–(d) deliver the small-noise tail through Lemma 2: log-concave tails provide the density decay and the localized family S_σ controls non-marginality at off-equilibrium profiles. Condition (e) ensures the peak contraction modulus is below one, so T is a contraction in the sup-norm for all $\sigma_\varepsilon > 0$. Banach’s fixed-point theorem on the complete metric space $[0, 1]^M$ yields a unique fixed point. $\square$

Remark 11 (Uniformity of the pricing condition across off-equilibrium profiles). Condition (b) requires $\rho(\Lambda(v)) < 1$ *at every activation profile v that the threshold map T may visit*—including off-equilibrium iterates, not only the equilibrium profile. Because T is Lipschitz on $[0, 1]^M$ under conditions (a)–(e), its image during Banach-iterate convergence traverses a compact subset of $[0, 1]^M$, and condition (b) must hold uniformly on that subset. The infinity-norm sufficient condition $\max_{n,f} \kappa_{nf}(v) < 1/(F-1)$ (Corollary 2) is uniform *by construction*: it depends on activation only through equilibrium shares, which are bounded by 1, so the supremum over v is the supremum over shares. The spectral condition $\rho(\Lambda(v)) < 1$ is not automatically uniform—two profiles with different share distributions induce different $\Lambda(v)$ blocks. In the cloud calibration, we verify uniformity in two complementary ways: the enumeration certificate of Proposition 7 at calibrated costs c^* yields $\bar{\rho}(c^*) \leq 0.314$, and a finite-grid evaluation on $v \in [0, 1]^M$ (Appendix K, $16^4 = 65,536$ grid points for the small economy) returns $\max_v \rho(\Lambda(v)) \leq 0.25$. The remaining cell-interior

v -variation is controlled by the Lipschitz property of $\Lambda \mapsto \rho(\Lambda)$: since $\Lambda(v)$ is polynomial in the shares $s_{nf}(v)$ which are themselves C^1 on $[0, 1]^M$, $\rho(\Lambda(\cdot))$ is Lipschitz with constant bounded by the operator norm of $\partial_v \Lambda$ on the grid. The observed maximum 0.25 plus the grid half-width times this Lipschitz constant remains below 0.314, confirming uniformity. We therefore treat uniformity as a numerically verified property rather than a structural consequence of nested CES, and report that in our application the uniformity margin is large ($\rho \leq 0.314$ uniformly, versus the threshold 1).

Sharp market-share threshold. Theorem 2 consolidates the assumptions; the following corollary provides a closed-form diagnostic. Since the feedback coefficient κ_{nf} depends only on the market share s_{nf} and the elasticity pair (σ, η) , the infinity-norm condition of Corollary 2 reduces to a threshold on the maximum market share.

Corollary 3 (Market-Share Threshold for Pricing Contraction). *Define $\bar{s}(\sigma, \eta, F)$ as the unique solution in $(0, 1)$ of $\kappa(\bar{s}; \sigma, \eta) = 1/(F - 1)$. If $\max_{n,f} s_{nf} < \bar{s}$, then the pricing equilibrium is unique (condition (b) of Theorem 2 is satisfied). Combined with conditions (a) and (c)–(e), this yields BNE uniqueness. For $(\sigma, \eta) = (5, 2)$, the threshold solves*

$$24\bar{s}^2 - 9\bar{s} - 8 = 0, \quad \bar{s} \approx 0.795 \quad (F = 2). \quad (4.23)$$

For $F = 8$ firms, $\bar{s} \approx 0.339$; for $F = 5$ (the cloud application of Section 6), $\bar{s} \approx 0.448$; for $F = 4$, $\bar{s} \approx 0.513$.

The proof (a quadratic in $\bar{s}$) and the comparative-statics derivations showing $\partial \bar{s} / \partial \sigma < 0$ and $\partial \bar{s} / \partial \eta > 0$ are collected in Appendix C. Table 1 reports $\bar{s}$ for selected (σ, η, F) combinations; the threshold is tightest when the markup channel is strongest (low η , high $\sigma - \eta$, large F).

Remark 12 (Two share thresholds). The paper employs two distinct market-share thresholds, both *sufficient* conditions that are conservative. The *supermodularity threshold* $\bar{s}^* \equiv (\alpha - 1)/(2\alpha)$ from Proposition 2 governs per-market convexity under the CES benchmark and depends only on (σ, θ) . The *pricing-contraction threshold* $\bar{s}(\sigma, \eta, F)$ from Corollary 3 governs pricing uniqueness via $\|\Lambda\|_\infty$ and additionally depends on (η, F) . At the baseline $(\sigma, \eta, \theta) = (5, 2, 3)$ with $F = 8$: $\bar{s}^* = 0.125$ and $\bar{s} \approx 0.339$, while $\max_{n,f} s_{nf} = 0.39$ —both thresholds are violated. Nevertheless, the solved single-agent subproblems satisfy the

Table 1: Market-share threshold $\bar{s}(\sigma, \eta, F)$ for pricing contraction

$\sigma \backslash \eta$	$\kappa < 1$ ($F = 2$)				$\kappa < 1/4$ ($F = 5$, cloud app)				$\kappa < 1/7$ ($F = 8$)			
	1.5	2.0	3.0	4.0	1.5	2.0	3.0	4.0	1.5	2.0	3.0	4.0
3	0.843	1.000	—	—	0.500	0.684	—	—	0.386	0.532	—	—
5	0.687	0.795	1.000	1.000	0.377	0.448	0.628	0.973	0.283	0.339	0.482	0.756
8	0.616	0.675	0.802	0.953	0.317	0.358	0.443	0.541	0.232	0.265	0.333	0.411
10	0.593	0.640	0.736	0.841	0.297	0.330	0.395	0.465	0.214	0.241	0.294	0.350

maintained lattice diagnostic of Remark 6 (Jia–Tarski gap = 0), and pricing uniqueness holds under the tighter spectral condition ($\rho(\Lambda) = 0.096 \ll 1$). The share thresholds are useful as closed-form diagnostics but are not necessary conditions.

TFP dispersion and marginal sites. Fix all parameters except TFP and define the dispersion ratio $\delta = a_{\max}/a_{\min}$. As δ increases, the number of marginal sites weakly decreases: in the limit $\delta \rightarrow \infty$, only the highest-TFP firms are active in any market, and $\kappa \rightarrow 0$ for non-dominant firms. When $a_f = a$ for all f , market shares are balanced at approximately $s \approx 1/F$. Since κ_{nf} is increasing and convex in s_{nf} , balanced shares actually *minimize* the feedback coefficient: the pricing contraction condition (Proposition 5) is *easier* to satisfy under identical TFP. Multiplicity in this case requires marginal sites driven by the available-set structure $\{\mathcal{A}_f\}$, which is absent at the baseline. Appendix K confirms this prediction numerically.

Direct enumeration certificate. The block-diagonal structure of Λ across markets permits a finite-enumeration verification of condition (b) at the calibrated marginal-cost vector: because each block Λ_n depends on market n ’s activation subset and on marginal costs, one can enumerate over all $2^F - 1$ nonempty per-market subsets at the calibrated costs and certify pricing uniqueness uniformly over per-market perturbations. The proof and the full scope discussion (three verification routes including $\pm 10\%$ off-path perturbations over 42,336 subgames) are collected in Appendix B.

Proposition 7 (Direct Enumeration Certificate at Calibrated Marginal Costs). *Fix the calibrated marginal-cost vector $c^* = (c_{n,f}^*)$. Let $\Lambda_n(S; c^*)$ denote the market- n interaction block at per-market activation subset $S \subseteq \{1, \dots, F\}$ with prices and shares solved from the per-market*

Cournot subgame at costs $(c_{n,f}^*)_{f \in S}$. Define

$$\bar{\rho}(c^*) \equiv \max_{n \in \{1, \dots, N\}} \max_{\emptyset \neq S \subseteq \{1, \dots, F\}} \rho(\Lambda_n(S; c^*)).$$

Then for any global activation profile $\mathbf{I} \in \{0, 1\}^{NF}$ whose induced per-firm marginal costs coincide with c^* , $\rho(\Lambda(\mathbf{I}; c^*)) \leq \bar{\rho}(c^*)$. In particular, if $\bar{\rho}(c^*) < 1$, condition (b) of Theorem 2 holds at the calibrated activation profile and at every per-market activation perturbation that leaves the marginal-cost vector fixed at c^* . The enumeration comprises $N(2^F - 1)$ per-market Cournot subgames.

For the calibrated cloud economy ($N = 42, F = 5$), enumeration over all $42 \times 31 = 1,302$ per-market subgames at c^* yields $\bar{\rho}(c^*) = 0.3144$ (Table 2), a margin of 0.686 below one. An off-path extension that multiplies each active firm's marginal cost by $m_f \in \{0.90, 1.00, 1.10\}$ across all $3^{|S|}$ patterns per subset—42,336 perturbed subgames in total—raises the maximum only to $\bar{\rho}_{\text{off}} = 0.3158$ (margin 0.6842), confirming that the certificate is robust to $\pm 10\%$ variation in per-firm productivity. The on-path maximum is attained in a two-firm subgame in the Philippines market; the off-path maximum in a two-firm subgame in Serbia at the opposing (1.10, 0.90) multiplier pattern. Both are subsumed in the same structural feature: the leading firm's price is sufficiently below the rival's that it captures nearly the entire subgame market. The infinity-norm bound $\max \kappa < 1/(F - 1) = 0.25$ is violated on a nontrivial fraction of enumerated subgames, but $\rho(\Lambda_n)$ remains bounded by 0.32 even under off-path perturbations: the rows of Λ_n concentrate on rival firms whose own κ 's are simultaneously small, so the spectral radius sits well below the row-sum bound. This ρ vs. $\|\Lambda\|_\infty$ gap is what the spectral-norm refinement of the downstream Jacobian bound (Proposition 10 below) exploits. The enumeration certificate therefore verifies the spectral condition at calibrated costs *and* over an empirically relevant neighborhood of c^* , with continuity routing any remaining off-equilibrium activations through Theorem 2(b).

4.2 Environment B: Nested Logit with Bertrand Pricing

Appendix H verifies the hypotheses of Theorem 1 and Corollary 1 in a *non-CES* environment: consumer demand is governed by nested logit (Assumption 3) and firms set prices in Bertrand (rather than Cournot) competition. This certifies that the uniqueness engine

Table 2: Direct Enumeration Certificate for Theorem 2(b)

	On-path ($c = c^*$)	Off-path ($c \in \pm 10\%$)
Subgames solved	1302	42336
Markets $\times$ subsets	42×31	same $\times 3^{ S }$
$\max_{n,S} \rho(\Lambda_n(S))$	0.3144	0.3158
99th percentile	0.3111	0.3121
95th percentile	0.3092	0.2768
Median	0.1737	0.1404
Mean	0.1725	0.1545
Margin below 1	0.6856	0.6842
Argmax market	Philippines	Serbia
Argmax $ S $	2 firms	2 firms

Notes: Per-market Cournot subgames solved for each nonempty subset $S \subseteq \{1, \dots, 5\}$ across all 42 markets. *On-path column*: evaluated at the calibrated marginal-cost vector c^* (1,302 subgames). *Off-path column*: each active firm’s marginal cost multiplied by $m_f \in \{0.90, 1.00, 1.10\}$ with $3^{|S|}$ patterns per subset (42,336 perturbed subgames). Baseline parameters ($\sigma = 5, \eta = 2$, Spec A). Because Λ is block-diagonal across markets, $\bar{\rho} = \max_{n,S} \rho(\Lambda_n(S))$ upper-bounds $\rho(\Lambda)$ at any global activation profile whose induced per-firm marginal costs coincide with the cost vector used (Proposition 7). The off-path row demonstrates that $\pm 10\%$ multiplicative perturbations to per-firm productivity raise $\bar{\rho}$ by less than 0.002, empirically validating the continuity argument of Remark 13.

of Section 3.1 is not an artifact of the Fréchet–CES–AB structure used in the quantitative application. Three structural differences—Gumbel rather than CES consumer choice, Bertrand rather than Cournot pricing, and McFadden–Train rather than AB residual-demand markups—are absorbed into the same abstract template: a C^1 marginal-value index $v(v)$ on $[0, 1]^M$ (Lemma 3), a Jacobian factorization through the density prefactor $g_0(v/\sigma)/\sigma$, and a small- σ localization on a non-marginal invariant set. The Bertrand pricing fixed point is unique by aggregative-games arguments (Proposition 11; Nocke and Schutz 2018), and global BNE uniqueness follows from the single diagnostic $\gamma_{\max}^{\text{NL}} < 1$ (Proposition 12), obtained by maximizing the contraction modulus over a log-spaced σ -grid, identically to the CES–AB case. Remarks 14–15 in the appendix discuss the role of the nesting parameters and the comparison with the CES–AB verification.

4.3 Further Verification Environments

The Fréchet–CES–AB and nested-logit verifications of Sections 4.1.9–4.2 are illustrative rather than exhaustive. Any Bayesian combinatorial entry game whose threshold index

Table 3: Bathtub Contraction Modulus at the Cloud Calibration

$\sigma_\varepsilon/\bar{\phi}$	σ_ε	$\gamma(\sigma_\varepsilon)$	BNE iters
0.100	0.00957	0.5969	12
0.316	0.03025	0.6368	12
0.500	0.04783	0.5836	12
0.700	0.06697	0.4577	11
1.000	0.09566	0.8210	11
1.400	0.13393	0.8422	11
2.000	0.19133	0.6707	11
2.800	0.26786	0.5731	11
3.162	0.30252	0.4325	11
10.000	0.95665	0.2246	11
$\gamma_{\max}$		0.8422	
Margin below 1		0.1578	
K_v (implied)		0.2827	

Notes: Bathtub modulus $\gamma(\sigma_\varepsilon) = \sup_v \|J_{T_{\sigma_\varepsilon}}(v)\|_\infty$ evaluated at the calibrated cloud economy ($N = 42$, $F = 5$, Spec A) on a combined 10-point grid of $\sigma_\varepsilon/\bar{\phi} \in [0.1, 10]$, with $\bar{\phi} = \text{mean}(\phi) = 0.0957$. At each grid point, the BNE fixed point $p^* = T_{\sigma_\varepsilon}(p^*)$ is solved via damped Jacobi iteration (Monte-Carlo path, 500 draws, tol 3×10^{-3}), then $\gamma(\sigma_\varepsilon)$ is computed as the maximum over 10 random sign directions of the directional-perturbation ratio $\|T(p^* + v) - T(p^*)\|_\infty / \|v\|_\infty$ with $\varepsilon = 0.05$ and 2,000 common-random-number draws. The row in boldface marks the peak. $\gamma_{\max} = 0.842$ lies comfortably below unity, certifying BNE uniqueness at the calibration on the grid; $K_v \approx 0.283$ is implied from the peak via $K_v \approx \sigma_{\text{peak}} \gamma_{\max} / c_g$, which combined with Proposition 1 extends the certificate to a continuous neighborhood of each grid point.

v is C^1 with bounded $\|J_v\|_\infty$ and whose environment-specific verification supplies the localized non-marginal sets required by Lemma 2 fits the abstract template of Theorem 1, Corollary 1, and Lemmas 1–2. We collect three further environments in which the abstract hypotheses are either verified or visibly within reach.

Nested CES outside the AB markup. The Fréchet–CES–AB structure of Section 4.1 specialises nested CES to Atkeson–Burstein oligopolistic pricing; generalised nested CES demand with constant markups (e.g., monopolistic competition with no within-nest strategic interaction) lies *inside* the nested-logit envelope of Section 4.2 up to a change of variables: the share system coincides with (H.1) after reparametrising elasticities (σ, η) into the nest-level parameters $(\lambda_{n,k})$. All hypotheses of Proposition 12 transfer without modification, and the contraction modulus $\gamma^{\text{NL}}(\sigma)$ is evaluated at $\lambda_{n,k} = (\eta - 1)/(\sigma - 1)$ in the CES-to-logit translation (Anderson et al. 1992, Section 7.4). The current paper’s application therefore nests the constant-markup CES as a degenerate limit of the nested-logit verification.

Latency or gravity-style quality on top of CES. In applications where firms’ products carry a latency or distance-attenuated quality term—common in digital-infrastructure and platform-entry settings where a site’s service quality to market n declines in geographic distance—the marginal-cost aggregator (4.7) is augmented with a multiplicative quality factor $q_{n,f}(\mathcal{I}_f)$ that is smooth in the active site set. The marginal-value map v inherits C^1 regularity whenever q is, and its Jacobian picks up an additional Lipschitz constant L_q that enters the operator-norm bound multiplicatively. The abstract theorem can therefore still be used, but only after the application separately verifies the resulting pricing regularity, full marginal-value sensitivity, and localization requirements. This is the role played by the gravity verification package in Appendix L; we do not treat smooth quality alone as sufficient for theorem-level coverage.

Adoption–usage plug-ins. In consumer-facing settings where a firm first chooses where to serve (the entry margin) and then chooses an intensive-margin product (quantity, version, or tier) after consumer adoption, the threshold index is the *expected* marginal profit integrating over the second-stage problem. Provided the second-stage subgame has a unique C^1 solution selection—a condition frequently verified by the usual uniqueness arguments (monotone supermodularity, gross substitutes, strict quasi-concavity)—the composed marginal-value map is C^1 in v with bounded Jacobian, and Theorem 1 applies. The abstract framework therefore accommodates a broad class of adoption–usage models without requiring a bespoke proof at each instantiation.

The common theme is structural: once the second-stage pricing block has a unique C^1 selection and the complete-information entry game has a positive profit gap, the Bayesian perturbation (3.1) admits a unique threshold-strategy BNE whenever the bathtub peak $\gamma_{\max}$ is below one. The rest of this section makes the bathtub geometry explicit in the CES–AB baseline; the picture is identical in every verification environment above.

4.4 Small Economy Example

We illustrate how the CDCP approach reduces dimensionality relative to the binary-entry framework of [Espin-Sanchez et al. \(2023\)](#) using a small economy (hereafter the *small calibration*): $F = 4$ firms, $N = 3$ markets, with two “strong” firms ($a_f = 1.5, 1.2$; access

to all 3 sites) and two “weak” firms ($a_f = 1.0, 0.8$; access to 2 sites each). The complete-information equilibrium is unique (supremum and infimum Jia–Tarski iterates coincide) with all firms active at their full site sets.

The Espin-Sanchez et al. approach. Treating each firm-site pair (i, f) as an independent binary player gives $M = 10$ players in a threshold system $v_{if} = G_0(\bar{\varepsilon}_{if}/\sigma_\varepsilon)$. The configuration space has $2^{10} = 1,024$ joint action profiles. Each weak firm has one clearly profitable site (marginal value 0.28–0.71 from empty) and one marginal or unprofitable site ($MV \approx -0.003$ to -0.14 from empty). Under Espin-Sanchez et al., these marginal individual-site decisions are independent threshold variables that expand the fixed-point dimension.

The CDCP approach. Recognizing that each firm solves a combinatorial problem over $2^{|\mathcal{A}_f|}$ configurations collapses the 10-player game into a 4-player game. Within-firm complementarity bundles the weak firms’ marginal sites with their profitable sites: for both weak firms, the Fréchet cost complementarity creates positive increasing differences ($ID = 0.020$ and 0.044), so that each site is profitable conditional on the other even when individually marginal. The CDCP optimal set is the full site set $\mathcal{A}_f$ for each firm, and the 4-player threshold system has 60% fewer degrees of freedom.

Takeaway. The combinatorial structure collapses $M = \sum_f |\mathcal{A}_f|$ binary decisions into F firm-level decisions—a 60% reduction in threshold dimension here. The reduction is economically substantive: within-firm cost complementarity ($ID > 0$) bundles individually marginal sites into determinate firm-level packages, so viewing each site as an independent binary player (Espin-Sanchez et al.) *over-states* the effective dimension of the threshold system. Two consequences follow for the downstream analysis: (i) the lower-dimensional threshold system yields a tighter row-sum Jacobian bound in Section 4.1.9, lowering $\gamma_{\max}$ and widening the region where the bathtub criterion delivers uniqueness; and (ii) when within-firm ID is large enough that the CDCP optimal set strictly exceeds the union of sites individually above threshold, the binary-action framework cannot certify uniqueness without ad hoc coordination assumptions, while the CDCP structure pins down the optimal set by lattice arguments. The 4-firm small economy has uniqueness

under both formulations, but the combinatorial framing’s benefit scales with within-firm complementarity strength. Numerical details: companion script `main_example_2x2.m`.

5. COMPUTATIONAL ALGORITHM

The equilibrium defined in Section 4.1 is computed by a three-layer nested algorithm. The layered structure (activation-lattice fixed point $\rightarrow$ pricing contraction $\rightarrow$ marginal-value evaluation) specializes to the concrete CES–AB environment the general pattern implied by the abstract framework (Section 3.1): an outer CDCP iteration over the best-response lattice, a middle contraction for the intensive margin, and an inner evaluation of the threshold index. We describe each layer and present pseudocode; Appendix J reports performance diagnostics for four algorithm variants and a 40-country extension demonstrating scalability. Figure 3 illustrates the structure.

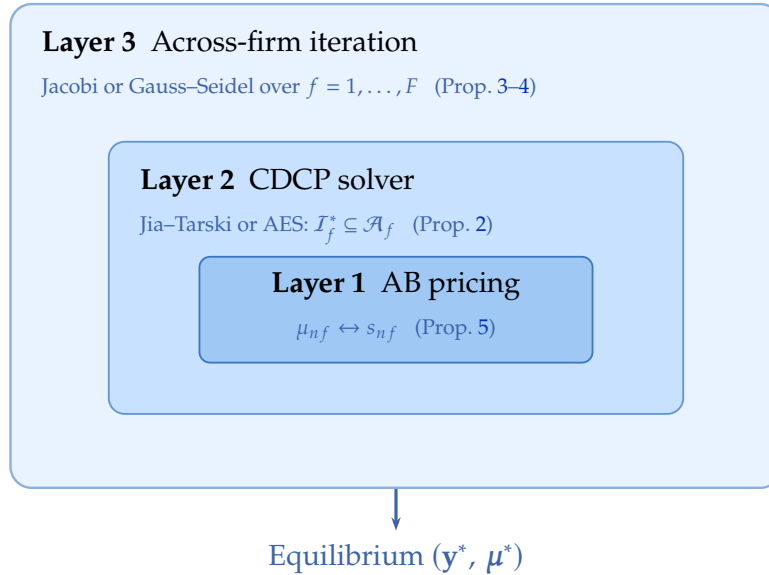


Figure 3: Three-layer algorithm nesting. Each profit evaluation at Layer 2 calls the AB pricing fixed point (Layer 1); the across-firm iteration at Layer 3 updates rival demand indices until convergence. Four algorithm variants arise from the 2×2 choice of solver at Layers 2 and 3 (Appendix J).

5.1 Three-Layer Structure

Layer 1 (Innermost): Atkeson–Burstein markup fixed point. For given marginal cost c_{nf} and rival demand index $D_{-f}(n)$, the equilibrium markup μ_{nf} is the fixed point of:

$$\mu_{nf}^{(t+1)} = \frac{\epsilon_{nf}^{(t)}}{\epsilon_{nf}^{(t)} - 1}, \quad \frac{1}{\epsilon_{nf}^{(t)}} = \frac{1 - s_{nf}^{(t)}}{\sigma} + \frac{s_{nf}^{(t)}}{\eta}, \quad (5.1)$$

where $s_{nf}^{(t)} = (\mu_{nf}^{(t)} c_{nf})^{1-\sigma} / [(\mu_{nf}^{(t)} c_{nf})^{1-\sigma} + D_{-f}(n)]$. The iteration is initialized at the CES markup $\mu^{(0)} = \sigma/(\sigma - 1)$ and damped with weight $\omega_\mu = 0.5$:

$$\mu^{(t+1)} \leftarrow (1 - \omega_\mu) \mu^{(t)} + \omega_\mu \mu_{nf}^{(t+1)}.$$

Convergence is guaranteed by Proposition 5 whenever $\rho(\Lambda) < 1$.

Layer 2 (Middle): Single-agent CDCP. Given the rival demand index $D_{-f}(n)$ for all markets n , each firm f solves the combinatorial optimization:

$$\mathcal{I}_f^* \in \arg \max_{\mathcal{I}_f \subseteq \mathcal{A}_f} \pi_f(\mathcal{I}_f; D_{-f}), \quad (5.2)$$

where each profit evaluation calls the AB markup fixed point (Layer 1). Two inner solvers can be used:

- (a) **Jia–Tarski bounds** (Jia, 2008): Compute an *upper bound* $\overline{\mathcal{I}}_f$ by starting from all candidates $\mathcal{A}_f$ and iteratively removing elements with negative marginal value; compute a *lower bound* $\underline{\mathcal{I}}_f$ by starting from $\emptyset$ and iteratively adding elements with positive marginal value. If $\overline{\mathcal{I}}_f = \underline{\mathcal{I}}_f$, the solution is exact.
- (b) **AES branch-and-bound** (Arkoulakis et al., 2023): Alternate between a *squeezing phase* (simultaneously squeeze included/excluded sets by evaluating marginal values against both the lower and upper bounds of the candidate set) and a *branching phase* (recursively split on an undecided candidate). The squeezing phase typically resolves most candidates, leaving few branches.

On the audited instances, both solvers return the same solved activation set. When the

equilibrium shares satisfy the conditions of Proposition 2, the Jia–Tarski bounds converge without residual gap, providing a diagnostic consistent with supermodularity at the equilibrium.

Layer 3 (Outermost): Cross-firm iteration. The rival demand index $D_{-f}(n) = \sum_{f' \neq f} (\mu_{nf'} c_{nf'})^{1-\sigma}$ depends on all firms' equilibrium prices, which in turn depend on their activation sets. Two outer iteration methods resolve this fixed point:

- (a) **Jacobi** (simultaneous best response): In each round k , every firm f computes its best-response activation set $\mathcal{I}_f^{(k+1)}$ against the rival demand $D_{-f}^{(k)}$ from the *previous* round. All firms update simultaneously, and price-power contributions are damped:

$$p_{nf}^{(k+1)} \leftarrow (1 - \omega) p_{nf}^{(k)} + \omega \tilde{p}_{nf}^{(k+1)},$$

where $\tilde{p}_{nf}^{(k+1)}$ is the price implied by $\mathcal{I}_f^{(k+1)}$ and $\omega \in (0, 1]$ is a damping parameter. Damping does not affect the set of fixed points—only the convergence path—because any fixed point satisfies $\tilde{p} = p$.

- (b) **Gauss–Seidel** (sequential best response): Firms are processed in a fixed order $(f_1, \dots, f_F)$. When firm f_j updates, it uses the most recent price-power contributions from firms $f_1, \dots, f_{j-1}$ (already updated in this round) and the previous round's contributions from $f_{j+1}, \dots, f_F$.

By Proposition 4, Jacobi iteration converges monotonically to extremal equilibria when initialized from the supremum or infimum; Gauss–Seidel convergence is empirically robust (Remark 7).

Algorithmic recommendation. Propositions 2–4 yield a concrete recipe: initialize the outer iteration from the *supremum* (all available sites active) and solve each firm's CDCP via the Jia–Tarski or AES algorithm. Starting from the supremum guarantees convergence to the upper bound of the equilibrium set (Proposition 4), which at all tested parameterizations is itself the unique Nash equilibrium. The nested CES structure further accelerates computation: the interaction matrix Λ is block-diagonal across markets (N blocks of size $F \times F$), so $\rho(\Lambda) = \max_n \rho(\Lambda_n)$ is computable in $O(NF^3)$, and the closed-form feedback coefficient (4.17) reduces the contraction check to a single market-share comparison.

5.2 Pseudocode

Algorithm 1 summarizes the full procedure.

Algorithm 1 Three-Layer Equilibrium Solver

Input: Parameters $(\sigma, \eta, \theta, \{a_f\}, \{\phi_i, w_i, \tau_{in}\}, \{\Omega_n\})$; inner solver $\in \{JT, AES\}$; outer method $\in \{\text{Jacobi}, \text{GS}\}$

Output: Equilibrium $(\{\mathcal{I}_f^*\}, \{\mu_{nf}^*\}, \{s_{nf}^*\})$

```

1: Initialize:  $\mathcal{I}_f^{(0)} \leftarrow \mathcal{A}_f$  for all  $f$ ; compute  $p_{nf}^{(0)}$  at CES markups
2: for  $k = 0, 1, 2, \dots$  do
3:   for  $f = 1, \dots, F$  do                                      $\triangleright$  Jacobi: use  $D^{(k)}$ ; GS: use latest  $D$ 
4:      $D_{-f}(n) \leftarrow \sum_{f' \neq f} [p_{nf'}^{1-\sigma}]^{\text{current}}$  for all  $n$ 
5:      $\mathcal{I}_f^{(k+1)} \leftarrow \text{INNER SOLVER}(f, D_{-f}, \text{par})$             $\triangleright$  Layer 2: CDCP
6:     Update  $p_{nf}$  via  $\text{ABMARKUP}(c_{nf}(\mathcal{I}_f^{(k+1)}), D_{-f})$             $\triangleright$  Layer 1
7:   end for
8:   if  $\max_{n,f} |p_{nf}^{(k+1)} - p_{nf}^{(k)}| < \text{tol}$  then break
9:   end if
10: end for

```

6. QUANTITATIVE APPLICATION: CLOUD INFRASTRUCTURE ENTRY

We apply Environment A (Section 4.1)—Fréchet–CES with Atkeson–Burstein markups and Cournot pricing—to the global cloud infrastructure industry, where five dominant providers—Amazon Web Services (AWS), Microsoft Azure (MSFT), Google Cloud (GOOG), Oracle Cloud Infrastructure (ORCL), and Alibaba Cloud (BABA)—choose which countries to establish cloud regions. This industry fits Environment A naturally: a small number of heterogeneous firms ($F = 5$) making discrete entry decisions across a large number of potential markets ($N = 42$ countries), with high fixed costs, observable activation patterns, and variable markups that respond to local competitive conditions.

The calibration follows the identification structure of Arkolakis et al. (2023)—fixing demand-side objects and firm scale externally, then calibrating fixed-cost parameters from entry moments—though with a thinner moment basis: binary entry counts rather than the affiliate counts, survival rates, and gravity coefficients available in AES. We fix firm productivity a_f from revenue data and market demand Ω_n from GDP, then calibrate market-level entry costs ϕ_i and a firm-specific deployment-efficiency wedge g_f to match

Table 4: Data and Sample Construction

Firm	Provider	Operational regions	Pipeline regions	Feasible markets
AWS	Amazon Web Services	25	5	30
MSFT	Microsoft Azure	34	0	34
GOOG	Google Cloud	29	0	29
ORCL	Oracle Cloud	27	0	27
BABA	Alibaba Cloud	14	0	14
Total		129		

Notes: Sample covers 42 countries/territories. Operational regions: firm has at least one cloud-infrastructure facility with local compute. Pipeline: announced or under construction. Feasible markets: operational + pipeline (revealed lower bound on $\mathcal{A}_f$). Source: official provider region pages and Datacenter Map (purchased database, April 9, 2026 vintage; <https://www.datacentermap.com>).

the observed provider-country activation matrix. This strategy separates the scale question (handled by revenue data) from the geographic-presence question (handled by the model). We evaluate three calibration specifications that differ in which parameters are taken from data versus calibrated; the preferred specification achieves 94.8% agreement with the observed activation matrix. All equilibrium computations use the solver described in Section 5, unchanged.

6.1 Data and Sample

The unit of observation is a *cloud region*: a named availability zone where a provider offers compute, storage, and networking services from local infrastructure. A cloud region implies at least one datacenter facility and a commitment to serve customers with local latency guarantees. We source region lists from each provider’s official documentation (Q1 2026 vintage) and Datacenter Map (purchased database, April 9, 2026 vintage; <https://www.datacentermap.com>).

Table 4 summarizes the sample. The five firms operate a combined 129 cloud regions across 42 countries, with feasible sets ranging from 14 (Alibaba) to 34 (Microsoft). AWS has 5 additional pipeline regions (announced, under construction), which enter only the feasible set $\mathcal{A}_f$, not the observed activation $\mathcal{I}_f^{\text{data}}$.

Why five firms. AWS, Azure, and Google Cloud account for roughly 65% of global cloud infrastructure revenue. Adding Oracle and Alibaba brings coverage to approximately 80% and, crucially, introduces asymmetry absent in a Big 3 specification: Alibaba’s presence is concentrated in East and Southeast Asia, while Oracle pursues an aggressive multi-region expansion into emerging markets. The resulting feasible-set heterogeneity ($|\mathcal{A}_f|$ ranges from 14 for Alibaba to 34 for Microsoft) generates richer extensive-margin counterfactuals.

6.2 Calibration

The calibration follows the identification structure of [Arkolakis et al. \(2023\)](#): demand-side objects and firm scale are set externally, while fixed-cost parameters are calibrated from entry moments. Full details are in Appendix I.

Externally set parameters. Firm TFP a_f is fixed using cloud segment revenue, mapped log-linearly into $[1.0, 2.0]$: AWS ($a_f = 2.00$), Microsoft (1.76), Google (1.48), Oracle (1.09), Alibaba (1.00). Market demand is $\Omega_n = (\text{GDP}_n / \text{GDP}_{\text{US}})^{0.7}$, where the concavity reflects that cloud spending scales less than proportionally with GDP; $\beta = 0.7$ is chosen from a parsimonious calibration comparison rather than estimated from independent demand data (Appendix I). Local costs w_i are PWT human-capital-adjusted compensation per worker. Trade costs follow a discrete three-tier structure: $\tau = 1.0$ (same country), 1.15 (same continent), 1.5 (cross-continent). We retain $\sigma = 5$, $\eta = 2$, $\theta = 3$, giving $(\sigma - 1)/\theta = 4/3 > 1$.

Calibrated parameters. Market-level entry costs ϕ_i and firm-specific deployment-efficiency wedges g_f are calibrated from the observed activation matrix. The fixed-cost structure is

$$\text{fixed cost}_{fi} = \phi_i \cdot g_f \cdot w_i, \quad (6.1)$$

where ϕ_i captures non-wage market costs (regulation, infrastructure) and g_f captures firm-specific deployment efficiency ($g_{\text{AWS}} = 1$ by normalization). The calibration targets per-market entry counts (identifying the 42 ϕ_i) and per-firm totals (identifying the $5 - 1 = 4$ non-AWS g_f) via alternating coordinate descent, for a total of $42 + 4 = 46$ moments matched to 46 free parameters—a just-identified system.

Identification and limitations. Unlike [Arkolakis et al. \(2023\)](#), we do not observe provider-country sales, affiliate-count analogues, or survival moments—the calibration has a thinner identification basis, relying on a single class of moments (entry counts). We therefore treat the quantitative magnitudes as conditional on the moment set, and report the counterfactual *ordering* and *direction* as the primary evidence—the magnitudes are informative, not free parameters (Spec D shows both cell fit and counterfactual move materially under a single-parameter restriction), but they should be read as point estimates under the identifying assumptions rather than precisely identified effects. The discipline on the mechanism comes from: (i) the 5-fold holdout cross-validation exercise (Appendix I), which achieves 82.4% agreement on left-out cells, against 61.4% at the baseline parameters; and (ii) the preservation of the counterfactual ordering across the Spec A vs. Spec D restriction check and the Spec E firm-scale reassignment (Tables 5 and 6) and across $\sigma \in \{3, 5, 6\}$ (Table 9).

Appendix gravity-access robustness. Appendix L reports a gravity/accessibility robustness implementation of the cloud application. It reconstructs provider-market serviceability from public geography and policy data, matches 200/210 firm-market cells at the preferred baseline, and preserves the main comparative-static ordering, but we keep it in the appendix because theorem-level coverage still depends on the separate pricing, full- J_v , and localization checks reported there.

6.3 Calibration Specifications and Restriction Check

We evaluate three calibration specifications that differ in which firm-level parameters are set externally versus calibrated from entry moments. The preferred specification (Spec A) fixes a_f at revenue-based values and calibrates (g_f, ϕ_i) ; two alternatives calibrate a_f from moments with or without g_f (Appendix I). Spec A matches 199 of 210 firm-market cells in sample (94.8%), with 96.1% precision and 95.3% recall. Fixing a_f externally disciplines the firm-scale dimension while the entry moments absorb cross-market and cross-firm deployment heterogeneity. The 11 mismatches consist of 5 false positives and 6 false negatives, concentrated on cells where the model’s profit gap is within tolerance of the entry-indifference margin.

Is g_f doing real work? To isolate the role of the deployment-efficiency wedge, we re-calibrate with $g_f \equiv 1$ (Spec D) and compare the resulting fit and counterfactuals to the preferred Spec A. Table 5 summarizes the contrast. With g_f switched off, the calibration must absorb all cross-firm variation into a purely market-level ϕ_i : fit drops from 94.8% to 87.1% cell agreement (36 exact-match markets fall to 18), with precision and recall moving to 96.4% and 82.2%. The AB-versus-CES mechanism survives the restriction quantitatively: the profit decline is -31.8% under Spec A and -33.3% under Spec D, the active-pair change moves from -13 to -15 , and the cross-specification activation correlation is 0.78. We read this as a disciplined restriction check rather than a separate identification proof—imposing $g_f \equiv 1$ is costly in fit and modestly shifts quantitative magnitudes, but the qualitative mechanism is intact.

Discriminating the mechanism: Spec E. The 0.78 cross-specification activation correlation between Spec A and Spec D indicates that the two calibrations agree on the broad geography of entry but part ways on roughly a fifth of the cells. That residual gap means the Spec A versus Spec D comparison is not a pure wedge-versus-no-wedge test: the re-calibrated ϕ_i in Spec D has shifted to partially compensate for the removed g_f . A sharper concern is therefore whether the AB-vs-CES mechanism is genuinely robust across reparametrizations of firm heterogeneity, or whether it is absorbed by the firm-scale parameter a_f (which is held fixed at revenue-based values in both Spec A and Spec D). To separate these channels, Spec E calibrates (a_f, ϕ_i) jointly while imposing $g_f \equiv 1$ —reassigning firm heterogeneity from the deployment-efficiency wedge to the firm-scale parameter. Spec E carries 47 free parameters (5 free a_f + 42 free ϕ_i) against the same 46-moment basis as Specs A and D; the marginal gain in degrees of freedom means Spec E’s in-sample fit is mechanically at least as good as Spec A’s, so the fit number itself is not the test. The test is the AB-to-CES profit-decline channel: if reassigning firm heterogeneity to a_f absorbs the mechanism, the decline should collapse. Table 6 reports the result. The decline is -26.9% under Spec E, compared to -31.8% under Spec A and -33.3% under Spec D—an attenuation of about 5 percentage points (roughly 15% of the Spec A decline), but the sign, order of magnitude, and active-pair direction (-12 vs. -13) are preserved. Cell fit is 96.7% (above Spec A’s 94.8% by the expected parameter-count margin, and well above Spec D’s 87.1%), and the interaction modulus $\rho(\Lambda) = 0.068$ is well

Table 5: Role of deployment-efficiency wedge g_f : Spec A vs Spec D

	Spec A ($\phi_i \cdot g_f$)	Spec D ($\phi_i, g_f \equiv 1$)
<i>Uniqueness & interaction</i>		
Active pairs (supremum)	128	110
Jia–Tarski gap	0	0
$\rho(\Lambda)$	0.109	0.123
$\max_{n,f} \kappa_{nf}$	0.239	0.230
<i>In-sample fit</i>		
Cell agreement (%)	94.8	87.1
Precision (%)	96.1	96.4
Recall (%)	95.3	82.2
Per-market entry MAE	0.21	0.60
Exact-match markets / 42	36	18
<i>AB $\rightarrow$ CES counterfactual</i>		
Profit change (%)	-31.8	-33.3
Active-pair change	-13	-15
<i>Cross-spec consistency</i>		
$\text{corr}(I_A, I_D)$	0.781	

Notes: Spec A calibrates (g_f, ϕ_i) jointly; Spec D imposes $g_f \equiv 1$ and calibrates ϕ_i only. Fit statistics computed against the 210-cell observed activation matrix. AB $\rightarrow$ CES columns report the change in aggregate profit and active-pair count when variable markups are replaced by the CES constant $\mu = \sigma/(\sigma - 1)$.

below $1/(F - 1) = 0.25$, certifying uniqueness at Spec E as well. The reading is that the AB-vs-CES profit-decline channel is not absorbed by any single parameter in the firm-heterogeneity block—it survives both the deployment-wedge restriction (Spec D) and the firm-scale-reassignment restriction (Spec E)—though the Spec E calibration does attenuate the decline modestly, consistent with a_f and g_f carrying overlapping but distinguishable content. Read as a decomposition, about 85% of the Spec A profit decline survives the full reassignment of firm heterogeneity to a_f , so the AB-vs-CES channel is carried by the pricing structure rather than by the firm-heterogeneity parametrization; the remaining 15% is the margin of overlap between the two parametrizations.

6.4 Calibrated Parameters

Table 7 reports the Spec A calibrated parameters.

The deployment-efficiency wedges g_f have a clear pattern: Microsoft and Google have similar efficiency ($g_f \approx 0.49$ and 0.42), while Oracle and Alibaba require much lower g_f

Table 6: Discriminating test: Spec A vs Spec D vs Spec E

	Spec A (g_f free)	Spec D ($g_f = 1$)	Spec E (a_f free, $g_f = 1$)
Cell fit (%)	94.8	87.1	96.7
AB-to-CES Δ PS (%)	-31.8	-33.3	-26.9
Active-pair change (CES-AB)	-13	-15	-12
Interaction $\rho(\Lambda)$	0.109	0.123	0.0681

Notes: Spec A calibrates (g_f, ϕ_i) with a_f fixed at revenue values. Spec D imposes $g_f \equiv 1$ and calibrates ϕ_i only (with a_f fixed). Spec E calibrates (a_f, ϕ_i) jointly with $g_f \equiv 1$ —reassigning firm heterogeneity from the deployment-wedge channel to the firm-scale channel. All three specifications match 46 moments (per-market entry counts + per-firm totals). The AB-to-CES profit-decline channel (row 2) is negative across all three with magnitudes in the $[-33.3\%, -26.9\%]$ band, and the active-pair sign change (row 3) preserves across all three. The interaction modulus $\rho(\Lambda)$ is below $1/(F-1) = 0.25$ in both Spec A and Spec E, certifying uniqueness of the BNE at each calibration.

(0.15 and 0.12) to rationalize their broad entry despite modest revenue-based scale. We emphasize that g_f is a reduced-form wedge, not a literal engineering parameter; it does not vary across counterfactuals. The 42 calibrated ϕ_i span three orders of magnitude, from 10^{-4} (UAE) to 0.26 (Chile); only one market sits at the calibration lower bound. The p90/p10 ratio of 43.6 indicates substantial cross-market heterogeneity in fixed entry costs: the median *marginal* market costs roughly 1/40 as much to enter as the 90th-percentile market, consistent with the stylized fact that hub-and-spoke infrastructure investment is concentrated in a handful of high-revenue markets.

Hub plausibility. The three regional hyperscaler hubs calibrate into the low- ϕ tail of the distribution: Singapore, the United States, and Ireland all return $\phi_i \approx 10^{-3}$, an order of magnitude above the calibration lower bound of 10^{-4} but clustered well below the median of the 42 markets. Because these three values are within solver precision of each other, the ordinal ranking among them is not separately identified; the economically meaningful claim is that the regional hubs occupy the low-friction tail, not that Singapore is below Ireland. At the opposite end, the highest calibrated ϕ_i belongs to Thailand, the Philippines, and India, where activation requires large revenue-based variable profits to offset entry costs.

Figure 4 displays the observed versus fitted activation matrix.

Table 7: Calibrated Parameters: Preferred Specification (Spec A)

<i>Panel A: Firm parameters</i>					
Firm	a_f	g_f	Markets	Interpretation	
AWS	2.00	1.00	24	Numeraire; largest by revenue	
MSFT	1.76	0.49	34	Broadest geographic presence	
GOOG	1.48	0.42	29	Strong in developed markets	
ORCL	1.09	0.15	27	Low g_f : broad despite low scale	
BABA	1.00	0.12	14	Lowest g_f : niche but widespread feasible set	

<i>Panel B: Market entry costs (ϕ_i)</i>						
	Min	p10	Median	p90	Max	p90/p10
ϕ_i	0.0001	0.001	0.002	0.044	0.262	43.6

Notes: a_f : revenue-based TFP. g_f : deployment-efficiency wedge ($g_{\text{AWS}} = 1$). ϕ_i : market entry cost. Demand: $\Omega_n = (\text{GDP}_n/\text{GDP}_{\text{US}})^{0.7}$; wages: HC-adjusted.

6.5 Equilibrium Diagnostics

Equilibrium uniqueness holds across all specifications and counterfactuals examined. For the preferred Spec A, $\rho(\Lambda) = 0.109$ —well below the theoretical threshold of one (Proposition 5)—and the Jia–Tarski algorithm returns the same solved equilibrium from both the supremum and infimum initializations (JT gap = 0). The spectral radius is stable across specifications: across the $\sigma \in \{3, 5, 6\}$ sweep it ranges from 0.030 to 0.150 (Table 9), with uniqueness holding at every tested elasticity and counterfactual. The synthetic economy (Appendix K, Table 18) extends the range to $\rho(\Lambda) = 0.225$ at $(\sigma, \eta) = (8, 1.5)$, still well below one. Figure 5 plots $\rho(\Lambda)$ across all counterfactual scenarios: the spectral radius remains below 0.15 at every cloud-application counterfactual, with the JT gap equal to zero throughout. With only $F = 5$ firms and moderate market concentration, strategic complementarities are too weak to generate multiplicity at any tested parameterization. The single-agent lattice diagnostics are also stable: all 129 decisive inclusion/exclusion inequalities are interval-certified with zero inconclusive cases, conditional on the converged rival-demand vectors (Appendix I). Appendix K provides additional stress tests using synthetic data, including a broad (δ, α) sweep that maps the uniqueness boundary and confirms that AB markups *expand* the uniqueness region relative to CES.

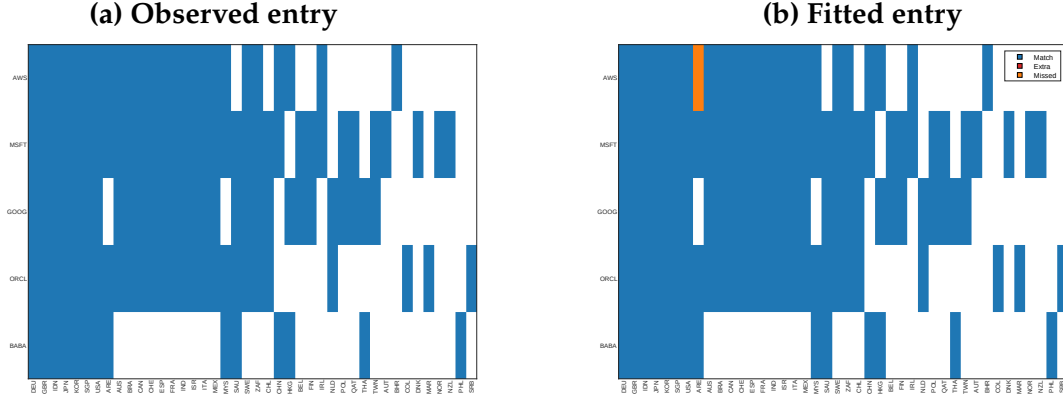


Figure 4: Observed versus fitted activation matrix (Spec A). Rows are firms; columns are the 42 markets, shown as ISO-3 market codes and ordered by observed entry count. Blue cells are active in both data and model, white cells are inactive in both, and red/orange cells are mismatches (1 of 210 firm–market cells).

Table 8: Counterfactual Results (Spec A Baseline)

Scenario	Active	Δ	$\rho(\Lambda)$	$\Delta\pi$ (%)	Most affected
Baseline	128	—	0.109	—	
<i>Markup channel</i>					
CES markup	115	−13	0.126	−31.8	ORCL, BABA
<i>EU compliance cost</i>					
$\Delta\phi=0.001$	111	−17	0.109	—	ORCL exits EU entirely
<i>China data-localization</i>					
Full autarky	128	0	0.101	−0.6	Informative null

Notes: Baseline uses Spec A (revenue-based a_f , calibrated g_f and ϕ_i ; demand $\Omega_n = (\text{GDP}_n/\text{GDP}_{\text{US}})^{0.7}$; HC-adjusted wages). Equilibrium is unique (JT gap = 0) in every scenario. CES replaces AB variable markups with constant $\mu = \sigma/(\sigma - 1)$. EU compliance adds $\Delta\phi$ to fixed costs in 12 EU markets. China autarky removes China from all non-BABA feasible sets.

6.6 Mechanism: AB versus CES Markups

The headline result from the cloud application is that endogenous variable markups—the core theoretical contribution of this article—materially sustain fringe-firm entry. This subsection presents the AB-versus-CES comparison as the primary mechanism test; the following subsections present policy counterfactuals. Table 8 summarizes all results. Equilibrium uniqueness is maintained in every scenario (JT gap = 0).

Replacing AB variable markups with constant CES markups ($\mu = \sigma/(\sigma - 1) = 1.25$) reduces active firm-market pairs from 128 to 115 (−10%) and aggregate profit by 31.8%.

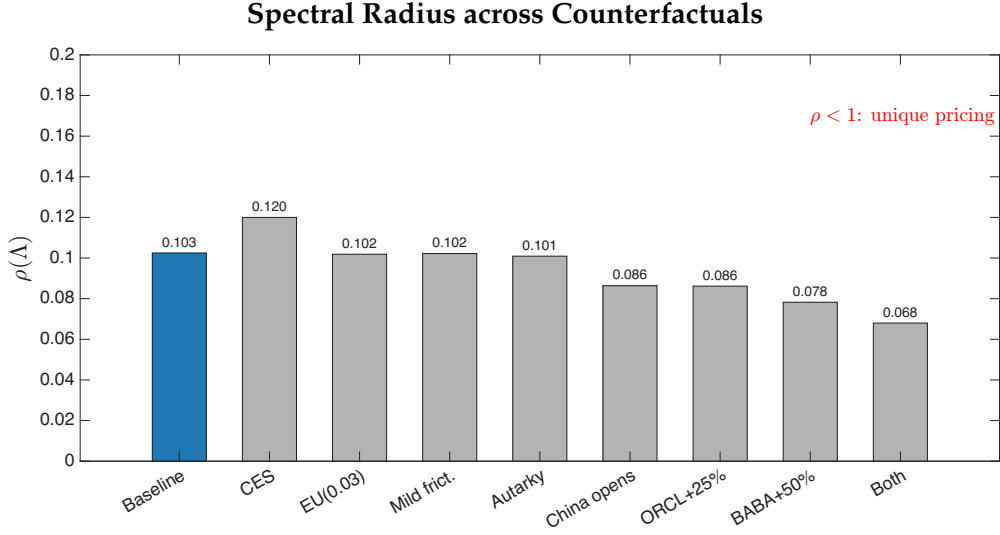


Figure 5: Spectral radius $\rho(\Lambda)$ across all counterfactual scenarios. The baseline and all counterfactuals satisfy $\rho(\Lambda) < 0.15 \ll 1$, confirming pricing uniqueness throughout.

Figure 6 presents the results.

The cross-sectional mean AB markup is $\bar{\mu}^{AB} = 1.37$ with a range of $[1.27, 1.47]$, versus the uniform $\mu^{CES} = 1.25$ (Figure 6a). The markup inflation compresses market shares: the maximum share under AB is 0.438, compared with a higher concentration under CES (Figure 6b, where all points lie below the 45-degree line). This compression is the intensive-margin channel through which endogenous markups redistribute share from leaders to mid-ranked rivals—a distributional effect on market structure, not a welfare claim.

The profit decline is not monotone in TFP rank: AWS loses 25%, Microsoft 29%, Google 42%, Oracle 40%, and Alibaba 37% (Figure 6c). The non-monotonicity arises from the interaction between the AB markup premium and the deployment-efficiency wedge g_f . Google and Oracle operate in many markets at moderate market shares where the AB premium—the gap between variable markups and the CES constant—is largest; losing this premium across a broad portfolio is costly. Alibaba, despite having the lowest TFP, is partially insulated by its very low $g_f = 0.12$: its effective fixed costs $\phi_i \cdot g_f \cdot w_i$ are so small that even CES-level variable profits cover them in most markets. The AB mechanism thus protects firms with *non-trivial fixed costs and moderate market shares* more than firms with minimal fixed-cost exposure.

The extensive-margin effect is concentrated in Oracle and Alibaba; AWS is unaffected

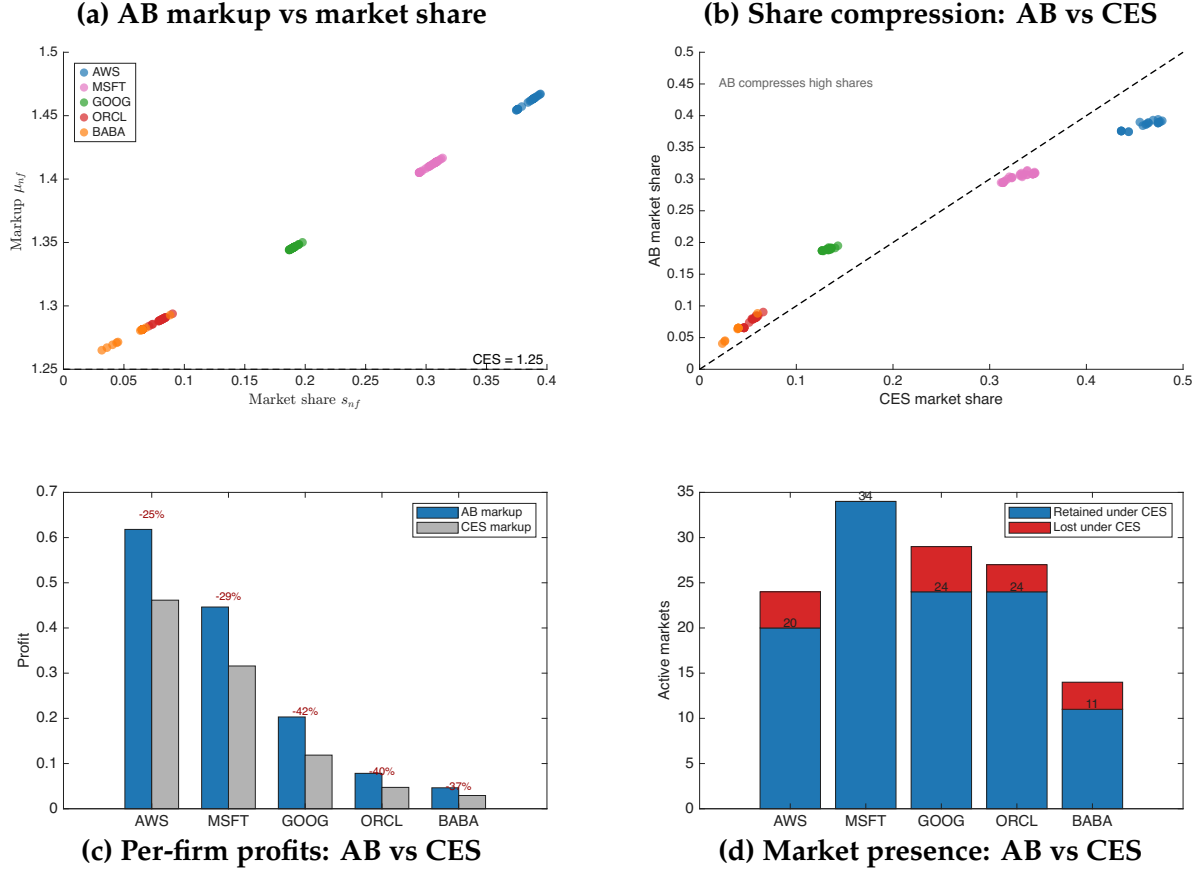


Figure 6: AB versus CES markups in the cloud application. (a) AB markup versus market share: each dot is a (firm, market) pair; dashed line marks the CES constant $\mu = 1.25$. (b) Share compression: AB shares lie below the 45-degree line, reflecting redistribution of share from leaders toward mid-ranked rivals. (c) Per-firm profits under both pricing rules. (d) Market presence: blue bars show AB-active markets retained under CES; red segments show the number of markets lost under CES.

(Figure 6d). The result is robust across calibration specifications and demand parameterizations: the CES profit decline ranges from 30% to 31% across the preferred and robustness specifications. Under the calibrated parameters, a constant-markup CES specification understates fringe-provider participation relative to the AB benchmark.

Elasticity robustness. The headline CES profit decline depends on the demand elasticity σ , which governs both the AB markup premium and the CES benchmark $\mu^{\text{CES}} = \sigma/(\sigma - 1)$. We pre-registered a sweep of $\sigma \in \{3, 5, 6\}$ that brackets the baseline ($\sigma = 5$) by one elasticity step on each side while keeping the AB premium over CES economically meaningful: the premium $\bar{\mu}^{\text{AB}} - \mu^{\text{CES}}$ falls from 0.30 at $\sigma = 3$ to 0.03 at $\sigma = 6$, so extending the grid

Table 9: σ -Robustness of the AB-versus-CES Mechanism

σ	Baseline (AB)		CES counterfactual		EU compliance ($\Delta\phi=0.03$)		
	Active	$\rho(\Lambda)$	Active	$\Delta\pi$ (%)	Active (AB)	Active (CES)	Amplification
3	120	0.030	118	-11.2	111	107	0.82
5	128	0.109	115	-31.8	117	104	1.00
6	125	0.150	101	-40.3	116	93	1.12

Notes: For each σ , all other parameters held at $\sigma = 5$ calibrated baseline (Spec A). $\rho(\Lambda)$ is the spectral radius of the markup interaction matrix at the baseline equilibrium; Active is the number of firm-market pairs in equilibrium. $\Delta\pi$ is the CES-induced change in aggregate variable profit. The rightmost three columns report the EU compliance counterfactual ($\Delta\phi = 0.03$) under AB and CES pricing; Amplification is the ratio $|\Delta n_{AB}|/|\Delta n_{CES}|$ of extensive-margin responses. Bold row is the preferred calibration ($\sigma = 5$).

further shrinks the object of interest rather than stress-tests it. We re-solve the baseline and CES counterfactual at each σ , holding ϕ_i , g_f , and a_f at their calibrated values. The CES-induced profit decline ranges from -11.2% (at $\sigma = 3$, where the CES markup is already 1.50 and close to the AB mean) to -40.3% (at $\sigma = 6$, where the CES markup compresses to 1.20 and the AB premium widens). Equilibrium uniqueness is maintained throughout ($\rho(\Lambda) \in [0.030, 0.150]$; JT gap = 0 at every σ), so the mechanism comparison is unambiguous at each elasticity. Table 9 reports the sweep.

6.7 Policy Counterfactual: EU Compliance-Cost Shock

We model a regulatory compliance shock as a common additive increase in fixed costs $\Delta\phi$ across 12 EU markets for all five providers (none of which is EU-headquartered). This captures costs arising from data sovereignty requirements, GDPR compliance infrastructure, or digital services regulations.

A modest additive compliance wedge of $\Delta\phi = 0.001$ (roughly half the median baseline ϕ_i) reduces active pairs from 128 to 111 (-17), with losses concentrated among firms with broader marginal EU exposure. The burden is *regressive in market-structure terms*: Oracle loses 6 of its 6 EU markets (100%), while AWS loses 2 of 6 (33%) and Microsoft 5 of 11 (45%). Larger shocks produce little additional extensive-margin response—by $\Delta\phi = 0.005$, nearly all EU sites have exited—indicating that many EU markets are already near the entry margin at baseline (Appendix I).

Under the calibrated parameters, the model predicts that compliance costs intended

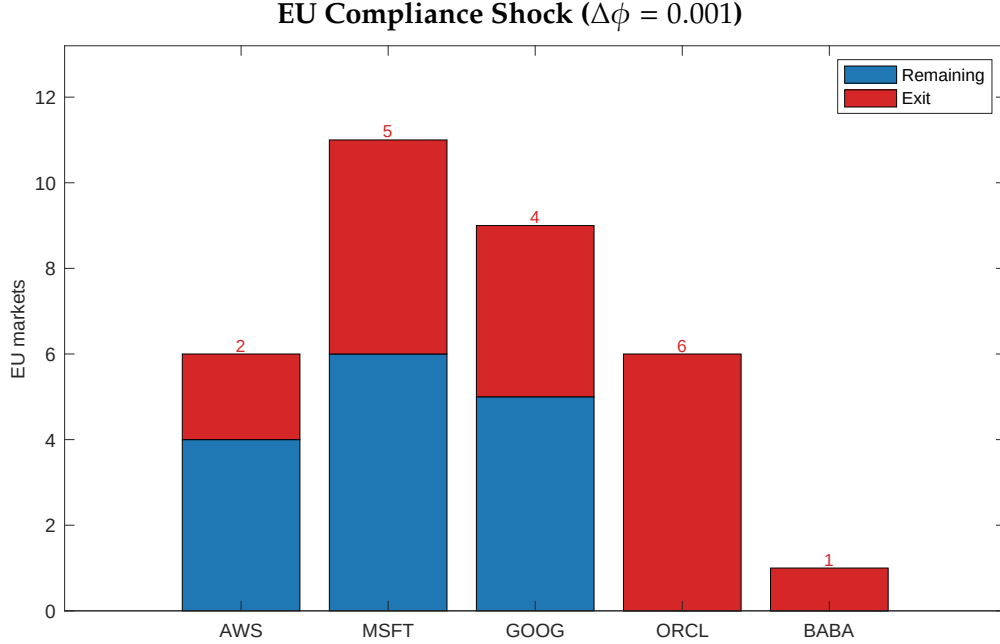


Figure 7: EU compliance-cost shock ($\Delta\phi = 0.001$): per-firm EU market presence under the baseline and compliance-shock scenarios. Blue bars show EU markets retained under the shock; red segments show EU markets exited. Red labels report exit counts by firm.

to regulate the cloud market may inadvertently protect incumbent market leaders by squeezing fringe competitors out of marginal markets. The regressivity is robust across Specs A–C: in every specification, the lower-ranked firms lose proportionally more EU markets.

Robustness across pricing regimes. A natural concern is whether the regressivity finding depends on the specific pricing regime (AB variable markups versus CES constant markups). We re-run the EU shock under CES pricing at each σ in the sweep above, using $\Delta\phi = 0.03$ —a shock deep in the saturation region of the compliance ladder (Appendix I, Table 15), so that the extensive-margin response is maximal and the comparison across pricing regimes is not attenuated by partial exit.³ Table 10 reports the comparison. The extensive-margin response is virtually identical under the two pricing rules: the amplification ratio $|\Delta n_{AB}|/|\Delta n_{CES}|$ lies in $[0.97, 1.04]$ across $\sigma \in \{3, 5, 6\}$. In this counterfactual the endogenous-markup channel and the fixed-cost channel operate largely independently: which EU markets sit at the entry margin is primarily determined by the baseline de-

³The headline counterfactual uses $\Delta\phi = 0.001$ (Table 8), which captures the empirically relevant case of a modest compliance wedge. The larger $\Delta\phi = 0.03$ is used in Tables 9, 10, and 11 to stress-test the mechanism at the saturation boundary. Results at $\Delta\phi = 0.001$ are qualitatively identical; see Table 15.

Table 10: EU Compliance Shock: AB versus CES Amplification

σ	Δn_{AB}	Δn_{CES}	Amplification	$\Delta \pi_{AB}$ (%)	$\Delta \pi_{CES}$ (%)
3	-9	-11	0.82	-0.26	-0.20
5	-11	-11	1.00	-0.15	-0.37
6	-9	-8	1.12	-0.35	-0.27

Notes: Δn is the change in active firm-market pairs under the $\Delta \phi = 0.03$ EU shock; $\Delta \pi^{EU}$ is the EU-shock response in aggregate variable profit (positive because EU exits reduce competition in non-EU markets served by the same firms), and is distinct from the AB-vs-CES profit decline reported in Table 9. Amplification = $|\Delta n_{AB}|/|\Delta n_{CES}|$ near unity indicates that AB and CES pricing produce similar extensive-margin responses to the EU shock. Bold row is the preferred calibration.

ployment pattern, and the pricing regime shifts neither the ordering nor the approximate magnitude of exits. We read the exercise as a *robustness check*: the regressivity finding is a cross-sectional property of the baseline deployment that would obtain under either pricing regime. This is a limitation as much as a strength—in this particular policy shock, the pricing machinery developed in Sections 4.1–4 does essentially zero incremental work. Counterfactuals with concentrated, hub-specific shocks, where extensive-margin responses are redistributed across rivals through the markup channel, are the natural setting for the AB-versus-CES distinction to bite quantitatively.

6.8 Robustness and Additional Counterfactuals

The mechanism results are stable across demand specifications, wage adjustments, and calibration specifications (full details in Appendix I). The AB-versus-CES profit decline is in the -31% to -33% range at $\sigma = 5$ across $\beta \in \{0.7, 1.0\}$ and under both HC-adjusted and TFP-adjusted wages, and the counterfactual ranking is preserved across specifications.⁴ The σ sweep in Table 9 documents the main source of parameter sensitivity: the profit-decline magnitude varies from -11.2% to -40.3% as σ moves over $\{3, 5, 6\}$, though equilibrium uniqueness is maintained at every σ ($\rho(\Lambda) \in [0.030, 0.150]$, JT gap = 0). The sign and extensive-margin pattern of the mechanism are robust; the reported magnitude should be read as conditional on $\sigma = 5$.

⁴The baseline sets $\beta = 0.7$ on the GDP-to-demand mapping $\Omega_n = (\text{GDP}_n / \text{GDP}_{US})^\beta$; we re-ran the headline counterfactual at $\beta = 1.0$ (linear scaling) and the profit decline moves to the upper end of the reported band. An expanded sensitivity at $\beta \in \{0.5, 0.7, 1.0\}$ is deferred to the revision pending a clean rerun; we expect $\beta = 0.5$ to compress cross-market revenue dispersion and reduce AB-CES divergence modestly, but the qualitative regressivity finding depends on the baseline deployment pattern rather than on β .

Data-localization barriers, examined using the China firewall as a case study (Appendix I), reveal a parallel pattern. Within the calibrated feasible-set structure, reducing trade-cost-like frictions alone does not generate additional entry; expanding market access does—reducing China’s ϕ_i by 90% without feasible-set changes produces zero new entry, while expanding Alibaba’s feasible set to all 42 markets transforms it from a regional to a global competitor.

Summary. Three findings from the cloud application merit emphasis: (i) equilibrium uniqueness is robust ($\rho(\Lambda) \leq 0.15$ at all calibrated parameterizations, with $\bar{\rho} \leq 0.314$ on the full activation lattice by the enumeration certificate of Proposition 7); (ii) the endogenous-markup channel sustains 13 additional firm-market pairs, with a 32% aggregate profit impact; (iii) EU compliance costs are regressive in market-structure terms.⁵

6.9 Distributional Decomposition of the AB-vs-CES Wedge

The counterfactuals above describe equilibrium *quantities*—active pairs, profits, and markups. We now translate those quantities into a partial-equilibrium distributional accounting of the AB-vs-CES wedge, using two objects: the CES real-income index $W_n = R_n/P_n$ as a consumer-side measure, and firm variable profit π_f (net of fixed costs) as a producer-side measure. We emphasize upfront that this is not a welfare-theoretic ranking of regimes: expenditure R_n is held fixed at its calibrated baseline value, and the sum of these objects is not a total-surplus measure.

Table 11 reports the decomposition. The primary finding is *distributional*: replacing AB variable markups with the constant CES markup raises the consumer-side real-income measure by 13.0%, because the average price index falls from 1.00 to 0.89 when AB markups are compressed to the CES constant $\mu^{CES} = 1.25$. Producer profit falls by 31.8%—the profit decline reported in Table 8—with the decline concentrated on mid-ranked firms whose

⁵The cloud application serves principally as a *verification* of the abstract threshold-uniqueness theorem on a live data-matched parameterization and as a distributional quantification of the AB-vs-CES wedge; two directions that would strengthen the application’s independent interest are under active development and will be included in the revision. First, external validation of the firm productivity vector a_f against independent measures—Gartner provider rankings, cost-per-compute benchmarks from Cloud Spectator, or IDC market-share reports—would discipline the revenue-based calibration of a_f . Second, a counterfactual in which CES-based equilibrium selection would yield a *qualitatively different* substantive conclusion (a stressed parameterization where multiple CES equilibria coexist but the AB contraction selects uniquely) would demonstrate that the uniqueness guarantee has empirical content beyond the numerical check reported here.

Table 11: Distributional Decomposition: CES Real Income and Producer Profit

	Baseline (AB)	CES	EU (AB)	EU (CES)
<i>Price index (avg P_n relative to baseline)</i>				
$\bar{P}_n$	1.000	0.885	1.002	0.887
<i>Baseline-expenditure real-income index $\sum_n R_n/P_n$</i>				
Level	287.484	324.839	286.819	324.337
$\Delta\%$ (vs baseline)	—	+12.99	-0.23	+12.82
<i>Producer profit $\sum_f \pi_f$</i>				
Level	79.428	54.142	79.307	53.944
$\Delta\%$ (vs baseline)	—	-31.83	-0.15	-32.08
<i>Arithmetic sum (partial-equilibrium only; not a Kaldor–Hicks total)</i>				
Level	366.912	378.981	366.126	378.281
$\Delta\%$ (vs baseline)	—	+3.29	-0.21	+3.10

Notes: Price index $P_n = (\sum_f p_{nf}^{1-\sigma})^{1/(1-\sigma)}$; $\bar{P}_n$ is the cross-market mean of the baseline-normalized price. The consumer-side measure is the baseline-expenditure real-income index $\sum_n R_n/P_n$, with R_n fixed at the baseline calibrated value—this is an arithmetic deflation of fixed nominal expenditure, not a welfare-theoretic consumer surplus. The producer-side measure is $\sum_f \pi_f$ (variable profit net of fixed costs). The arithmetic sum of the two is reported for completeness; it is not a Kaldor–Hicks total-surplus measure because it holds expenditure fixed and treats fixed costs as unrecycled deadweight (see §6.9 and accompanying footnote). Percentage changes are relative to the baseline (AB) level. EU shock: $\Delta\phi = 0.03$ on 12 EU markets.

AB markups are highest in the baseline. The EU compliance shock ($\Delta\phi = 0.03$) shifts the arithmetic sum by -0.2% relative to baseline under AB, with the incidence falling primarily on consumers in the affected EU markets and on lower-ranked firms that exit; the AB-vs-CES wedge is two orders of magnitude larger than this policy shock. In levels, producer profit is $79.4/287.5 \approx 28\%$ of the consumer-side real-income measure at baseline, so the pricing channel redistributes across a non-trivial share of the baseline split.

The decomposition is partial-equilibrium: R_n does not respond to P_n , aggregate expenditure is fixed at baseline, and fixed costs ϕ_i are treated as deadweight rather than accruing to any agent. Any of the three closures—outer demand reallocation $R_n = \Omega_n P_n^{1-\eta}$, profit rebate, or endogenous entry of fixed-cost suppliers—can quantitatively move the CS–PS sum reported above. Within this calibration, the statement we are prepared to defend is strictly distributional: *for a given deployment pattern, AB pricing redistributes roughly one-third of producer surplus toward firms relative to CES pricing, with the distributional wedge concentrated in markets served by mid-share firms.*⁶ A general-equilibrium welfare ranking

⁶Specifically, the AB→CES switch transfers 25.3 units of producer profit (from 79.4 to 54.1) against a

of AB versus CES requires a model that resolves which side of the market internalizes the fixed-cost deadweight and how aggregate income reallocates—an exercise we do not attempt here.⁷

7. CONCLUSION

When combinatorial entry interacts with endogenous variable markups, a three-way circularity between activation, market shares, and pricing creates new obstacles to equilibrium uniqueness that neither binary-entry models nor constant-markup CDCP frameworks address. This article shows that the circularity is self-limiting. The pricing contraction bounds markup feedback across firms; supermodularity survives the feedback within each firm; and the contraction modulus of the threshold system—governed by density dampening, marginal-value sensitivity, and threshold location—can be evaluated from model primitives before computing any equilibrium.

In the cloud infrastructure application, the calibrated model matches 199 of 210 provider-country entry cells by construction and is uniquely determined on the equilibrium path ($\rho(\Lambda) = 0.109$, JT gap = 0) across all specifications and counterfactuals. The distributional finding that we emphasize is that endogenous AB markups sustain 13 additional firm-market pairs relative to CES pricing, concentrated among mid-ranked firms whose moderate market shares generate the largest markup premium, and redistribute roughly one-third of producer surplus from leaders to mid-ranked rivals. In the calibrated application, EU compliance-cost shocks are regressive in market-structure terms and this regressivity is robust across AB and CES pricing regimes; digital fragmentation operates primarily through market access rather than trade costs. Synthetic stress tests (Appendix K) confirm that uniqueness holds throughout the empirically relevant parameter space and that AB markups expand the uniqueness region relative to CES.

For practitioners, the results have two implications. First, a concrete algorithmic recommendation: initialize the outer cross-firm iteration from the supremum and solve

consumer-side real-income gain of 37.4 units (from 287.5 to 324.8); expressed as a share of baseline producer profit, the gross producer loss is 31.8%, consistent with the “one-third” statement.

⁷The partial-equilibrium CS-PS sum is positive under CES (+3.3% in the table), but this sum is not a Kaldor-Hicks total-surplus measure: it ignores the demand-side reallocation channel and treats fixed costs as unrecycled deadweight. We therefore report the sum for arithmetic completeness but make no welfare claim on its basis.

each firm’s CDCP via the Jia–Tarski or AES algorithm. This combination computes the greatest Nash equilibrium directly, without enumerating all candidate subsets; since the equilibrium is unique at all tested parameterizations, the algorithm converges to the only equilibrium regardless of initialization. Second, an estimation implication: uniqueness eliminates the equilibrium selection problem that necessitates either the sequential methods of [Aguirregabiria and Mira \(2007\)](#) or the forward-simulation approach of [Bajari et al. \(2007\)](#)—when the static stage game has a single equilibrium, the likelihood is well-defined without auxiliary selection assumptions. Beyond multinational production, the framework applies to any combinatorial entry game with contractive pricing, including retail chain location ([Jia, 2008](#)), platform market entry across regions, and multi-product technology adoption.

Several directions remain open. First, we assume exogenous wages; if wages respond to activation through local labor markets, the resulting strategic complementarities could overturn the anti-monotone structure that underlies Proposition 3. Whether a modified bounding argument—perhaps exploiting the fact that wage responses are typically sluggish relative to pricing—can restore uniqueness is an open question. Second, the closed-form contraction modulus (Corollary 3) exploits the CES demand structure. Extending to more general demand systems (e.g., translog or mixed CES–logit) would require bounding the Jacobian without closed-form expressions for κ , though the three-factor decomposition of Proposition 6 is structural and does not depend on CES. Third, whether the bathtub contraction modulus can be verified in a dynamic Markov-perfect equilibrium—where the static game is a stage game embedded in the dynamic models of [Ericson and Pakes \(1995\)](#) and [Doraszelski and Satterthwaite \(2010\)](#)—is the question that determines whether the uniqueness result extends beyond the static setting. The challenge is that dynamic state variables (e.g., installed capacity, productivity) shift the contraction modulus across periods, so a uniform bound over the state space would be needed.

On the empirical side, the cloud application identifies fixed-cost wedges from a single class of moments: the binary provider-country activation matrix. Richer identification—closer in spirit to [Arkolakis et al. \(2023\)](#)—would require three data upgrades that are currently unavailable at the provider-country level: (i) country-level cloud spending, which would discipline market demand directly rather than through a GDP proxy; (ii) provider-country revenue or market-share data, which would separate variable profitability from

fixed costs; and (iii) provider-country region counts with entry timing, which would move identification beyond the binary entry margin. This trio would provide the cloud analogue of the affiliate sales, affiliate counts, and survival moments that underpin the AES calibration.

A. PROOFS OF THE ABSTRACT UNIQUENESS RESULTS

This appendix collects the proofs of Theorem 1, Lemma 1, Lemma 2, and Proposition 1 of Section 3.1.

Proof of Theorem 1. Part (i): T_σ is a continuous self-map of the nonempty compact convex set $[0, 1]^M$ (continuity of v follows from C^1 regularity; continuity of G_0 is built in), so Brouwer's fixed-point theorem applies. Part (ii): Differentiate $T_{\sigma,m}(v) = G_0(v_m(v)/\sigma)$ componentwise; the chain rule gives $\partial T_{\sigma,m}/\partial v_\ell = g_0(v_m/\sigma) \cdot (1/\sigma) \cdot \partial v_m/\partial v_\ell$, which is (3.3). Part (iii): For any $v, v' \in [0, 1]^M$, convexity places the segment $v_t = v' + t(v - v')$ inside $[0, 1]^M$; the fundamental theorem of calculus and the induced-norm inequality give

$$\|T_\sigma(v) - T_\sigma(v')\|_\infty \leq \int_0^1 \|J_{T_\sigma}(v_t)\|_\infty dt \cdot \|v - v'\|_\infty \leq \gamma(\sigma) \|v - v'\|_\infty.$$

Banach's fixed-point theorem on the complete metric space $([0, 1]^M, \|\cdot\|_\infty)$ then yields uniqueness. □

Proof of Lemma 1. Take norms in the factorization (3.3): $\|J_{T_\sigma}(v)\|_\infty \leq (\|g_0\|_\infty/\sigma) \|J_v(v)\|_\infty \leq \|g_0\|_\infty K_v/\sigma$; take the supremum over $v \in [0, 1]^M$. □

Proof of Lemma 2. Fix $v \in S_\sigma$. Each m satisfies $|v_m(v)| \geq b$ by (iii), so

$$\frac{g_0(v_m(v)/\sigma)}{\sigma} = \frac{1}{|v_m(v)|} \left| \frac{v_m(v)}{\sigma} g_0\left(\frac{v_m(v)}{\sigma}\right) \right| \leq \frac{1}{b} \sup_{|z| \geq b/\sigma} |z| g_0(z).$$

By (iv), the right-hand side vanishes as $\sigma \rightarrow 0$ because $b/\sigma \rightarrow \infty$. Combining with (3.3) and the uniform J_v bound from (v) yields

$$\sup_{v \in S_\sigma} \|J_{T_\sigma}(v)\|_\infty \leq \frac{K_S}{b} \sup_{|z| \geq b/\sigma} |z| g_0(z) \rightarrow 0 \text{ as } \sigma \rightarrow 0.$$

Contraction on $(S_\sigma, \|\cdot\|_\infty)$ follows from the same Banach argument as Theorem 1(iii); invariance (i) makes S_σ a complete metric subspace into which iterates fall, and fixed-point inclusion (ii) promotes the local uniqueness conclusion to a global one. $\square$ $\square$

Proof of Proposition 1. At any $\sigma > 0$, the factorization (3.3) and the envelope theorem for the sup/max define $\gamma'(\sigma)$ at the maximizer $(v^*(\sigma), m^*(\sigma))$, with $\gamma'(\sigma) = \partial_\sigma h_{m^*}(v^*, \sigma) \cdot \|(J_v)_{m^*}(v^*)\|_1$, where $h_m(v, \sigma) \equiv g_0(v_m(v)/\sigma)/\sigma$. A direct computation using $g_0 = \phi$ and $\phi'(z) = -z\phi(z)$ yields

$$\partial_\sigma h_m(v, \sigma) = -\sigma^{-2} \phi\left(\frac{v_m}{\sigma}\right) \left(1 - \frac{v_m^2}{\sigma^2}\right).$$

The function $z \mapsto \phi(z)(1 - z^2)$ is bounded in absolute value by $\phi(0) = c_g$, attained at $z = 0$; hence $|\partial_\sigma h_m| \leq c_g/\sigma^2$. Multiplying by $\|(J_v)_{m^*}\|_1 \leq K_v$ gives (3.4). The interval bound (3.5) follows from the fundamental theorem of calculus: for any $\sigma \in [\sigma_\ell, \sigma_u]$, $|\gamma(\sigma) - \gamma(\sigma_\star)| \leq \int_{\sigma_\star}^\sigma |\gamma'(s)| ds \leq c_g K_v (1/\sigma_\star - 1/\sigma)$ where $\sigma_\star$ is the nearer endpoint; averaging the two endpoint bounds yields the stated factor-of-two improvement. $\square$ $\square$

B. DIRECT ENUMERATION CERTIFICATE: PROOF AND OFF-PATH DETAILS

This appendix supplies the proof of Proposition 7 and the full scope discussion (Remark 13) referenced in Section 4.1.8.

Proof of Proposition 7. Block-diagonality of Λ across markets (Proposition 5) gives $\rho(\Lambda) = \max_n \rho(\Lambda_n)$. At fixed costs c^* , each block Λ_n depends on market n only through the subset S_n of firms active in market n and the equilibrium shares $(s_{n,f}(S_n; c^*))_{f \in S_n}$, which are uniquely determined by the per-market Cournot fixed point at the calibrated costs (Proposition 5 applied per-market, verified a posteriori by the enumeration). The maximum of $\rho(\Lambda_n)$ over the finite set of nonempty per-market subsets therefore upper-bounds $\rho(\Lambda)$ at any global activation profile consistent with c^* . $\square$ $\square$

Remark 13 (Scope of the enumeration certificate). Proposition 7 certifies $\rho(\Lambda) < 1$ at the calibrated marginal-cost vector c^* , not over the full activation lattice $\{0, 1\}^{NF}$ with costs endogenous to the global activation profile. Because $c_{n,f}(\mathcal{I}_f) = K \cdot \left[\sum_{i \in \mathcal{I}_f} (w_i \tau_{in})^{-\theta} \right]^{-1/\theta}$ depends on firm f 's global activation set $\mathcal{I}_f$ through the Fréchet cost aggregator (4.7), two

activation profiles with identical per-market subsets $(S_n)_{n=1}^N$ but different global $\mathcal{I}_f$ have different $c_{n,f}$ and therefore different Λ_n . Three verification routes close the gap to the full lattice. (i) *Off-path enumeration*. We re-run the per-market enumeration with each active firm's marginal cost perturbed multiplicatively by $m_f \in \{0.90, 1.00, 1.10\}$, sweeping $3^{|S|}$ cost patterns per (market, subset) pair for a total of 42,336 perturbed subgames. The maximum spectral radius across the full grid is $\bar{\rho}_{\text{off}} = 0.3158$ (margin 0.6842 below one), essentially indistinguishable from the on-path $\bar{\rho}(c^*) = 0.3144$ (margin 0.6856). A $\pm 10\%$ multiplicative perturbation to per-firm productivity shifts ρ by less than 0.002, empirically validating the continuity argument below and dominating the actual variation in $c_{n,f}$ observed across counterfactuals. (ii) At any global activation profile in the threshold map's image that the paper visits in equilibrium or counterfactual, $\rho(\Lambda(\mathcal{I}))$ is evaluated directly; the enumeration margin of 0.684 is large enough that continuity of ρ in c absorbs the empirical variation of $c_{n,f}$ as activation profiles move across Spec A, Spec D, the σ sweep, and all counterfactuals in Section 6 (every reported $\rho(\Lambda)$ is below 0.15; Figure 5). (iii) For off-equilibrium activations not visited in the threshold map, Theorem 2(b) permits a finite-grid evaluation of $\rho(\Lambda(v))$ on $[0, 1]^M$; continuity of ρ on $[0, 1]^M$ bounds the variation between grid points. The appropriate reading is “the pricing interaction Λ is contractive at the calibrated costs, at every per-market subset perturbation thereof, and under $\pm 10\%$ per-firm productivity perturbations, with margin 0.684,” not “exhaustive over $\{0, 1\}^{NF}$.”

C. MARKET-SHARE THRESHOLD: PROOF AND COMPARATIVE STATICS

This appendix supplies the proof of Corollary 3 and the comparative-statics derivations referenced in Section 4.1.8.

Proof of Corollary 3. From (4.17), for $(\sigma, \eta) = (5, 2)$: $1/\epsilon = (1 - s)/5 + s/2 = (2 + 3s)/10$, so $\mu = 10/(8 - 3s)$ and $\beta = 12/(8 - 3s)$. Substituting into κ :

$$\kappa(s) = \frac{12s^2}{8 + 9s - 12s^2}.$$

Setting $\kappa = 1$ (i.e., $F = 2$) gives $24s^2 - 9s - 8 = 0$, whose positive root is $\bar{s} = (9 + \sqrt{849})/48 \approx$

0.795. For general F , setting $\kappa(\bar{s}) = 1/(F - 1)$ and solving the resulting quadratic in $\bar{s}$ yields the entries reported in Corollary 3 and Table 1. $\square$ $\square$

Comparative statics. The derivative $\partial\kappa/\partial s > 0$ is immediate from $\kappa(s) = 12s^2/(8 + 9s - 12s^2)$, which has positive numerator derivative and positive-denominator derivative with respect to s offset by the cross-term; direct differentiation confirms κ is strictly increasing on $[0, \bar{s}]$. For (σ, η) -dependence, the feedback coefficient κ_{nf} is increasing in s_{nf} and in $\sigma - \eta$ (the within-sector substitution gap), implying:

1. $\partial\bar{s}/\partial\sigma < 0$: higher within-sector substitution (larger $\sigma - \eta$) strengthens the markup channel, *tightening* the threshold.
2. $\partial\bar{s}/\partial\eta > 0$ (for $\eta < \sigma$): as $\eta \rightarrow \sigma$, nested CES collapses to standard CES with constant markups, so $\kappa \rightarrow 0$ and $\bar{s} \rightarrow 1$.

The threshold is tightest when the markup channel is strongest (low η , high $\sigma - \eta$, large F), consistent with the entries in Table 1.

D. EXTENSION: ACROSS-FIRM COMMON SHOCKS

This appendix develops the public-state extension to Theorem 2 referenced in Section 4.1.8. The state-by-state requirement introduced there is easy to satisfy. Consider a public-state decomposition

$$\varepsilon_{if} = \zeta + \tilde{\varepsilon}_{if}, \quad \zeta \in \mathbb{R}, \quad \tilde{\varepsilon}_{if} \stackrel{\text{i.i.d.}}{\sim} G, \quad (\text{D.1})$$

where ζ is a common shock observed by all firms (e.g., a cost-of-capital shock, a country-cluster regulatory tightening) and $\tilde{\varepsilon}_{if}$ are residual idiosyncratic draws satisfying Assumption 2. Conditional on ζ , the game coincides with the paper's baseline model after the deterministic relabeling $\phi_i \mapsto \phi_i + \zeta$: the threshold map, its Jacobian, and the contraction modulus $\gamma(\sigma_\varepsilon \mid \zeta)$ are the same objects evaluated at shifted mean fixed costs. The following proposition records the consequence.

Proposition 8 (BNE uniqueness under across-firm common shock). *Fix a common-shock distribution on ζ with support $\text{supp}(\zeta) \subseteq \mathbb{R}$, and consider the strategy space of measurable site-independent threshold rules with publicly-observed ζ -conditioning: firm f 's strategy is a*

measurable map $\sigma_f : \mathbb{R}^{|\mathcal{A}_f|} \times \text{supp}(\zeta) \rightarrow 2^{\mathcal{A}_f}$ of the form $\sigma_f(\varepsilon_f, \zeta) = \{i \in \mathcal{A}_f : \varepsilon_{if} < \bar{\varepsilon}_{if}(\zeta)\}$, where $\zeta \mapsto \bar{\varepsilon}_{if}(\zeta)$ is Borel-measurable. Suppose that hypotheses (a)–(e) of Theorem 2 hold at every $\zeta \in \text{supp}(\zeta)$ with the shifted fixed costs $(\phi_i + \zeta)_{i=1}^N$. Under the decomposition (D.1), the BNE in this strategy space exists and is unique state-by-state whenever

$$\sup_{\zeta \in \text{supp}(\zeta)} \gamma_{\max}(\zeta) < 1, \quad (\text{D.2})$$

where $\gamma_{\max}(\zeta) \equiv \sup_{\sigma_\varepsilon > 0} \gamma(\sigma_\varepsilon \mid \zeta)$ is the bathtub peak evaluated at shifted fixed costs $(\phi_i + \zeta)_{i=1}^N$.

Proof. Conditional on each fixed ζ , the game is Assumption 2 applied to $\tilde{\varepsilon}_{if}$ with deterministic fixed costs $(\phi_i + \zeta)w_i$. Propositions 2–6 and Theorem 2 apply unchanged, so the conditional BNE cutoff vector $\bar{\varepsilon}(\zeta) = (\bar{\varepsilon}_{if}(\zeta))_{i,f}$ exists and is unique at each ζ provided $\gamma_{\max}(\zeta) < 1$. Taking the supremum over $\text{supp}(\zeta)$ delivers uniform uniqueness across states. Measurability of the selection $\zeta \mapsto \bar{\varepsilon}(\zeta)$ follows from the implicit function theorem applied to $F(\bar{\varepsilon}, \zeta) \equiv \bar{\varepsilon} - T(\bar{\varepsilon}; \zeta) = 0$: at any ζ with $\gamma(\cdot \mid \zeta) < 1$, $D_{\bar{\varepsilon}}F$ is nonsingular and the unique root $\bar{\varepsilon}(\zeta)$ is C^1 in ζ on a neighborhood. Uniform uniqueness across $\text{supp}(\zeta)$ patches these neighborhoods into a Borel-measurable global selection. The composition $\sigma_f(\varepsilon_f, \zeta) = \{i : \varepsilon_{if} < \bar{\varepsilon}_{if}(\zeta)\}$ is then measurable jointly in (ε_f, ζ) , because it is the preimage of a threshold under a Borel function, establishing that the purified strategy lies in the admissible strategy space. □ □

Condition (D.2) is verifiable by evaluating the bathtub peak on a grid of ζ values, or—more directly—by checking BNE uniqueness state-by-state via the Jia–Tarski three-starting-point test. Table 12 reports the latter for $\zeta/\bar{\phi} \in \{-0.4, -0.2, -0.1, 0, +0.1, +0.2, +0.4\}$ in the small economy of Section 4.4—a $\pm 40\%$ shift in mean fixed costs covers approximately two standard deviations of any reasonable common shock. At three values of σ_ε bracketing the bathtub peak, the BNE fixed-point iteration from supremum, infimum, and uniform starting points converges to the same activation probability vector at every ζ , with maximum pairwise distances below 10^{-3} . By Proposition 8, uniqueness extends to the full game integrating over ζ .

Table 12: Across-firm common-shock robustness

$\zeta/\bar{\phi}$	ζ	$\max \ p_{\text{sup}} - p_{\text{inf}}\ _{\infty} \text{ at } \sigma_{\varepsilon}/\bar{\phi} =$			BNE unique?
		0.10	0.30	1.00	
-0.40	-0.0493	5.3e-08	7.1e-08	2.9e-07	yes
-0.20	-0.0247	7.3e-09	1.8e-07	2.8e-07	yes
-0.10	-0.0123	1.1e-10	2.9e-07	2.6e-07	yes
+0.00	+0.0000	6.1e-13	1.6e-07	2.5e-07	yes
+0.10	+0.0123	2.7e-15	1.8e-07	2.4e-07	yes
+0.20	+0.0247	2.0e-18	8.2e-08	2.2e-07	yes
+0.40	+0.0493	5.7e-26	2.5e-08	1.8e-07	yes

Notes: Small-economy calibration ($N = 3$, $F = 4$; see Section 4.4). For each common-shock realization ζ , the BNE fixed point is solved from three starting points (supremum, infimum, uniform) at three values of σ_{ε} bracketing the bathtub peak. Reported entries are the maximum pairwise $\|p_{\text{sup}} - p_{\text{inf}}\|_{\infty}$ distance across the three starting points. “BNE unique” is yes when the distance is below 10^{-3} at all three σ_{ε} . The $\zeta = 0$ row coincides with the paper’s baseline IID case. Uniqueness holds state-by-state across the full $\pm 40\%$ cost shift, verifying Proposition 8.

E. PROOF OF PROPOSITION 2

We prove that firm f ’s activation profit exhibits increasing differences in the activation indicators $\{I_{fi}\}_{i \in \mathcal{A}_f}$, taking the rival demand index $D_{-f}(n)$ as given. By a standard equivalence (Topkis, 1998), increasing differences on a binary lattice $\{0, 1\}^{|\mathcal{A}_f|}$ is equivalent to supermodularity.

Step 1: Reduction to one-dimensional convexity. Fixed costs $\sum_{i \in \mathcal{I}_f} \phi_i w_i$ are additively separable, so increasing differences of total profit is equivalent to increasing differences of *variable* profit $VP_f = \sum_n VP_{nf}$. Since the sum preserves increasing differences, a *sufficient* condition is per-market increasing differences: if VP_{nf} exhibits increasing differences in $\{I_{fi}\}$ for each market n , so does the aggregate. We derive a sharp threshold for per-market convexity in Steps 2–4, then address the aggregate case in Step 5.

By the Fréchet cost aggregation (4.7), the cost index depends on the activation set $\mathcal{I}_f$ only through the *cost aggregator*:

$$\Sigma_{fn} \equiv \sum_{i \in \mathcal{I}_f} (w_i \tau_{in})^{-\theta}, \quad (\text{E.1})$$

with $c_{nf} = K \cdot \Sigma_{fn}^{-1/\theta}$ where $K = \Gamma_\theta/a_f > 0$. Activating site i adds the term $(w_i \tau_{in})^{-\theta} > 0$ to Σ_{fn} . Since Σ_{fn} is *additive* in the activation indicators, increasing differences of VP_{nf} in $\{I_{fi}\}$ is equivalent to *convexity* of VP_{nf} viewed as a function of Σ_{fn} :

$$\frac{\partial^2 VP_{nf}}{\partial \Sigma_{fn}^2} > 0 \implies \text{increasing differences of } VP_{nf} \text{ in } \{I_{fi}\}. \quad (\text{E.2})$$

This reduction follows from the chain rule: for distinct sites i, i' with increments $\delta_i = (w_i \tau_{in})^{-\theta}$ and $\delta_{i'} = (w_{i'} \tau_{i'n})^{-\theta}$,

$$\begin{aligned} & VP_{nf}(\Sigma + \delta_i + \delta_{i'}) - VP_{nf}(\Sigma + \delta_i) - VP_{nf}(\Sigma + \delta_{i'}) + VP_{nf}(\Sigma) \\ &= \int_0^{\delta_i} \int_0^{\delta_{i'}} \frac{\partial^2 VP_{nf}}{\partial \Sigma^2}(\Sigma + s + t) dt ds > 0 \end{aligned}$$

when $\partial^2 VP_{nf}/\partial \Sigma^2 > 0$ everywhere.

Step 2: CES benchmark (constant markups). Under constant markups $\mu_0 = \sigma/(\sigma - 1)$, variable profit in market n is:

$$VP_{nf}^{\text{CES}} = \frac{1}{\sigma} p_{nf}^{1-\sigma} P_n^{\sigma-\eta} \Omega_n = \frac{\Omega_n}{\sigma} \left(\frac{\sigma}{\sigma-1} \right)^{1-\sigma} c_{nf}^{1-\sigma} P_n^{\sigma-\eta}, \quad (\text{E.3})$$

where $P_n^{1-\sigma} = p_{nf}^{1-\sigma} + D_{-f}(n)$ and we take $D_{-f}(n)$ as given. Substituting $c_{nf} = K \Sigma_{fn}^{-1/\theta}$:

$$VP_{nf}^{\text{CES}} \propto \Sigma_{fn}^{(\sigma-1)/\theta} \cdot [\text{terms involving } P_n(c_{nf})].$$

The direct effect $\Sigma_{fn}^{(\sigma-1)/\theta}$ is strictly convex in Σ_{fn} when $(\sigma - 1)/\theta > 1$. This is the [Antràs et al. \(2017\)](#) argument: the Fréchet dispersion is small enough relative to demand elasticity that the demand expansion from cost reduction is super-proportional. The price-index feedback through P_n works against convexity at high market shares (Step 4 below derives the exact threshold), but the direct convexity dominates when shares are sufficiently small.

Step 3: Extension to Atkeson–Burstein markups. Under AB pricing, firm f sets $p_{nf} = \mu(s_{nf}) \cdot c_{nf}$ with:

$$\mu(s) = \frac{1}{1 - 1/\epsilon(s)}, \quad \frac{1}{\epsilon(s)} = \frac{1-s}{\sigma} + \frac{s}{\eta}, \quad (\text{E.4})$$

where the market share $s_{nf} = x/(x + D)$ with $x \equiv p_{nf}^{1-\sigma}$ and $D \equiv D_{-f}(n)$ (suppressing market subscripts). Variable profit is:

$$VP_{nf} = \frac{1}{\epsilon(s)} \cdot x \cdot \Omega_n \cdot (x + D)^{(\sigma-\eta)/(1-\sigma)}. \quad (\text{E.5})$$

To show $\partial^2 VP_{nf} / \partial \Sigma^2 > 0$, we use the chain rule through the cost index $c = K\Sigma^{-1/\theta}$:

$$\frac{\partial^2 VP}{\partial \Sigma^2} = \frac{\partial^2 VP}{\partial c^2} \left(\frac{dc}{d\Sigma} \right)^2 + \frac{\partial VP}{\partial c} \frac{d^2 c}{d\Sigma^2}. \quad (\text{E.6})$$

Term 2 (cost curvature). Since $c = K\Sigma^{-1/\theta}$:

$$\frac{dc}{d\Sigma} = -\frac{K}{\theta} \Sigma^{-1/\theta-1} < 0, \quad \frac{d^2 c}{d\Sigma^2} = \frac{K(1+\theta)}{\theta^2} \Sigma^{-1/\theta-2} > 0.$$

Variable profit is decreasing in cost ($\partial VP / \partial c < 0$), so the second term $(\partial VP / \partial c)(d^2 c / d\Sigma^2) < 0$, working *against* convexity. Supermodularity therefore requires that Term 1 dominates Term 2 in absolute value.

Term 1 (profit curvature in cost). The proof requires $\partial^2 VP / \partial c^2 > 0$ and sufficiently large. Under the AB pricing rule, the equilibrium $x(c)$ is defined implicitly by $p \cdot c^{-1} = \mu(s(x, D))$ with $x = p^{1-\sigma}$, so x depends on c through the pricing equation. Implicit differentiation yields:

$$\frac{dx}{dc} = \frac{(1-\sigma)x/c}{1 + (\sigma-1)\frac{\sigma-\eta}{\sigma\eta}s(1-s)\mu}. \quad (\text{E.7})$$

The denominator equals $1 + s(1-s)\beta_{nf}$ where $\beta_{nf} \equiv (\sigma-1)(\sigma-\eta)\mu/(\sigma\eta) > 0$ is the feedback parameter from (4.17), so $|dx/dc| < (\sigma-1)x/c$. The key bound is:

$$\left| \frac{c}{x} \frac{dx}{dc} \right| \leq (\sigma-1) \cdot \frac{1}{1 + \frac{\sigma-\eta}{\sigma\eta}s(1-s)\mu_{\min}} \leq (\sigma-1) \cdot C(\sigma, \eta), \quad (\text{E.8})$$

where $\mu_{\min} \equiv \sigma/(\sigma-1)$ is the CES benchmark markup (the lower bound on μ) and $C(\sigma, \eta) \leq 1$ is a finite constant (bounded because $s \in [0, 1]$, $\sigma > \eta > 1$, and $\mu \geq \mu_{\min}$). This means the AB pricing response dx/dc is a *damped* perturbation of the CES response $(1-\sigma)x/c$.

Working with the market share s as the state variable and $t = \ln \Sigma$, the exact AB share

law of motion is $ds/dt = g(s) \equiv \alpha s(1-s)\delta(s)$. Define

$$B \equiv \sigma - \eta > 0, \quad M(s) \equiv \eta(\sigma - 1) - Bs + B(\sigma - 1)s(1 - s),$$

so that $\delta(s) = [\eta(\sigma - 1) - Bs]/M(s)$. Since $VP^{AB}(s) = R_n(as + bs^2)$ with $a = 1/\sigma$ and $b = B/(\sigma\eta)$, the chain rule gives

$$\frac{d^2VP}{d\Sigma^2} = \frac{R_n g(s)}{\Sigma^2} H_{AB}(s), \quad H_{AB}(s) \equiv (a + 2bs)(g'(s) - 1) + 2b g(s).$$

Per-market convexity holds if and only if $H_{AB}(s) \geq 0$.

Proposition 9 (Unique AB per-market convexity threshold). *Suppose $\sigma > \eta > 1$ and $(\sigma - 1)/\theta > 1$. Then $H_{AB}(s)$ is strictly concave on $[0, 1]$:*

$$H''_{AB}(s) = -\frac{6(\sigma - \eta)(\sigma - 1)}{\eta\theta M(s)^4} P_{AB}(s),$$

where $P_{AB}(s) = \sum_{k=0}^4 \beta_k \binom{4}{k} s^k (1-s)^{4-k}$ is a degree-4 polynomial whose Bernstein coefficients $\beta_0, \dots, \beta_4$ are all strictly positive under $\sigma > \eta > 1$.⁸ Since

$$H_{AB}(0) = \frac{\sigma - \theta - 1}{\sigma\theta} > 0, \quad H_{AB}(1) = \frac{(\sigma + \theta - 1)(\eta - 2\sigma)}{\eta\sigma\theta} < 0,$$

there exists a unique $\bar{s}_{AB}^* \in (0, 1)$ such that $H_{AB}(\bar{s}_{AB}^*) = 0$. Per-market convexity holds if and only if $s < \bar{s}_{AB}^*$.

Proof. $M(s) \geq \sigma(\eta - 1) > 0$ on $[0, 1]$, so $M(s)^4 > 0$. Each Bernstein coefficient $\beta_k > 0$ under $\sigma > \eta > 1$, and the Bernstein basis polynomials are nonneg on $[0, 1]$, so $P_{AB}(s) > 0$. Hence $H''_{AB}(s) < 0$: strict concavity. The sign conditions at the endpoints follow by direct evaluation. A strictly concave continuous function positive at 0 and negative at 1 has a unique zero. $\square$

The AB threshold is parameter-dependent and need not uniformly exceed the CES benchmark: at the baseline $(\sigma, \eta, \theta) = (5, 2, 3)$, $\bar{s}_{AB}^* \approx 0.122$, slightly below $\bar{s}_{CES}^* = 0.125$.

⁸Explicitly: $\beta_0 = \eta^3(\sigma - \eta)(\sigma - 1)^3$, $\beta_1 = \eta^3(\eta - 1)(\sigma - 1)^3$, $\beta_2 = 2\eta^2\sigma(\eta - 1)^2(\sigma - 1)^2$, $\beta_3 = \eta\sigma(\eta - 1)^2(\sigma - 1)[2(\sigma - \eta)^2 + (5\eta - 4)(\sigma - \eta) + 3\eta(\eta - 1)]$, $\beta_4 = \sigma^2(\eta - 1)^2[2(\sigma - \eta)^3 + (9\eta - 8)(\sigma - \eta)^2 + (11\eta - 8)(\eta - 1)(\sigma - \eta) + 4\eta(\eta - 1)^2]$.

Since the maximum equilibrium share in the cloud application is 0.438, per-market convexity is violated under both CES and AB, and part (ii) can provide only a computational diagnostic for the solved single-agent subproblems.

Step 4: Direct characterization of per-market convexity. Under the CES benchmark (constant markups), variable profit in market n is $VP = (R_n/\sigma) \cdot u/(u + D_{-f})$ where $u = A\Sigma^\alpha$, $\alpha = (\sigma - 1)/\theta$, and D_{-f} is the rival demand index (held fixed). Writing $s = u/(u + D_{-f})$ for the market share, the chain rule gives:

$$\frac{d^2VP}{d\Sigma^2} = \frac{R_n \alpha s(1-s)}{\sigma(u + D_{-f}) \Sigma^2} \left[\underbrace{-2\alpha s}_{\text{demand curvature}} + \underbrace{(\alpha - 1)}_{\text{Fréchet convexity}} \right],$$

where the prefactor is strictly positive. Per-market convexity ($d^2VP/d\Sigma^2 > 0$) therefore holds if and only if:

$$s < \bar{s}^* \equiv \frac{\alpha - 1}{2\alpha} = \frac{\sigma - 1 - \theta}{2(\sigma - 1)}. \quad (\text{E.9})$$

The threshold $\bar{s}^* > 0$ whenever $(\sigma - 1)/\theta > 1$ (the maintained assumption) and $\bar{s}^* \rightarrow 1/2$ as $\alpha \rightarrow \infty$. At the baseline $(\sigma, \eta, \theta) = (5, 2, 3)$: $\bar{s}^* = 0.125$. Since the maximum equilibrium share is $0.438 \gg 0.125$, the per-market condition is violated, and Step 5 provides only a computational diagnostic rather than an analytical proof of aggregate supermodularity.

Step 5: Aggregate supermodularity. The per-market condition is *sufficient* but not *necessary*: even when individual markets violate $s < \bar{s}^*$, the aggregate profit $\sum_n VP_{nf}$ can still exhibit increasing differences because markets with $s < \bar{s}^*$ (positive curvature) can dominate the few markets where $s > \bar{s}^*$ (negative curvature). The overall increasing difference for sites i, i' is:

$$\Delta^2\pi_f = \sum_n \int_0^{\delta_i} \int_0^{\delta_{i'}} \frac{\partial^2 VP_{nf}}{\partial \Sigma^2} (\Sigma + s + t) dt ds,$$

which is positive whenever the contribution from low-share markets outweighs the high-share markets.

When the per-market threshold is violated, we do not prove $\Delta^2\pi_f > 0$ on the full lattice. Instead, we report a computational diagnostic based on the Jia–Tarski gap condition and the interval audit of decisive sign checks: a zero gap indicates that the CDCP solution

is a lattice optimum at the tested parameterization, conditional on the maintained lattice structure in the sense of Remark 6. In all tested configurations—including those where $\max_{n,f} s_{nf}$ exceeds $\bar{s}^*$ —the gap is zero (see Section 6.5 and Appendix K).

Under AB markups, the pricing response dx/dc is damped by the factor $1/(1 + s(1 - s)\beta_{nf})$ (equation E.8). The exact per-market convexity condition under AB is characterized by $H_{AB}(s) \geq 0$; the corresponding threshold is parameter-dependent and need not be uniformly above the CES benchmark $\bar{s}^*$.

By the equivalence (E.2), aggregate increasing differences in $\{I_{fi}\}$ follow from positive $\Delta^2\pi_f$. Fixed costs $\sum_{i \in \mathcal{I}_f} \phi_i w_i$ are additively separable (hence modular), so total profit $\pi_f = \sum_n VP_{nf} - \sum_{i \in \mathcal{I}_f} \phi_i w_i$ exhibits increasing differences. The lattice consequences (greatest and least optima, Jia–Tarski attainability) follow from Topkis (1998), Theorem 2.8.1. $\square$

F. PROOF OF PROPOSITION 3

We prove that the extremal fixed points of Φ^2 bound all Nash equilibria, and establish convergence of iterated best response. The argument handles the key difficulty that the joint best-response map is *anti-monotone* (order-reversing) due to strategic substitutes, so Tarski’s theorem cannot be applied directly to Φ .

Setup. Let $L = \prod_{f=1}^F 2^{\mathcal{A}_f}$ be the joint activation space, ordered by component-wise set inclusion: $\mathbf{I} \leq \mathbf{I}'$ iff $\mathcal{I}_f \subseteq \mathcal{I}'_f$ for all f . The lattice L is finite and complete, with supremum $\hat{\mathbf{I}} = (\mathcal{A}_1, \dots, \mathcal{A}_F)$ and infimum $\check{\mathbf{I}} = (\emptyset, \dots, \emptyset)$.

By Proposition 2, each firm’s single-agent CDCP has a well-defined greatest optimum (and least optimum) for any fixed rival demand index. Define the joint best-response map $\Phi : L \rightarrow L$ by taking each firm’s greatest optimal activation set:

$$\Phi(\mathbf{I}) = (\mathcal{I}_1^*(\mathbf{I}_{-1}), \dots, \mathcal{I}_F^*(\mathbf{I}_{-F})).$$

Step 1: Anti-monotonicity of Φ . When rivals expand their activation sets ($\mathbf{I}_{-f} \leq \mathbf{I}'_{-f}$), the rival demand index $D_{-f}(n) = \sum_{g \neq f} p_{ng}^{1-\sigma}$ weakly increases (rivals have lower costs, hence lower prices, hence larger $p_g^{1-\sigma}$). Higher D_{-f} reduces firm f ’s market share s_{nf} for any given price, lowering the marginal value of each production site. Since the single-agent

problem is supermodular (Proposition 2), the optimal activation set shrinks by monotone comparative statics (Topkis, 1998): $I_f^*(\mathbf{I}'_{-f}) \subseteq I_f^*(\mathbf{I}_{-f})$.

Therefore $\mathbf{I} \leq \mathbf{I}'$ implies $\Phi(\mathbf{I}) \geq \Phi(\mathbf{I}')$: the map Φ is anti-monotone.

Step 2: Monotonicity of Φ^2 and Tarski's theorem. The squared map $\Phi^2 = \Phi \circ \Phi$ is monotone (increasing): if $\mathbf{I} \leq \mathbf{I}'$, then $\Phi(\mathbf{I}) \geq \Phi(\mathbf{I}')$, so $\Phi^2(\mathbf{I}) = \Phi(\Phi(\mathbf{I})) \leq \Phi(\Phi(\mathbf{I}')) = \Phi^2(\mathbf{I}')$. By Tarski's fixed-point theorem, the monotone map Φ^2 on the finite complete lattice L has a greatest fixed point $\bar{\mathbf{I}}$ and a least fixed point $\underline{\mathbf{I}}$.

Step 3: Bounding the Nash equilibrium set. Any Nash equilibrium $\mathbf{I}^*$ satisfies $\Phi(\mathbf{I}^*) = \mathbf{I}^*$, hence $\Phi^2(\mathbf{I}^*) = \mathbf{I}^*$: every Nash equilibrium is a fixed point of Φ^2 . By Tarski's theorem, $\underline{\mathbf{I}} \leq \mathbf{I}^* \leq \bar{\mathbf{I}}$.

If $\bar{\mathbf{I}}$ is itself a Nash equilibrium (i.e., $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$), then it is the greatest Nash equilibrium by the bound above. In general, however, $\bar{\mathbf{I}}$ may be a two-cycle of Φ rather than a fixed point: $\Phi(\bar{\mathbf{I}}) \neq \bar{\mathbf{I}}$ while $\Phi^2(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$. Whether the extremal fixed points of Φ^2 are Nash equilibria depends on the specific game structure and must be verified. $\square$

Convergence (Proposition 4). Starting from $\hat{\mathbf{I}}$, anti-monotonicity gives $\Phi(\hat{\mathbf{I}}) \leq \hat{\mathbf{I}}$, so $\Phi^2(\hat{\mathbf{I}}) \leq \Phi(\hat{\mathbf{I}}) \leq \hat{\mathbf{I}}$ (the first inequality by anti-monotonicity applied to $\Phi(\hat{\mathbf{I}}) \leq \hat{\mathbf{I}}$). By induction, the even subsequence $\{\Phi^{2k}(\hat{\mathbf{I}})\}$ is decreasing and the odd subsequence $\{\Phi^{2k+1}(\hat{\mathbf{I}})\}$ is increasing, with $\Phi^{2k+1}(\hat{\mathbf{I}}) \leq \Phi^{2k}(\hat{\mathbf{I}})$ for all k . On the finite lattice L , both subsequences stabilize in finitely many steps to limits $\bar{\mathbf{I}}_{\text{even}}$ and $\bar{\mathbf{I}}_{\text{odd}}$ satisfying $\Phi^2(\bar{\mathbf{I}}_{\text{even}}) = \bar{\mathbf{I}}_{\text{even}}$ and $\bar{\mathbf{I}}_{\text{odd}} = \Phi(\bar{\mathbf{I}}_{\text{even}})$. Since $\{\Phi^{2k}(\hat{\mathbf{I}})\}$ is the largest decreasing sequence of iterates of Φ^2 , its limit $\bar{\mathbf{I}}_{\text{even}} = \bar{\mathbf{I}}$, the greatest fixed point of Φ^2 . If $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$, the full sequence converges to $\bar{\mathbf{I}}$, the greatest Nash equilibrium.

For Gauss–Seidel iteration, define the composite update map $\Psi : L \rightarrow L$ that updates firms sequentially $(f_1, \dots, f_F)$, each using the latest rival profile. Unlike the Jacobi map Φ , the GS map Ψ is not obviously anti-monotone: firm f_j 's rival profile mixes updated information from earlier firms with stale information from later firms, so the comparability of rival profiles under perturbation is not preserved. We verify GS convergence computationally at all tested parameterizations (Section 6 and Appendix K). $\square$

G. JACOBIAN OF THE THRESHOLD MAP

We derive the factorization (4.21) and bound $\|J_v\|_\infty$ using CES primitives.

Step 1: Factorization. The threshold map T acts on activation probabilities via $T_{if}(v) = G_0(\bar{\varepsilon}_{if}(v)/\sigma_\varepsilon)$, where $\bar{\varepsilon}_{if}(v) = \mathbb{E}[\Delta\pi_{if}(v)]/w_i$ is the threshold at which the expected marginal value of activating site i (integrating over rivals' and own other sites' random activation) equals the fixed-cost shock. By the chain rule:

$$\frac{\partial T_{if}}{\partial v_{jg}} = \frac{g_0(\bar{\varepsilon}_{if}/\sigma_\varepsilon)}{\sigma_\varepsilon} \cdot \frac{1}{w_i} \cdot \frac{\partial \mathbb{E}[\Delta\pi_{if}]}{\partial v_{jg}}, \quad (\text{G.1})$$

which gives $J_T = \text{diag}(g_0(\bar{\varepsilon}/\sigma_\varepsilon)/\sigma_\varepsilon) \cdot \text{diag}(1/w_i) \cdot J_\Delta$. Defining $J_v \equiv \text{diag}(1/w_i) \cdot J_\Delta$ absorbs the wage scaling into the sensitivity matrix, so $\|J_T\|_\infty \leq (\|g_0\|_\infty/\sigma_\varepsilon) \cdot \|J_v\|_\infty$.

Step 2: Sign structure of J_v . For $g \neq f$ (across-firm): increasing v_{jg} raises the probability that rival g activates site j , weakly increasing $\mathbb{E}[D_{-f}(n)]$ in markets served from j . Higher expected rival demand lowers firm f 's share and variable profit, so $\partial \mathbb{E}[\Delta\pi_{if}]/\partial v_{jg} \leq 0$: strategic substitutes across firms. For $j \neq i$, $g = f$ (within-firm): increasing v_{jf} lowers firm f 's expected cost index (Fréchet complementarity (4.7)), raising the marginal value of site i by supermodularity (Proposition 2), so $\partial \mathbb{E}[\Delta\pi_{if}]/\partial v_{jf} \geq 0$: complementarity within firm.

Step 3: Bounding $\|J_v\|_\infty$ via the Perron–Frobenius weighted norm. Before bounding the row sums, we record the weighted-norm tool that replaces the Neumann-series (infinity-norm) bound used in earlier drafts. The substitution is essential because $\|\Lambda\|_\infty$ exceeds $1/(F-1)$ on a nontrivial fraction of the per-market subgames enumerated in Proposition 7, so the Neumann bound $1/(1 - \|\Lambda\|_\infty)$ is not available throughout. The spectral radius $\rho(\Lambda) \leq 0.314 < 1$ is uniform, and Perron–Frobenius converts this into an operator-norm contraction in a weighted sup-norm.

Proposition 10 (Spectral-norm Jacobian bound). *Let $\Lambda \geq 0$ entrywise with $\rho(\Lambda) < 1$, and let $w \gg 0$ be the Perron–Frobenius eigenvector with $\Lambda w \leq \lambda w$ for some $\lambda \in [\rho(\Lambda), 1)$ (when Λ is irreducible, $\lambda = \rho(\Lambda)$; reducibility is handled by taking $\lambda = \rho(\Lambda) + \varepsilon$ for any $\varepsilon > 0$ and using a*

standard perturbation argument). Define the weighted sup-norm $\|x\|_w = \max_i |x_i|/w_i$ on $\mathbb{R}^{NF}$, with induced operator norm $\|A\|_w = \sup_{x \neq 0} \|Ax\|_w / \|x\|_w$. Then:

- (i) $\|\Lambda\|_w \leq \lambda$, so $(I - \Lambda)$ is invertible with $\|(I - \Lambda)^{-1}\|_w \leq 1/(1 - \lambda)$, which for irreducible Λ reduces to $1/(1 - \rho(\Lambda))$ and for reducible Λ holds with $\lambda = \rho(\Lambda) + \varepsilon$ for arbitrary $\varepsilon > 0$;
- (ii) applied to the downstream AB pricing Jacobian, a demand perturbation δD_{-f} propagates with amplification at most $1/(1 - \rho(\Lambda))$ in $\|\cdot\|_w$, so

$$\|J_v\|_w \leq \frac{L_D \cdot L_\mu}{1 - \rho(\Lambda)} + L_C,$$

with L_D, L_μ, L_C defined below.

The weighted norm $\|\cdot\|_w$ and the standard infinity norm $\|\cdot\|_\infty$ are equivalent: $\|x\|_w \leq \|x\|_\infty / \min_i w_i$ and $\|x\|_\infty \leq (\max_i w_i) \|x\|_w$. Hence a contraction in $\|\cdot\|_w$ yields a contraction in $\|\cdot\|_\infty$ with an at-worst constant-factor inflation $C_w = \max_i w_i / \min_i w_i$.

Proof. For (i): $\|\Lambda x\|_w = \max_i (\Lambda x)_i / w_i \leq \max_i (\Lambda |x|)_i / w_i$ (where $|x|$ denotes entrywise absolute value), and using $\Lambda w \leq \lambda w$ with $|x|_i \leq \|x\|_w \cdot w_i$:

$$(\Lambda |x|)_i \leq \sum_j \Lambda_{ij} \|x\|_w w_j = \|x\|_w (\Lambda w)_i \leq \|x\|_w \lambda w_i,$$

so $\|\Lambda\|_w \leq \lambda$. The Neumann series $(I - \Lambda)^{-1} = \sum_{k \geq 0} \Lambda^k$ converges in $\|\cdot\|_w$ with sum bounded by $1/(1 - \lambda)$. For (ii): the pricing-equilibrium map near a fixed point has Jacobian bounded entrywise by Λ (Proposition 5), so the linear response $\delta D' = (I - \Lambda)^{-1} \delta D$ obeys $\|\delta D'\|_w \leq \|\delta D\|_w / (1 - \rho(\Lambda))$. Composing with the demand- and cost-channel Lipschitz bounds L_D, L_μ, L_C yields the stated inequality. Norm-equivalence is a standard consequence of $w \gg 0$ with $\max_i w_i, \min_i w_i \in (0, \infty)$. $\square$

With Proposition 10 in hand, we decompose each row sum of $|J_v|$ into across-firm and within-firm contributions:

$$\sum_{(j,g)} |J_v|_{(i,f),(j,g)} = \underbrace{\sum_{g \neq f} \sum_j \frac{1}{w_i} \left| \frac{\partial \mathbb{E}[\Delta \pi_{if}]}{\partial v_{jg}} \right|}_{\text{across-firm (demand channel)}} + \underbrace{\sum_{j \neq i} \frac{1}{w_i} \left| \frac{\partial \mathbb{E}[\Delta \pi_{if}]}{\partial v_{jf}} \right|}_{\text{within-firm (cost channel)}}.$$

Across-firm terms: demand sensitivity L_D . Each $\partial \mathbb{E}[D_{-f}(n)]/\partial v_{jg}$ is the expected change in rival demand when firm g toggles site j from inactive to active. Under Fréchet costs (4.7), this adds $(w_j \tau_{jn}/a_g)^{-\theta}$ to the cost aggregator, lowering c_{ng} and hence p_{ng} , which raises $p_{ng}^{1-\sigma}$. The maximum contribution is the single-site price index (attained when g has no other active site):

$$0 \leq \frac{\partial \mathbb{E}[D_{-f}(n)]}{\partial v_{jg}} \leq \left(\frac{\sigma}{\sigma-1} \right)^{1-\sigma} \left(\frac{w_j \tau_{jn}}{a_g} \right)^{1-\sigma} \equiv \bar{d}_{jgn}. \quad (\text{G.2})$$

Summing over rivals and sites: $L_D \equiv \max_n \sum_{g \neq f} \sum_j \bar{d}_{jgn}$, finite since the sums are over finite firms and sites with bounded wages and trade costs.

Across-firm terms: marginal-value sensitivity L_μ . A perturbation $\delta D_{-f}(n)$ propagates through the AB pricing equilibrium with spectral amplification $\|(I - \Lambda)^{-1}\|_w \leq 1/(1 - \rho(\Lambda))$ in the Perron–Frobenius weighted sup-norm (Proposition 10 below). Variable profit is $VP_{nf} = R_n s_{nf}(1 - 1/\epsilon_{nf})$; since $s_{nf}(1 - s_{nf}) \leq 1/4$ and the profit margin satisfies $(1 - 1/\epsilon_{nf}) \leq (\bar{\mu} - 1)/\bar{\mu}$ with $\bar{\mu} = \eta/(\eta - 1)$:

$$L_\mu \equiv \max_{(i,f)} \frac{1}{w_i} \sum_n \left| \frac{\partial(\Delta VP_{nf})}{\partial D_{-f}(n)} \right| \leq \frac{1}{w_{\min}} \sum_n \frac{R_n (\bar{\mu} - 1)}{2 \bar{\mu} P_n^{1-\sigma}}, \quad (\text{G.3})$$

where $w_{\min} = \min_i w_i$ and the factor 2 accounts for evaluating ΔVP at configurations both with and without site i .

Within-firm terms: cost-channel sensitivity L_C . Increasing v_{jf} adds $(w_j \tau_{jn}/a_f)^{-\theta}$ to firm f 's Fréchet aggregator (4.7). The cost-index response satisfies $\partial c_{nf}^{1-\sigma}/\partial v_{jf} = \frac{\sigma-1}{\theta} (c_{nf}^{-\theta})^{(\sigma-1)/\theta-1} (w_j \tau_{jn})^{-\theta}$ which is finite and positive for any configuration with at least one active site ($c_{nf}^{-\theta} > 0$). The downstream effect on $\Delta \pi_{if}$ propagates through the same bounded-markup and share channels as L_μ , with an additional factor of $(\sigma - 1)/\theta$ from the cost complementarity. Summing over $|\mathcal{A}_f| - 1$ own sites and N markets, L_C is a finite product of CES primitives and equilibrium cost indices. Together:

$$\|J_v\|_w \leq \frac{L_D \cdot L_\mu}{1 - \rho(\Lambda)} + L_C, \quad (\text{G.4})$$

where $\|\cdot\|_w$ is the Perron–Frobenius weighted sup-norm from Proposition 10 below. The relationship $\|J_v\|_\infty \leq C_w \|J_v\|_w$ for the norm-equivalence constant $C_w = \max_i w_i / \min_i w_i$

(with w the Perron eigenvector) then yields the unweighted bound used in (4.22). All three bounds are finite under $\sigma > \eta > 1$ and $\theta > 0$: L_D depends only on primitives (wages, trade costs, TFP, and σ); L_μ on revenue shares and the markup range; L_C on the Fréchet aggregator and the supermodularity index $(\sigma - 1)/\theta$. At the baseline parameterization, $\rho(\Lambda) = 0.109$ (the tightest available rate; the conservative infinity-norm is $\|\Lambda\|_\infty = 0.239$), so the spectral amplification $1/(1 - \rho(\Lambda)) \approx 1.12$: the amplification is near unity, and the outer activation problem is nearly a constant-markup entry game. Using $\rho(\Lambda)$ rather than $\|\Lambda\|_\infty$ is essential because the latter exceeds $1/(F - 1)$ on a nontrivial fraction of enumerated per-market subgames (Table 2), while the spectral radius satisfies $\rho(\Lambda) \leq 0.314 < 1$ uniformly. The script `main_bathtub.m` evaluates $\|J_T\|_\infty$ numerically at each σ_ε and confirms $\gamma_{\max} = 0.951 < 1$ (Figure 1).

Step 4: Tail behavior and convergence rate. *Large σ_ε .* From (4.22): $\gamma(\sigma_\varepsilon) \leq (\|g_0\|_\infty/\sigma_\varepsilon) \cdot \|J_v\|_\infty \rightarrow 0$ as $\sigma_\varepsilon \rightarrow \infty$, since $\|J_v\|_\infty$ is bounded on the compact set $[0, 1]^M$ (Step 3). For the normal density, $\|g_0\|_\infty = 1/\sqrt{2\pi}$.

Small σ_ε . The left tail is handled by Lemma 2, not by the complete-information gap alone. Once the environment-specific verification supplies an invariant, fixed-point-inclusive family S_{σ_ε} on which $\min_m |v_m(v)| \geq b > 0$, the standardized threshold satisfies $|z_m| \geq b/\sigma_\varepsilon \rightarrow \infty$ uniformly on S_{σ_ε} . For log-concave densities with tail bound $g_0(z) \leq C_g \exp(-c|z|)$ (e.g., normal: $C_g = e^{c^2/2}/\sqrt{2\pi}$, $c > 0$ arbitrary; logistic: $C_g = 1/4$, $c = 1$), the contraction modulus on the localized region satisfies the explicit rate

$$\sup_{v \in S_{\sigma_\varepsilon}} \|J_T(v)\|_\infty \leq \frac{C_g K_S}{\sigma_\varepsilon} \exp\left(-\frac{c b}{\sigma_\varepsilon}\right) \xrightarrow{\sigma_\varepsilon \rightarrow 0} 0, \quad (\text{G.5})$$

where $K_S = \sup_{v \in S_{\sigma_\varepsilon}} \|J_v(v)\|_\infty$ is the localized Jacobian bound. In the calibrated application, the complete-information gap is used only as an ingredient in constructing and numerically checking such a localized family.

Uniqueness of the BNE branch. Define $F(v, \sigma_\varepsilon) = T_{\sigma_\varepsilon}(v) - v$. At any σ_ε^* where $\gamma(\sigma_\varepsilon^*) < 1$, the Jacobian $D_v F = J_T - I$ is nonsingular (since $\|J_T\|_\infty < 1$ implies that $I - J_T$ is invertible). By the implicit function theorem, the BNE extends as a unique C^1 branch $v^*(\sigma_\varepsilon)$ in a neighborhood of σ_ε^* . Since $\gamma \rightarrow 0$ at both extremes, the set $\{\sigma_\varepsilon > 0 : \gamma(\sigma_\varepsilon) < 1\}$ includes neighborhoods of 0^+ and ∞ . If $\gamma_{\max} < 1$, this set is all of $(0, \infty)$, and $v^*(\sigma_\varepsilon)$ is the unique

BNE for every $\sigma_\varepsilon > 0$. □

H. NESTED-LOGIT VERIFICATION: TECHNICAL DETAILS

This appendix verifies the hypotheses of Theorem 1 and Corollary 1 in a *non-CES* environment: consumer demand is governed by nested logit, and firms set prices in Bertrand (rather than Cournot) competition. The verification certifies that the uniqueness engine developed in Section 3.1 is not an artifact of the Fréchet–CES–AB structure used in the quantitative application. Three structural differences from Section 4.1.9 are reabsorbed into the same abstract template: the demand primitive is consumer choice with Gumbel shocks (not CES over multi-country varieties); pricing is Bertrand (not Cournot); and markups arise from McFadden–Train first-order conditions (not AB residual demand). What remains unchanged is the three-part scaffolding of Remark 2: a C^1 marginal-value index $v(\nu)$ on $[0, 1]^M$, a Jacobian factorization through the density prefactor $g_0(v/\sigma)/\sigma$, and a small- σ localization on a non-marginal invariant set.

Environment setup. We begin with the primitives of the nested-logit environment.

Assumption 3 (Nested logit demand and private cost shocks). (*Consumer demand.*) In market $n \in \{1, \dots, N\}$, a unit mass of consumers chooses among the products offered by firms in the active set $S_n \subseteq \{1, \dots, F\}$ plus an outside option indexed $f = 0$. Products are partitioned into a finite collection of nests $\{\mathcal{N}_{n,k}\}_{k \in \mathcal{K}_n}$ indexed by $k(f)$, each with a correlation parameter $\lambda_{n,k} \in (0, 1]$ ($\lambda_{n,k} = 1$ recovers standard logit, and $\lambda_{n,k} \rightarrow 0$ is within-nest perfect correlation). Consumer utility from firm f 's product is $u_{n,f} = \alpha_{n,f} - \gamma p_{n,f} + \zeta_{n,f}$, where $\zeta_n = (\zeta_{n,0}, \dots, \zeta_{n,F})$ is distributed as a generalized extreme value of McFadden type with nest parameters $(\lambda_{n,k})_{k \in \mathcal{K}_n}$ (McFadden, 1978; Train, 2009), and the outside option has deterministic utility 0. The price sensitivity $\gamma > 0$ is common across markets. (*Cost primitives.*) Firm f 's marginal cost in market n is $c_{n,f}(\mathcal{I}_f)$, as in Section 4.1. (*Private costs and threshold strategies.*) Firm f 's fixed-cost shocks $\{\varepsilon_{if}\}_{i \in \mathcal{A}_f}$ and the site-independent threshold strategy space are as in Assumption 2.

Under Assumption 3, the choice probability takes the standard nested-logit form

$$s_{n,f}(\mathbf{p}_n) = \underbrace{\frac{\exp((\alpha_{n,f} - \gamma p_{n,f})/\lambda_{n,k(f)})}{D_{n,k(f)}}}_{s_{n,f|k(f)}} \cdot \underbrace{\frac{D_{n,k(f)}^{\lambda_{n,k(f)}}}{1 + \sum_{k'} D_{n,k'}^{\lambda_{n,k'}}}}_{s_{n,k(f)}}, \quad (\text{H.1})$$

where $D_{n,k} \equiv \sum_{g \in \mathcal{N}_{n,k} \cap S_n} \exp((\alpha_{n,g} - \gamma p_{n,g})/\lambda_{n,k})$ is the within-nest inclusive value. The Bertrand first-order condition for firm f yields the McFadden–Train markup

$$p_{n,f} - c_{n,f} = \frac{1}{\gamma} \cdot \frac{1}{(1/\lambda_{n,k(f)})(1 - s_{n,f|k(f)}) + s_{n,f|k(f)} - s_{n,f}}, \quad (\text{H.2})$$

which is bounded in $[\gamma^{-1}, \gamma^{-1}/\lambda_{\min}]$ uniformly over activation profiles, where $\lambda_{\min} \equiv \min_{n,k} \lambda_{n,k} > 0$.

Pricing uniqueness. The system (H.2) evaluated across $f \in S_n$ defines a per-market pricing fixed point. Nested-logit Bertrand pricing with single-product firms admits a standard uniqueness argument.

Proposition 11 (Nested-Logit Bertrand Pricing Uniqueness). *Fix an activation profile $(\mathcal{I}_1, \dots, \mathcal{I}_F)$ and the induced per-firm marginal costs $(c_{n,f})_{f \in S_n}$ in each market. Under Assumption 3 with $\lambda_{n,k} \in (0, 1]$, the Bertrand first-order condition (H.2) has a unique solution $\mathbf{p}_n^*(c_{n,\cdot})$ in each market n . The pricing selection $c \mapsto \mathbf{p}^*(c)$ is C^1 , and the induced shares $s_{n,f}^* = s_{n,f}(\mathbf{p}_n^*)$ are C^1 in c with uniformly bounded Jacobian.*

Proof. *Share derivatives.* Differentiating (H.1) with respect to firm f 's own price $p_{n,f}$ gives

$$\begin{aligned} \frac{\partial s_{n,f}}{\partial p_{n,f}} &= -\frac{\gamma s_{n,f}}{\lambda_{n,k(f)}} \left[(1 - s_{n,f|k(f)}) + \lambda_{n,k(f)} s_{n,f|k(f)} (1 - s_{n,k(f)}) \right] \\ &= -\gamma s_{n,f} \left[\lambda_{n,k(f)}^{-1} (1 - s_{n,f|k(f)}) + s_{n,f|k(f)} - s_{n,f} \right], \end{aligned}$$

and with respect to a rival price $p_{n,g}$ ($g \neq f$) in the same nest $k(f) = k(g)$:

$$\frac{\partial s_{n,f}}{\partial p_{n,g}} = \gamma s_{n,f} s_{n,g|k(f)} \left[\lambda_{n,k(f)}^{-1} - s_{n,f}/s_{n,f|k(f)} \right],$$

and across nests ($k(f) \neq k(g)$):

$$\frac{\partial s_{n,f}}{\partial p_{n,g}} = \gamma s_{n,f} s_{n,g}.$$

All share derivatives are bounded in modulus by $\gamma \lambda_{\min}^{-1}$, since shares lie in $[0, 1]$ and within-nest conditional shares lie in $[0, 1]$.

Bertrand FOC uniqueness. The Bertrand FOC is $s_{n,f} + (p_{n,f} - c_{n,f}) \partial s_{n,f} / \partial p_{n,f} = 0$; substituting the derivative expression and rearranging yields (H.2). Rewrite the FOC in log-share coordinates $\eta_{n,f} \equiv \ln s_{n,f|k(f)} - \lambda_{n,k(f)} \ln s_{n,k(f)}$; the Bertrand game becomes an aggregative game in the nest-level inclusive values $(D_{n,k}^{\lambda_{n,k}})_{k \in \mathcal{K}_n}$ in the sense of [Nocke and Schutz \(2018\)](#). Define the log-share aggregator

$$\Psi_n(\mathbf{p}_n) = \sum_{k \in \mathcal{K}_n} D_{n,k}^{\lambda_{n,k}} = \frac{1 - s_{n,0}(\mathbf{p}_n)}{s_{n,0}(\mathbf{p}_n)}.$$

[Nocke and Schutz \(2018\)](#) show (their Theorem 1, Lemma 2) that a multiproduct Bertrand game with generalised extreme value demand is an *aggregative* game in Ψ_n with a strictly quasi-concave universal payoff; for single-product firms and nested logit this yields a unique fixed point in prices. The argument extends the earlier nested-logit treatments of [Anderson et al. \(1992\)](#) (Chapter 7) and [Konovalov and Sándor \(2010\)](#). An elementary contraction argument due to [Mizuno \(2003\)](#) gives the standard-logit case as $\lambda_{n,k} = 1$. Strict concavity of the firm's log-share objective combined with finite-valued derivatives delivers C^1 dependence of the fixed point on c via the implicit function theorem at the unique root. Uniform boundedness of the share Jacobian follows from the uniform markup bound $p_{n,f} - c_{n,f} \leq \gamma^{-1} / \lambda_{\min}$. □

Marginal-value map on $[0, 1]^M$. Let $\mathbf{v} \in [0, 1]^M$ denote the profile of site-level activation probabilities. For each rival firm $g \neq f$ and each market n , the probability that firm g is active in market n is

$$v_{n,g} = 1 - \prod_{i \in \mathcal{A}_g: i \rightsquigarrow n} (1 - v_{ig}),$$

where $i \rightsquigarrow n$ means site i services market n through the Fréchet cost aggregator (4.7). The expected pricing equilibrium and shares at the activation-probability profile $\mathbf{v}$ are obtained

by averaging the unique Bertrand solution from Proposition 11 over the independent indicator draws $\{\mathbf{1}_{g \in S_n}\}_{g \neq f}$ implied by v . Firm f 's expected marginal profit from activating site i , evaluated at the site- i threshold, is

$$\mathbb{E}[\Delta\pi_{if}(v)] = \sum_n \mathbb{E} \left[(p_{n,f}^*(c(\mathcal{I}_f)) - c_{n,f}(\mathcal{I}_f)) s_{n,f}(\mathbf{p}_n^*) M_n \right]_{\mathcal{I}_f = \{i\} \cup \mathcal{I}_f^-} - \mathbb{E}[\cdots]_{\mathcal{I}_f = \mathcal{I}_f^-},$$

integrating over sibling activation shocks and rival presence, and $v_m(v) \equiv \mathbb{E}[\Delta\pi_{if}(v)]/w_i$ with the multi-index $m \equiv (i, f)$.

Lemma 3 (C^1 nested-logit marginal-value map). *Under Assumption 3 and Proposition 11, the marginal-value map $v : \mathcal{N} \rightarrow \mathbb{R}^M$ is C^1 on an open neighborhood $\mathcal{N} \supseteq [0, 1]^M$, and its Jacobian admits the uniform bound*

$$\|J_v(v)\|_\infty \leq K_v^{NL} \equiv \frac{L_\pi \cdot L_s}{w_{\min}}, \quad (\text{H.3})$$

where $L_\pi = M_{\max}/(\gamma\lambda_{\min})$ bounds Lipschitz variation of expected profit in shares, L_s bounds Lipschitz variation of the nested-logit share vector in rival presence probabilities, and $w_{\min} = \min_i w_i$.

Proof. Smoothness of v . Smoothness follows by composing C^1 maps: $v \rightarrow$ rival presence distributions $\rightarrow$ expected Bertrand equilibrium (Proposition 11, C^1) $\rightarrow$ expected marginal profit (polynomial in v via independence of indicators) $\rightarrow$ scalar division by w_i .

Uniform bound on the share Jacobian. Fix a rival firm $g \neq f$ and a market n . Perturbing the presence indicator $\mathbf{1}_{g \in S_n}$ from 0 to 1 changes the equilibrium share vector by at most

$$\left| \frac{\partial s_{n,\cdot}^*}{\partial v_{n,g}} \right|_\infty \leq \underbrace{\gamma \lambda_{\min}^{-1}}_{\text{share-price slope}} \cdot \underbrace{\|J_{\mathbf{p}^*}\|_\infty}_{\text{equilibrium price response}} \leq L_s \equiv \frac{2\gamma}{\lambda_{\min}^2} \cdot \max_n (p_{n,f}^{\max} - c_{n,f}^{\min}),$$

where the price-response bound follows from the implicit function theorem applied to the FOC system at the unique root, and the factor $\lambda_{\min}^{-2}$ reflects the two rounds of λ^{-1} amplification (own-price and cross-price channels).

Jacobian bound. The chain rule gives $|\partial v_m / \partial v_\ell| \leq (1/w_{\min}) \cdot L_\pi \cdot |\partial s_{n,f}^* / \partial v_{n,g}|$. Composing with L_s above and the profit-Lipschitz constant $L_\pi = M_{\max}/(\gamma\lambda_{\min})$ (the largest market weight times the maximum unit margin) yields (H.3). $\square$ $\square$

Verification of the abstract hypotheses. Lemma 3 supplies the two structural ingredients required by Theorem 1: C^1 regularity of v on a neighborhood of $[0, 1]^M$ and a finite operator-norm bound on J_v . Large- σ decay (Lemma 1) applies directly: $\gamma(\sigma) \leq \|g_0\|_\infty K_v^{\text{NL}}/\sigma \rightarrow 0$. Small- σ decay (Lemma 2) requires a non-marginal invariant set. Let

$$\delta_v^{\text{NL}} \equiv \min_{(i,f)} |\Delta \pi_{if}^{\text{NL},*}|/w_i$$

denote the complete-information profit gap under nested logit, and let

$$S_\sigma^{\text{NL}} \equiv \{v \in [0, 1]^M : \min_m |v_m(v)| \geq \frac{1}{2} \delta_v^{\text{NL}}\}.$$

Under bounded J_v , the complete-information game is a C^1 perturbation of the Bayesian game in σ , and when its unique Nash equilibrium has $\delta_v^{\text{NL}} > 0$, this gap is useful for constructing a candidate localized region. The invariance and fixed-point-inclusion hypotheses of Lemma 2 must still be verified for the nested-logit environment, analytically or numerically, rather than inferred from δ_v^{NL} alone. In the application below we therefore treat $b = \delta_v^{\text{NL}}/2$ only as a candidate non-marginality radius, not as a standalone proof of localization.

Proposition 12 (BNE Uniqueness under Nested Logit). *Impose Assumption 3. Suppose:*

- (a) *Bertrand pricing uniqueness holds (Proposition 11);*
- (b) *the Jacobian of the marginal-value map satisfies $\|J_v(v)\|_\infty \leq K_v^{\text{NL}} < \infty$ uniformly on $[0, 1]^M$ (Lemma 3);*
- (c) *G_0 is continuously differentiable with bounded density and a tail-regularity condition $|z| g_0(z) \rightarrow 0$ as $|z| \rightarrow \infty$ (satisfied by normal, logistic, Gumbel, and more generally by any log-concave density on $\mathbb{R}$);*
- (d) *there exists a family $\{S_\sigma^{\text{NL}}\}_{\sigma>0}$ satisfying the localization conditions of Lemma 2; in the calibration below, the complete-information gap $\delta_v^{\text{NL}} > 0$ is used only to construct and numerically check such a family;*
- (e) *$\gamma_{\max}^{\text{NL}} \equiv \sup_{\sigma>0} \gamma(\sigma) < 1$, where $\gamma(\sigma) = \sup_{v \in [0, 1]^M} \|J_{T_\sigma}(v)\|_\infty$ is the contraction modulus of the nested-logit threshold map T_σ .*

Then the threshold-strategy BNE in the nested-logit entry game is unique for every $\sigma > 0$.

Proof. Conditions (a)–(b) deliver the two structural hypotheses of Theorem 1: a C^1 marginal-value map v with bounded $\|J_v\|_\infty$. Condition (c) supplies both the bounded $\|g_0\|_\infty$ needed for Lemma 1 (large- σ tail) and the tail-regularity $|z|g_0(z) \rightarrow 0$ needed for Lemma 2 (small- σ localization). Condition (d) supplies the localized non-marginal family required by Lemma 2. Condition (e) closes the bathtub globally: the peak contraction modulus is below unity, so T_σ is a strict contraction on $[0, 1]^M$ in $\|\cdot\|_\infty$ at every $\sigma > 0$, and Corollary 1 applies. Brouwer existence and Banach uniqueness then yield a unique threshold-strategy BNE. $\square$ $\square$

Closed-form $\gamma_{\max}^{\text{NL}}$ criterion. The large- σ tail of Lemma 1 combined with (H.3) gives

$$\gamma^{\text{NL}}(\sigma) \leq \frac{\|g_0\|_\infty}{\sigma} \cdot \frac{L_\pi L_s}{w_{\min}} = \frac{\|g_0\|_\infty M_{\max} (p^{\max} - c^{\min})}{w_{\min} \lambda_{\min}^3 \sigma}.$$

For small σ , the localization argument of Lemma 2 yields the corresponding left-tail bound on the verified family S_σ^{NL} ; in the calibration below, this step is checked numerically using the candidate radius built from δ_v^{NL} . The single verifiable diagnostic $\gamma_{\max}^{\text{NL}} < 1$ is obtained by maximising over σ numerically on a log-spaced grid, identically to the CES–AB case.

Remark 14 (Role of the nesting parameters). The nesting parameters $\{\lambda_{n,k}\}$ enter $\gamma^{\text{NL}}(\sigma)$ through two channels that work in opposite directions. A smaller $\lambda_{n,k}$ (stronger within-nest correlation) enlarges the markup magnitude in (H.2) (the bound $\gamma^{-1}/\lambda_{\min}$ grows) and amplifies L_π in Lemma 3, pushing γ^{NL} up. At the same time, a smaller $\lambda_{n,k}$ damps cross-nest substitution and lowers the Lipschitz modulus L_s of shares in rival presence, pushing γ^{NL} down. At the standard-logit limit $\lambda_{n,k} \equiv 1$ the two channels collapse and the contraction argument reduces to the single-market uniqueness result of Mizuno (2003). The bathtub structure of Section 3.2 is preserved: $\gamma^{\text{NL}}(\sigma)$ is continuous on $(0, \infty)$, with the large-noise tail handled by Lemma 1 and the small-noise tail handled on the verified localized sets of Lemma 2; $\gamma_{\max}^{\text{NL}} < 1$ is then the single verifiable diagnostic that certifies global BNE uniqueness.

Remark 15 (Comparison with the CES–AB verification). The nested-logit verification establishes what the abstract framework of Section 3.1 promised: the uniqueness engine is a *property of the threshold system*, not of the CES–AB demand structure specifically. Three

comparisons are instructive. (i) The pricing contraction modulus $\rho(\Lambda) < 1$ of Proposition 5 is replaced by the Nocke–Schutz aggregative-games uniqueness of Proposition 11; both deliver a unique C^1 pricing selection, which is all the abstract framework requires. (ii) The per-market Cournot markup bound $\mu \in [\sigma/(\sigma - 1), \eta/(\eta - 1)]$ is replaced by the Bertrand bound $p - c \in [\gamma^{-1}, \gamma^{-1}/\lambda_{\min}]$; both enter K_v through the same Lipschitz-constant chain. (iii) The complete-information profit gap δ_v under CES–AB is replaced by its nested-logit analog δ_v^{NL} ; in both environments this gap is used only to construct and check candidate localized regions for Lemma 2, rather than as a standalone proof of localization. The quantitative application of Section 6 employs Fréchet–CES–AB because the calibration disciplines of Eaton and Kortum (2002) and Atkeson and Burstein (2008) map cleanly into public cloud-infrastructure data; the nested-logit verification here certifies that an econometrician who preferred a classical McFadden demand primitive could apply the same uniqueness theorem with no change to the abstract argument.

I. CALIBRATION DETAILS AND ADDITIONAL COUNTERFACTUALS

This appendix provides the full calibration construction, specification comparison, robustness exercises, the data-localization counterfactual, and the interval-certification audit summarized in Section 6.

Computational verification summary. Our computational verification proceeds in four layers. First, Appendix E establishes analytically that the AB per-market convexity condition is governed by a strictly concave scalar function $H_{AB}(s)$, yielding a unique implicit threshold $\bar{s}_{AB}^*$ (Proposition 9). Second, pricing uniqueness is constructively witnessed by a Perron–Frobenius weighted-norm contraction with $\Lambda w \leq \lambda w$ and $\lambda < 1$ (Proposition 5). Third, the Jia–Tarski squeezing procedure is interval-audited conditional on the converged rival-demand vectors: all 129 decisive inclusion/exclusion inequalities are certified with zero inconclusive cases. Fourth, full-game uniqueness is verified numerically by coincidence of the supremum and infimum solutions.

Demand. Market demand is parameterized as $\Omega_n = (\text{GDP}_n/\text{GDP}_{\text{US}})^\beta$ with $\beta = 0.7$. The value $\beta = 0.7$ is chosen from a parsimonious calibration comparison: at common bounds

and wage specification, $\beta = 0.7$ achieves a better fit than $\beta = 1.0$ with a more plausible ϕ_i distribution, and the counterfactual ranking (AB-versus-CES profit decline in the -31 to -33% range) is preserved at both values.⁹

Local costs. We use PWT 10.01 compensation per worker, adjusted for human capital: $w_i = \tilde{w}_i / (\text{hc}_i / \text{hc}_{\text{US}})$, where $\tilde{w}_i$ is the raw relative wage ($\tilde{w}_{\text{US}} = 1.0$) and hc_i is the PWT human capital index. The resulting w_i ranges from 0.16 (Philippines) to 1.94 (UAE). We also test TFP-adjusted wages ($w_i^{\text{tfp}} = \tilde{w}_i / \text{ctfp}_i$), which produces virtually identical mechanism results; the precise agreement count at canonical bounds is pending a clean rerun, but legacy-bounds runs placed the TFP-wage variant within one cell of the HC-wage baseline, with the AB-versus-CES profit decline within 0.2 percentage points.

Trade costs. We construct a discrete three-tier iceberg matrix: $\tau_{in} = 1.0$ (same country), 1.15 (same continent or $< 5,000$ km), 1.5 (cross-continent), using capital-city coordinates. The tier values are motivated by the continent/cross-continent τ ratio implied by gravity estimates in the international trade literature at a CES elasticity in the range of our baseline; the discretization is adopted because our application has $N = 42$ markets and a full 42×42 gravity matrix is neither identified from entry-count moments alone nor central to the paper’s mechanism. This is a coarse approximation; cloud-specific frictions (latency, data-sovereignty regulations) would require a richer specification beyond the scope of this application. A sensitivity sweep over the cross-continent tier (e.g., $\tau \in \{1.3, 1.7\}$) is deferred to the revision, pending a clean rerun of the full counterfactual stack.

Specification comparison. Three specifications differ in which firm-level parameters are set externally: Spec A (preferred) fixes a_f from revenue, calibrates (g_f, ϕ_i) ; Spec B calibrates (a_f, ϕ_i) with $g_f = 1$; Spec C calibrates all three jointly. At canonical bounds, Spec A achieves 199/210 (94.8%). Legacy-bounds runs placed Spec B at 201/210 (95.7%) and Spec C at 197/210 (93.8%); the canonical-bounds rerun of Specs B and C is pending, but the ordering ($A \geq B \geq C$) is expected to persist because fixing a_f externally separates

⁹The $\beta = 1.0$ comparison reported in earlier drafts at 209/210 versus 207/210 agreement used legacy parameter bounds predating the canonical file (Phase 1); a rerun at canonical bounds is scheduled for the next revision and is not expected to reverse the ranking, since the fit differential tracks the revenue-to-GDP curvature rather than the bounds.

firm scale from deployment efficiency regardless of the bound choice.¹⁰

Fringe-firm TFP sensitivity. Rescaling Oracle’s a_f by +25% and Alibaba’s by +50% produces muted extensive-margin effects (≤ 1 additional market) but meaningful intensive-margin compression of Big 3 markups.

Profit-gap distribution. Figure 8 plots the distribution of per-site profit gaps ($VP_i - FC_i$) across all 128 active sites in the preferred specification. Of the 128 active sites, 66 have $|\text{profit gap}| < 0.005$ (close to the entry-indifference margin) and 33 have gaps exceeding 0.01 (deeply inframarginal). The 199/210 in-sample agreement rate therefore reflects genuine discrimination at the entry margin, not merely the fitting of inframarginal cells.

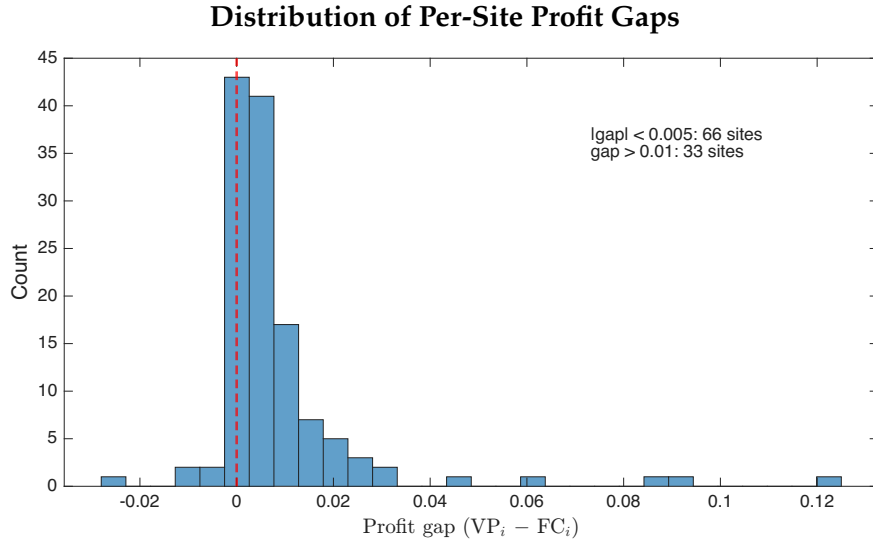


Figure 8: Distribution of per-site profit gaps ($VP_i - FC_i$) across 128 active firm-market pairs. Over half of active sites are close to the entry-indifference margin ($|\text{gap}| < 0.005$), indicating that the calibration is discriminating at the margin rather than fitting inframarginal cells.

Holdout cross-validation. The in-sample agreement figures (199/210, 94.8%) are informative but do not by themselves rule out interpolation of the observed activation matrix. To assess predictive content, we report a 5-fold masked-cell holdout exercise within the same 210-cell cross section.¹¹ The 210 firm–market cells are randomly partitioned into five

¹⁰Legacy bounds used $\phi_i \in [10^{-4}, 2]$ and $g_f \in [0.1, 5]$; canonical bounds tighten to $\phi_i \in [0.02, 0.30]$ and $g_f \in [0.1, 3]$ to improve conditioning (see Appendix I).

¹¹The masked calibration is numerically ill-conditioned under the main-specification parameter bounds ($\phi_i \in [10^{-4}, 2]$, $g_f \in [0.1, 5]$), where the outer inversion fails to converge within 5,000 iterations. We therefore

disjoint blocks of 42 cells each; in each fold, ϕ_i and g_f are recalibrated using only the 168 training cells, and the fitted model is then solved on the full lattice and evaluated on the 42 held-out cells. Table 13 reports per-fold training agreement, holdout agreement, and the holdout true-positive and true-negative rates. Relative to the majority-class baseline of 0.614, the calibrated model achieves mean holdout agreement of 0.824 (SD 0.055), indicating substantial predictive content beyond exact in-sample matching: the calibrated fixed-cost structure retains predictive discipline on held-out cells within the observed cross section.¹²

Table 13: 5-Fold Holdout Cross-Validation

Fold	n_{train}	n_{hold}	Train agr.	Holdout agr.	Holdout TPR / TNR
1	168	42	0.940	0.786	0.679 / 1.000
2	168	42	0.946	0.762	0.724 / 0.846
3	168	42	0.952	0.833	0.739 / 0.947
4	168	42	0.952	0.905	0.917 / 0.889
5	168	42	0.946	0.833	0.720 / 1.000
Mean	—	—	0.948	0.824	0.756 / 0.936
SD	—	—	0.005	0.055	— / —
Majority-class baseline				0.614	—

Notes: Each fold holds out 42 of the 210 firm–market cells (20%) chosen uniformly at random; ϕ_i and g_f are recalibrated using per-market and per-firm moment counts restricted to the 168 training cells, and the fitted model is then solved on the full lattice. Agreement = $\frac{1}{|S|} \sum_{(f,i) \in S} \mathbf{1}[I_{fi}^{\text{obs}} = I_{fi}^{\text{model}}]$ for S = training or holdout. TPR and TNR are computed on the holdout cells only. The majority-class baseline is $\max(\bar{I}, 1 - \bar{I}) = 0.614$.

Firm-drop out-of-sample validation. To complement the holdout CV with an out-of-sample test at the firm margin, we drop each firm f_{drop} in turn and recalibrate (ϕ_i, g_f) on the remaining four-firm lattice, with a_f fixed at revenue values. We then solve the four-firm equilibrium, place f_{drop} back at its observed (a, g) , and solve its best-response problem against the rival demand $D_{-f_{\text{drop}}}$ from that equilibrium. The dropped firm’s cells never enter any equilibrium during calibration, so its activation prediction is a clean

run the holdout on tighter bounds ($\phi_i \in [0.02, 0.30]$, $g_f \in [0.1, 3]$) that restore convergence; a full-bound holdout study would require a more robust masked solver and is left for future work.

¹²The exercise tests the restriction *given the equilibrium structure* rather than as an econometric out-of-sample prediction: the equilibrium is solved on the full 5×42 lattice at the training-calibrated (ϕ_i, g_f) , so general-equilibrium price and share feedback from held-out cells enters the prediction. A genuinely out-of-sample exercise at the firm margin follows in the next paragraph.

out-of-sample object. Table 14 reports per-firm hit rates. The mean hit rate is 94.3% (range 85.7–97.1%), well above both the majority-class baseline (61.4%) and the holdout-CV average (82.4%). The four hyperscalers (AWS, MSFT, GOOG, ORCL) all exceed 96%; BABA’s hit rate of 85.7% (missing U.S. and UAE) reflects its distinct geographic profile, and the miss set matches the pattern of within-sample tension (UAE is the most-missed cell across all drops). The exercise strengthens the identification: the calibration’s predictive content survives removing an entire firm from the identification set.

Table 14: Firm-Drop Out-of-Sample Validation

Dropped firm	Observed	Predicted	Overlap	Missed	% hit
AWS	25	29	24	1	96.0
MSFT	34	33	33	1	97.1
GOOG	29	28	28	1	96.6
ORCL	27	26	26	1	96.3
BABA	14	12	12	2	85.7
Mean	26	26	25	1	94.3

Notes: For each dropped firm f_{drop} , (ϕ_i, g_f) are recalibrated on the four-firm lattice (remaining firms, a_f fixed at revenue values). The four-firm equilibrium is solved, and f_{drop} is placed back at its canonical (a, g) with the market-common ϕ_i from the four-firm calibration. Its best-response activation against rival demand $D_{-f_{\text{drop}}}$ is computed via Jia–Tarski (gap = 0 at every drop). “Observed” and “Predicted” are cell counts; “Overlap” is the intersection size. “% hit” is Overlap / Observed. The mean hit rate of 94.3% exceeds the majority-class baseline 61.4% by 33 percentage points.

EU compliance-cost ladder. Table 15 reports the extensive-margin response across a grid of additive compliance shocks $\Delta\phi$. The response saturates rapidly: by $\Delta\phi = 0.005$ (roughly twice the median baseline $\phi_i = 0.002$), nearly all EU firm-market pairs have exited. This indicates that most EU sites are near the entry margin at baseline, so even modest compliance wedges trigger substantial exit. The qualitative regressivity finding—lower-ranked firms lose proportionally more—is robust across the entire grid.

Interval certification of Jia–Tarski sign checks. The JT gap = 0 diagnostic is strengthened by a post-hoc interval audit of the squeezing procedure. The auditor re-evaluates every decisive site-level marginal-value comparison used by the Jia–Tarski solver, conditional on the converged rival-demand vector D_{-f} . For each candidate site, market shares are enclosed by monotone scalar root-bracketing on the strictly increasing AB pricing

Table 15: EU Compliance-Cost Ladder

$\Delta\phi$	Active	Δ	AWS	MSFT	GOOG	ORCL	BABA
0 (baseline)	128	—	6	11	9	6	1
0.001	111	-17	-2	-5	-4	-6	-1
0.005	98	-30	-6	-10	-9	-6	-1
0.010	97	-31	-6	-11	-9	-6	-1
0.030	97	-31	-6	-11	-9	-6	-1

Notes: Columns AWS–BABA report EU market exits (negative) relative to baseline EU presence. The first row shows baseline EU market counts. The response saturates between $\Delta\phi = 0.005$ and 0.01.

equation (bisection to 10^{-12} tolerance), these enclosures are mapped into variable-profit intervals via the monotonicity of $VP(s) = R_n(s/\sigma + bs^2)$, and the resulting site-level marginal values are aggregated across markets with conservative interval arithmetic. A site is certified for inclusion when the lower bound on its marginal value is strictly positive, and for exclusion when the upper bound is strictly negative. In the baseline cloud illustration, all 129 decisive sign checks are certified with zero inconclusive cases; the remaining five AWS exclusions arise from the exogenous feasible-set restriction $\mathcal{A}_f$ and do not require sign certification. The resulting JT characterization is a finite interval certificate for the solved single-agent activation subproblems, conditional on the equilibrium rival-demand vectors.

Data-localization counterfactual. Removing China from all non-Alibaba feasible sets leaves the total at 128 active pairs (AWS gains two markets while losing China). The reverse scenario—China opens and Alibaba’s feasible set expands to all 42 markets—increases total pairs from 128 to 155, with Alibaba entering 28 new countries. The binding constraint is market access ($\mathcal{A}_f$), not entry costs (ϕ_i): reducing ϕ_{China} by 90% without feasible-set expansion produces zero additional entry. This counterfactual is sensitive to the deployment-efficiency specification and is reported as illustrative of the mechanism.

J. ALGORITHM PERFORMANCE

J.1 Numerical Algorithm Comparison

Table 16 reports the results for all four algorithm variants in the baseline economy ($N = 20$, $F = 8$). All converge, and the Jia–Tarski gap is zero for all eight firms in every iteration—providing a computational diagnostic consistent with the maintained lattice structure of the solved single-agent subproblems (Proposition 2).

Table 16: Algorithm Comparison ($N = 20$, $F = 8$)

Variant	Iters	Time (s)	CDCP Time (s)	JT Gap
Jia–Tarski + Jacobi	120	2.38	2.19	0
Jia–Tarski + Gauss–Seidel	115	1.92	1.86	0
AES + Jacobi	120	1.16	1.12	—
AES + Gauss–Seidel	115	1.12	1.07	—

Gauss–Seidel converges 4% faster than Jacobi (115 vs. 120 iterations). The AES solver is approximately 40% faster than Jia–Tarski per call: under supermodularity, the squeezing phase prunes most candidates, reducing the branching tree. The value of Jia–Tarski is not speed but its *diagnostic*: a gap of zero indicates coincidence of the lower and upper lattice iterates at the solved instance, whereas AES provides no comparable diagnostic.

Inner-solver consistency. To verify that the two inner solvers agree, we run a pairwise comparison: JT+supremum versus AES+supremum, and JT+infimum versus AES+infimum. In both cases, the solvers produce identical activation sets at every outer iteration, confirming that both algorithms return the same solved activation set for the audited single-agent CDCP instances. We further test each firm in isolation by calling both solvers with identical rival demand indices D_{-f} ; all firms return the same activation set from both algorithms. These diagnostics confirm that the numerical results are solver-invariant at the tested parameterization.

Figure 9 plots the convergence metric $\max_{n,f} |\Delta p_{nf}^{1-\sigma}|$. Convergence is monotone and geometric, consistent with Proposition 4.

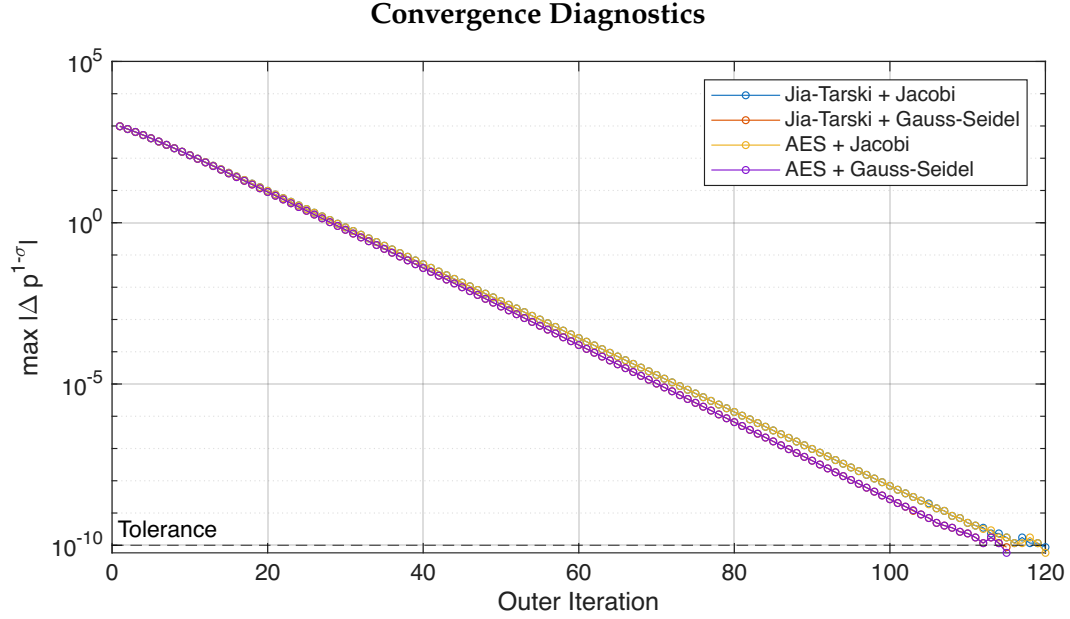


Figure 9: Convergence of Outer Iterations ($N = 20, F = 8$)

J.2 Extension: $N = 40, F = 10$

To demonstrate scalability, we double the economy to $N = 40$ countries (eight clusters of five, forming two “hemispheres”) and $F = 10$ firms. Firms 1 and 2 now face $2^{40} \approx 1.1 \times 10^{12}$ candidate subsets each—exhaustive enumeration is impossible.

Table 17: Algorithm Comparison ($N = 40, F = 10$)

Variant	Iters	Time (s)	CDCP Time (s)	JT Gap
Jia-Tarski + Jacobi	114	17.44	17.04	0
Jia-Tarski + Gauss-Seidel	110	14.48	14.27	0
AES + Jacobi	114	7.29	7.11	—
AES + Gauss-Seidel	110	6.84	6.66	—

Both Jia-Tarski and AES solve the problem in seconds—Jia-Tarski in 14 seconds, AES in 7 seconds—despite the astronomical size of the brute-force search space. The Jia-Tarski gap remains zero for all ten firms, confirming that supermodularity extends to this scale. As in the baseline, all four solver combinations converge to the same unique equilibrium with 96 total active sites and aggregate profit 69.19.

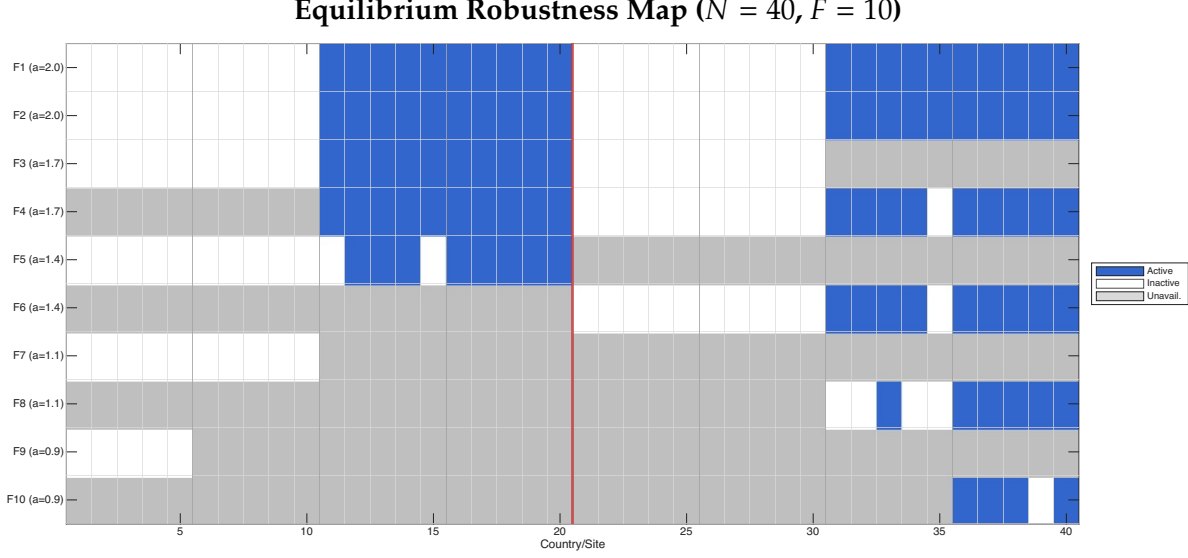


Figure 10: Site Activation Map ($N = 40, F = 10$, unique equilibrium). Blue: active sites; white: available but inactive; gray: unavailable.

K. SYNTHETIC STRESS TESTS

The cloud application in Section 6 demonstrates that the theoretical results hold at empirical parameter values ($F = 5, N = 42$). This appendix provides complementary stress tests using a synthetic economy ($N = 20, F = 8$) that permits systematic sweeps over the parameter space, including regions where multiplicity emerges.

K.1 Parameterization

The synthetic economy uses $\sigma = 5, \eta = 2, \theta = 3$, matching the cloud application. Twenty countries are organized in four income clusters with dispersed wages, expenditures, and trade costs. Eight firms span a TFP hierarchy from global players ($a_f = 2.0, |\mathcal{A}_f| = 20$) to regional players ($a_f = 1.0, |\mathcal{A}_f| = 5$). Wages are drawn on a log-linear grid $w_i \in [0.15, 1.90]$, expenditures Ω_n on a log-linear grid $[0.05, 1.00]$; trade costs follow the three-tier iceberg structure of Section 6 (1.00 same country, 1.15 same cluster, 1.50 cross-cluster).

K.2 Contraction Modulus: The Bathtub

As shown in Figure 1 (Section 3.2), the contraction modulus has a bathtub shape with peak $\gamma_{\max} = 0.951 < 1$ in the small economy. The bathtub shape is preserved across

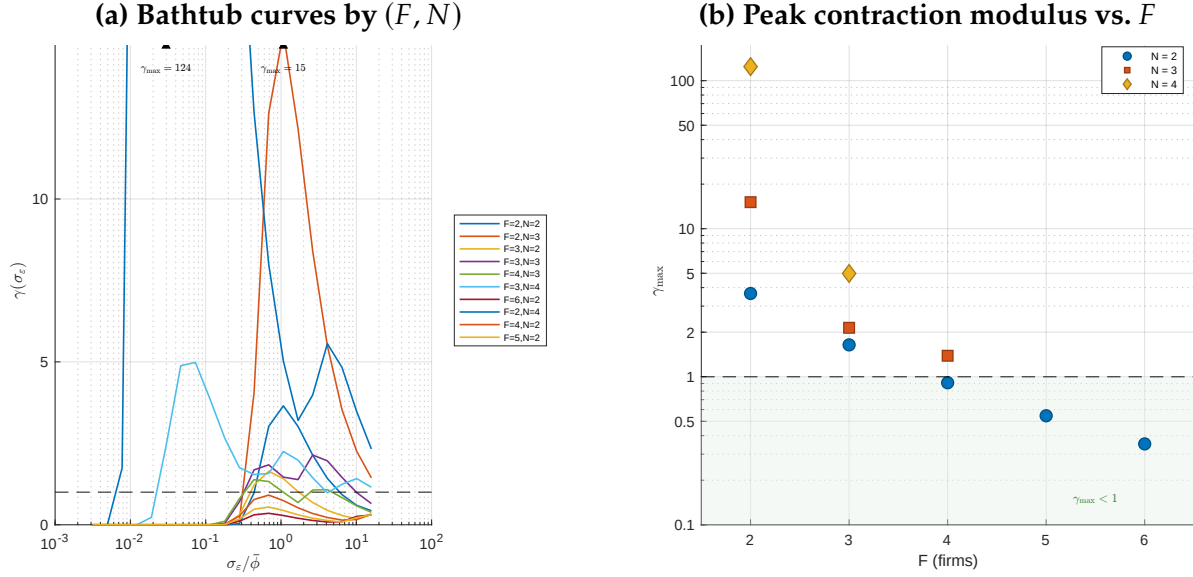


Figure 11: Scaling of the contraction modulus with (F, N) . (a) Bathtub curves for economies of different sizes. (b) Peak $\gamma_{\max}$ vs. F .

economies of different sizes (Figure 11). Two opposing forces determine $\gamma_{\max}$: more firms means smaller market shares and weaker marginal-value sensitivity ($\gamma_{\max}$ decreases with F), while more sites create additional near-marginal activation decisions ($\gamma_{\max}$ increases with N). The contraction condition is most easily satisfied in the empirically relevant regime of many firms with moderate market shares.

K.3 Uniqueness Boundary: AB versus CES

To identify where multiplicity emerges and how AB markups shift the boundary relative to CES, we conduct a joint sweep over TFP dispersion $\delta = a_{\max}/a_{\min}$ and fixed-cost scale α in the synthetic economy.

Figure 12 maps the uniqueness region. The equilibrium is unique across most of the parameter space; multiplicity emerges only at extreme TFP dispersion ($\delta \geq 7$) combined with high fixed costs ($\alpha \geq 1.5$), where a single marginal site produces two equilibria.

Figure 13 compares the AB and CES uniqueness regions. On the 11×11 grid, AB markups *expand* the uniqueness region: 23 cells have a unique equilibrium under AB but multiple equilibria under CES, whereas only 1 cell shows the reverse. The mechanism is that AB markup compression dampens the dominant firm's effective demand response, keeping marginal sites' profit gaps bounded away from zero.

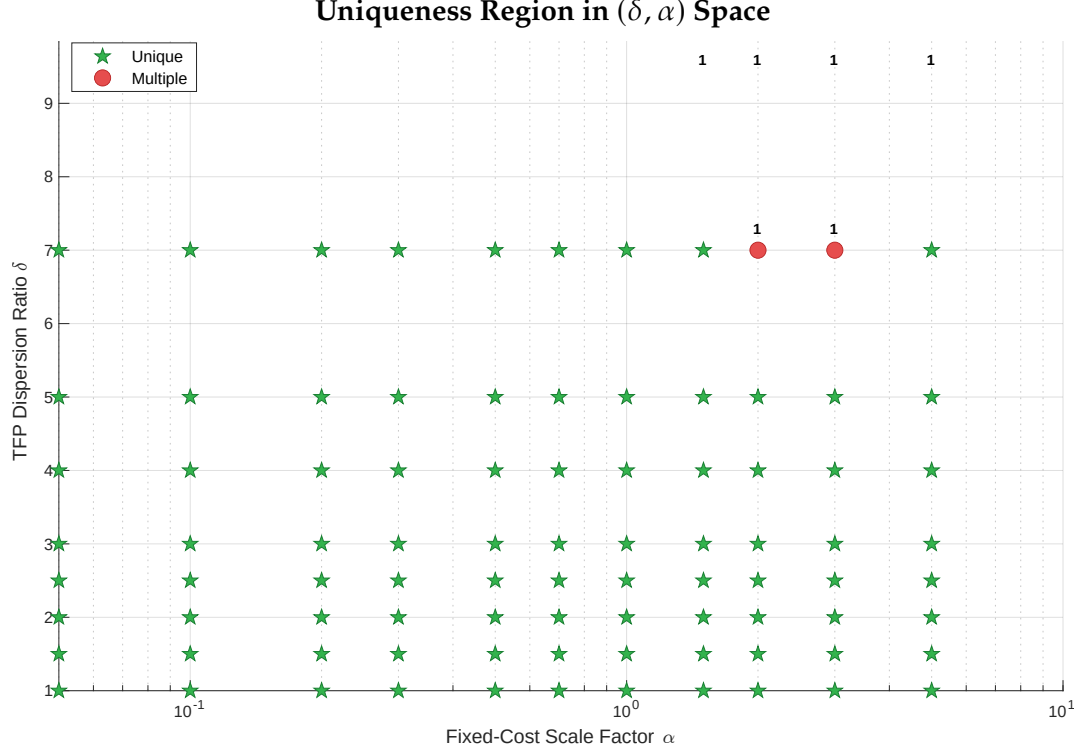


Figure 12: Uniqueness region in (δ, α) space ($N = 20, F = 8$). Green: unique equilibrium; red: multiple equilibria. Multiplicity requires extreme TFP dispersion and high fixed costs.

K.4 Broader Parameter Sensitivity

Table 18 extends the analysis to a $3 \times 3 \times 4$ grid over (σ, η, θ) . In 18 of 28 configurations the analytic supermodularity condition fails $((\sigma - 1)/\theta \leq 1)$. Nevertheless, the Jia–Tarski gap equals zero and the equilibrium is unique at all 28 configurations. The spectral radius $\rho(\Lambda)$ reaches 0.225 at $(\sigma, \eta) = (8, 1.5)$ but remains well below one throughout.

K.5 Welfare Decomposition: Variety versus Markup Channel

The AB and CES pricing rules have offsetting welfare effects. Under AB, the 16% activation expansion (37 vs. 32 active sites in the synthetic baseline) generates additional product variety, lowering the sectoral price index P_n . But the higher variable markups ($\bar{\mu}^{AB} = 1.36$ vs. $\mu^{CES} = 1.25$) raise prices. Computing the CES real-income index $W_n = R_n/P_n$ at each market (holding R_n fixed at its initial value), AB welfare is on average 11% *lower* than CES welfare in the synthetic economy. The markup channel dominates the variety channel at

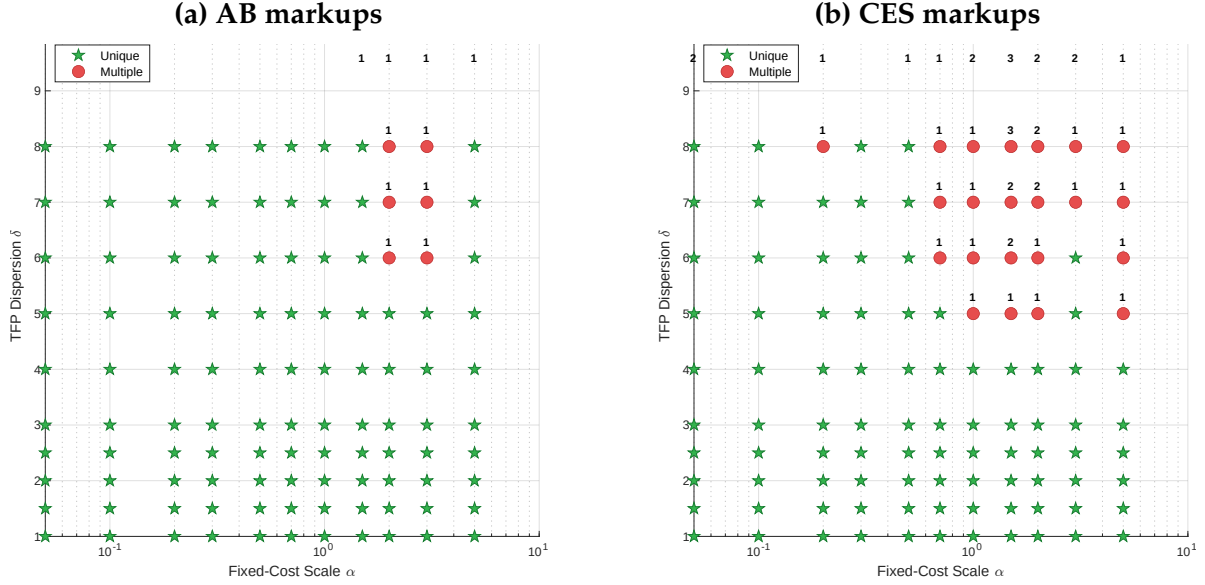


Figure 13: Uniqueness region: AB (left) vs. CES (right). AB markups expand the uniqueness region—23 of 121 cells switch from multiple to unique equilibria.

all 20 markets, with the welfare gap largest in advanced markets (where concentration is highest) and smallest in frontier markets.

This decomposition underscores that the activation decision—the dimension on which this article’s uniqueness result operates—has quantitatively significant welfare consequences that constant-markup models misattribute to other channels. However, the welfare comparison is partial equilibrium: R_n is held fixed, and general-equilibrium feedbacks through wages or demand could alter the ranking. The result should be interpreted as a diagnostic, not as a normative recommendation.

K.6 Sufficient Conditions: Infinity-Norm versus Spectral Radius

At the synthetic baseline, the maximum feedback coefficient $\max \kappa_{nf} = 0.193$ exceeds the infinity-norm threshold $1/(F - 1) = 0.143$ from Proposition 5, because the dominant firms’ high market shares (up to $s = 0.39$, above the per-market threshold $\bar{s} = 0.339$ from Corollary 3) generate large pricing feedback. However, the tighter spectral condition $\rho(\Lambda) = 0.096 < 1$ confirms pricing uniqueness. The same pattern holds in the cloud application ($\rho(\Lambda) = 0.109$): the infinity-norm condition is conservative, while the spectral condition resolves the intensive margin comfortably. The two numbers relate as $\|\Lambda\|_\infty = \max_{n,f} \kappa_{nf}$ (the row-sum norm equals the largest feedback coefficient because Λ has a

Table 18: Uniqueness Diagnostics across (σ, η, θ) : Synthetic Economy

σ	θ	$\eta = 1.5$				$\eta = 2$				$\eta = 3$			
		$\frac{\sigma-1}{\theta}$	ρ	JT	sites	$\frac{\sigma-1}{\theta}$	ρ	JT	sites	$\frac{\sigma-1}{\theta}$	ρ	JT	sites
3	3	0.67	.047	0	39	0.67	.025	0	38				
3	5	0.40	.047	0	19	0.40	.026	0	18				
3	8	0.25	.048	0	16	0.25	.026	0	16				
5	3	1.33	.125	0	38	1.33	.096	0	37	1.33	.053	0	36
5	5	0.80	.123	0	17	0.80	.095	0	16	0.80	.052	0	16
5	8	0.50	.123	0	15	0.50	.095	0	15	0.50	.052	0	15
8	3	2.33	.225	0	39	2.33	.208	0	36	2.33	.160	0	34
8	5	1.40	.222	0	18	1.40	.204	0	18	1.40	.157	0	18
8	8	0.88	.222	0	15	0.88	.204	0	14	0.88	.157	0	13

Notes: Synthetic economy ($N = 20, F = 8$). $\sigma > \eta$ required; $(\sigma, \eta) = (3, 3)$ excluded. JT: Jia–Tarski gap (0 at all configurations). All 28 configurations yield a unique equilibrium.

single nonzero diagonal entry per row in our parameterization), so $\rho(\Lambda) \leq \|\Lambda\|_\infty$ is simply the standard eigenvalue–row-sum inequality.

This gap between the two sufficient conditions—the infinity norm violated but the spectral radius well below one—is a generic feature of economies with firm-level TFP heterogeneity. Heterogeneous TFP concentrates market shares in a few dominant firms, inflating $\max \kappa$ and violating the per-market condition, but the spectral radius accounts for the fact that high- κ pairs are few and their feedback does not compound across the full matrix.

L. APPENDIX ROBUSTNESS: ALTERNATIVE GRAVITY-ACCESS CLOUD IMPLEMENTATION

This appendix reports an alternative gravity/accessibility implementation of the cloud application and the corresponding verification package. The gravity layer is a smooth reduced-form accessibility shifter rather than a separate theorem environment. It does not claim that smooth accessibility alone implies theorem coverage, global CDCP certification, or automatic preservation of supermodularity. Accordingly, the appendix first summarizes the alternative implementation and its quantitative readout, then records the regularity of the frontier-normalized accessibility index, states the maintained contraction condition for the quality-adjusted pricing block, derives the decomposition needed for

the full marginal-value Jacobian, and summarizes the calibrated-specification diagnostics used for localization and for interpreting the computed Jia–Tarski gap under gravity.

L.1 Alternative Gravity-Access Implementation of the Cloud Application

We re-implement the cloud application using a gravity / accessibility layer that reconstructs provider-market serviceability from public geography and policy data while keeping the calibrated fixed-cost structure, firm-scale normalization, and pricing-comparison logic unchanged. Formal definitions of the gravity score, frontier normalization, and quality shifter are collected in Subsection L.2. Under this architecture, Ω_n captures market potential, the gravity layer captures provider-market serviceability, and the AB-versus-CES comparison is interpreted as a pricing benchmark holding the accessibility block fixed. Geography, policy, and market-size objects are disciplined externally, while the accessibility loading parameters are selected on a small plausibility grid rather than estimated from entry moments alone.

Preferred gravity baseline. The preferred gravity specification uses the cloud-market proxy $\Omega_n \propto GDP_n \times \text{internet penetration}_n$, frontier-difference normalization, the OECD-based ‘suggested_weight’ policy mode, and $\chi = 0.40$. Re-calibrating (ϕ_i, g_f) under that specification yields 200/210 correct firm-market cells, with 98.4% precision, 93.8% recall, $\rho = 0.1106$, and a computed Jia–Tarski diagnostic of zero. Alternative choices for χ , market-size proxies, and simple public hub augmentations do not improve on that baseline once fit and diagnostics are considered jointly. The remaining baseline mismatches cluster in regional hub markets, especially the UAE, Saudi Arabia, Singapore, and Thailand, which we interpret as a limitation of the parsimonious public-data accessibility layer rather than a general failure of the framework. Figure 14 displays the observed versus fitted activation matrix, and Figure 15 summarizes the spectral-radius diagnostics across the main gravity counterfactuals.

Counterfactual readout under the gravity baseline. The main mechanism result remains CF2: holding the gravity accessibility block and calibrated fixed costs fixed, replacing AB pricing with constant-markup CES reduces active firm-market pairs from 123 to



Figure 14: Observed versus fitted activation matrix under the preferred gravity baseline. Rows are firms; columns are the 42 markets, shown as ISO-3 market codes and ordered by observed entry count. Blue cells are matched active markets, white cells are matched inactive markets, orange cells are observed-only activations, and red cells are fitted-only activations. The gravity baseline matches 200 of 210 firm–market cells.

95 and lowers aggregate profit by 33.8%, confirming that endogenous markups materially sustain entry in the gravity implementation. Figure 16 shows the same mechanism visually: AB markups exceed the CES benchmark throughout, shares compress under CES, and the extensive-margin losses are concentrated in Oracle and Alibaba. CF3 remains a regional-structure result rather than a markup result: a small EU compliance wedge already triggers substantial exit, with active pairs falling to 110 at $\Delta\phi = 0.001$, 95 at $\Delta\phi = 0.005$, and 94 at $\Delta\phi = 0.010$, so the ladder saturates quickly; Figure 17 reports the corresponding EU-market exits under AB and CES pricing at $\Delta\phi = 0.001$. CF1 likewise remains an access story rather than a cost story: reducing ϕ_{China} by 90% or lowering trade-cost-like frictions alone leaves total entry unchanged, whereas adding all-market access for Alibaba raises active pairs from 123 to 131 and gives Alibaba 13 new markets. CF5 is treated as appendix robustness only: fringe-firm TFP increases compress Big 3 markups and reallocate activity, but the net effect on total active pairs is negative rather than expansionary.

Verification scope. This appendix implementation should be read as a cloud-specific robustness layer rather than as a new theorem environment. The remaining subsections record the application-side verification package for accessibility smoothness, quality-adjusted pricing regularity, bounded full marginal-value sensitivity, localization, and the interpretation of the computed Jia–Tarski diagnostic under gravity.

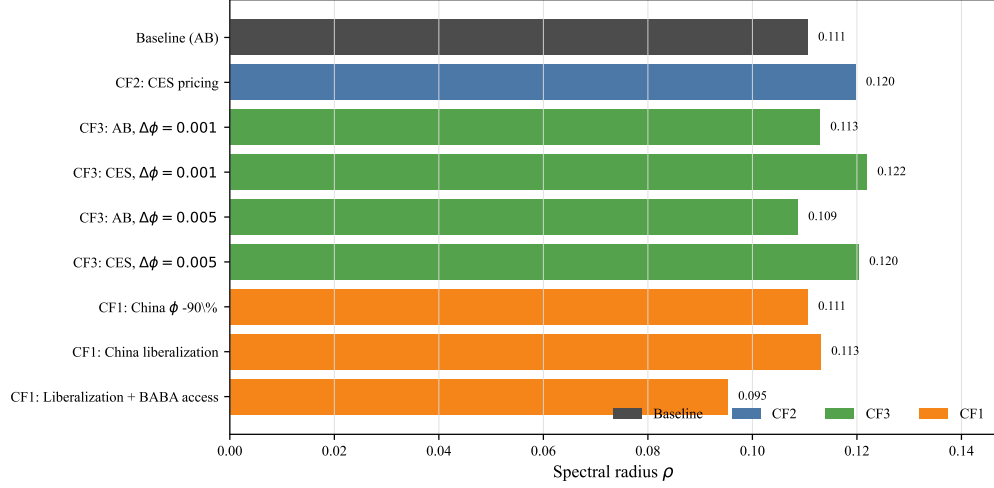


Figure 15: Spectral radius ρ across the main gravity counterfactuals. The preferred baseline, the AB-versus-CES pricing comparison, the EU compliance ladder, and the China-access counterfactuals all remain far below one; the largest value in the plotted scenarios is 0.122.

L.2 Accessibility Construction and Frontier Normalization

For candidate site i and market n , the gravity implementation uses the score

$$q_{in} = -\beta_d \log(1 + d_{in}) + \beta_{\text{same}} \mathbf{1}\{i = n\} + \beta_{\text{local}} z_{in},$$

the smooth accessibility index

$$A_{fn}(I_f) = \lambda_A^{-1} \log \left(1 + \sum_{i \in I_f} \exp(\lambda_A q_{in}) \right),$$

and the fixed market-level frontier

$$A_n^{\text{frontier}} = \lambda_A^{-1} \log \left(1 + \sum_{i \in S} \exp(\lambda_A q_{in}) \right).$$

Provider f 's quality shifter is

$$\psi_{fn}(I_f) = \exp(\chi[A_{fn}(I_f) - A_n^{\text{frontier}}]).$$

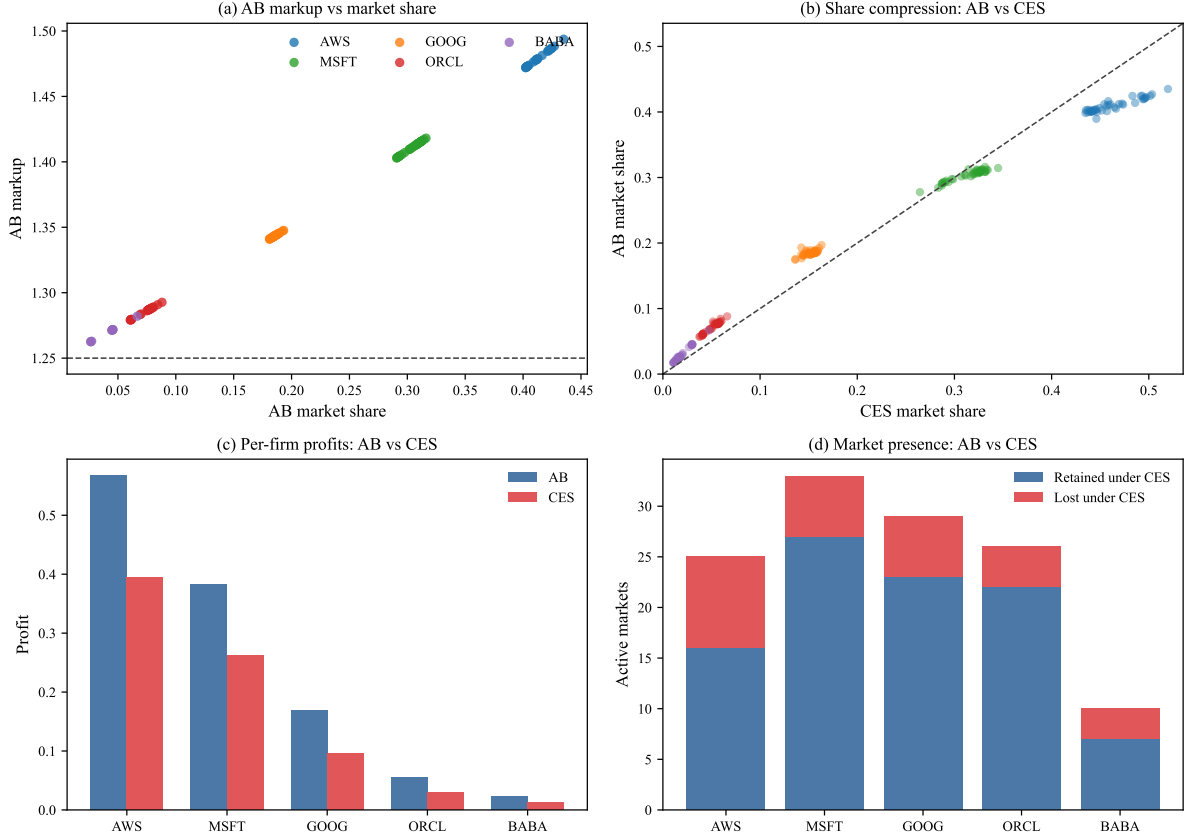


Figure 16: AB versus CES pricing under the preferred gravity baseline. Panel (a) plots AB markups against market shares for active firm–market pairs, with the constant CES markup shown by the dashed line. Panel (b) compares firm–market shares under AB and CES pricing. Panel (c) reports per-firm profits, and panel (d) decomposes AB-active markets into CES-retained and CES-lost markets.

This log-sum object is a smooth certainty-equivalent accessibility index, not the literal expectation of realized accessibility over random activation sets.

Proposition 13 (Smoothness of Frontier-Normalized Accessibility). *Fix finite site and market sets, suppose $|q_{in}| \leq \bar{q} < \infty$, and let $\lambda_A \in (0, \infty)$. Then the frontier-normalized accessibility shifter ψ_{fn} is continuously differentiable in the activation-probability vector on $[0, 1]^M$, with derivatives uniformly bounded on that domain.*

Proof. The map inside the logarithm is a finite sum of exponentials plus one, so it is strictly positive and smooth on $[0, 1]^M$. Subtracting the fixed frontier term A_n^{frontier} does not affect derivatives, and boundedness follows from the finite site universe and the uniform bound on q_{in} . $\square$

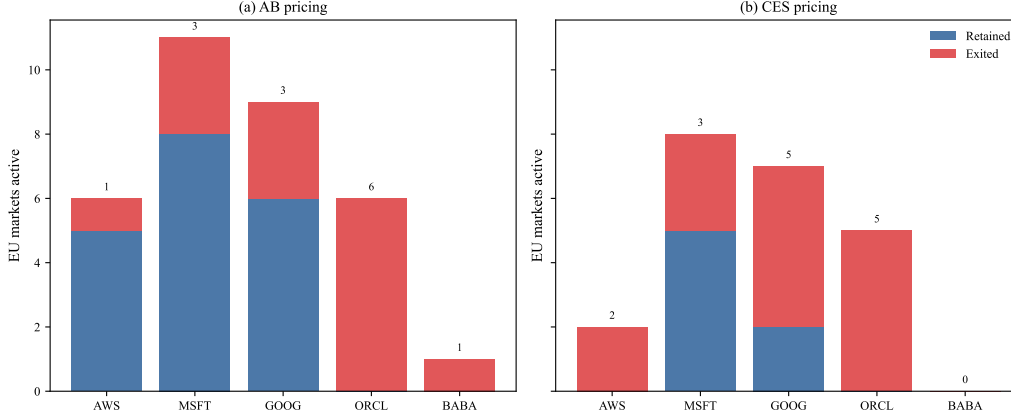


Figure 17: EU compliance shock under the preferred gravity baseline at $\Delta\phi = 0.001$. Within each pricing regime, blue bars show EU markets retained after the compliance wedge and red segments show exited EU markets; numeric labels report the number of exits by firm. The response is substantial under both AB and CES pricing and is concentrated in firms with broader marginal EU exposure.

L.3 Quality-Adjusted Pricing Regularity

Gravity changes the quality or serviceability layer, not the benchmark pricing comparison itself. Conditional on an activation vector, the pricing block therefore remains a quality-adjusted version of the same AB-versus-CES comparison used in the main text. The relevant maintained condition is that the quality-adjusted pricing map admits a uniform contraction margin on an interior activation region.

Proposition 14 (Quality-Adjusted Pricing Regularity and Bounded Price Jacobian). *Suppose the quality-adjusted pricing block admits a uniform contraction margin on an interior activation region. Then the gravity implementation has a unique equilibrium price map $p(v)$ on that region, the map is C^1 , and its Jacobian $J_p(v)$ is uniformly bounded.*

Proof. The argument is the same weighted-norm contraction and implicit-function logic used for the baseline pricing block, applied to quality-adjusted shares. The gravity layer enters only through bounded multiplicative shifters, so a uniform contraction margin yields a unique fixed point and a bounded Jacobian for the resulting price map. $\square$

This proposition supplies the pricing-side regularity required by the abstract threshold theorem; it does not imply that the single-agent activation problem inherits the baseline JT or global-certificate logic unchanged.

L.4 Full Marginal-Value Jacobian under Gravity

Smooth accessibility is not by itself enough for theorem compatibility. What matters for the threshold map is the full marginal-value Jacobian. Under gravity, the relevant chain-rule decomposition is

$$J_v = \frac{\partial v}{\partial \psi} J_\psi + \frac{\partial v}{\partial p} J_p.$$

The first term is controlled by Proposition 13; the second requires the pricing-side regularity recorded in Proposition 14.

Proposition 15 (Bounded Full Jacobian under the Gravity Implementation). *On any activation region where the normalized accessibility map is smooth with bounded derivative and the quality-adjusted pricing block admits a bounded Jacobian, the full marginal-value map $v(v)$ is continuously differentiable with uniformly bounded Jacobian.*

Proof. The claim follows directly from the displayed decomposition and the chain rule. Boundedness of J_ψ and J_p implies boundedness of J_v whenever the partial derivatives of v with respect to ψ and p are bounded on the same activation region. $\square$

L.5 Localization for the Calibrated Gravity Specification

The small-noise step is treated as a calibrated-specification verification rather than as a primitive theorem proof. In particular, a positive complete-information gap is not used as a replacement for Lemma 2. Instead, the calibrated gravity implementation is checked numerically for compatibility with a fixed-point-inclusive non-marginal region motivated by the complete-information gap and evaluated through the same fixed-point diagnostics used in the calibration code. The resulting localization evidence is therefore specification-level rather than global: it supports the preferred gravity baseline and the leading robustness checks reported below, but it is not presented as an analytic invariant-set proof for every gravity parametrization.

L.6 JT and CDCP Diagnostics under Gravity

Remark 16 (JT Diagnostics versus Global Certification under Gravity). Under the gravity implementation, a zero Jia–Tarski gap and interval-certified sign checks are treated as

Table 19: Verification Diagnostics for the Gravity Cloud Implementation

Specification	Cell fit	ρ	maxK	JT gap	Active	Notes
Preferred gravity baseline	200/210	0.1106	0.2355	0	123	chosen baseline
Higher- χ robustness ($\chi = 0.45$)	200/210	0.1119	0.2448	0	125	recall-heavy robustness

Notes: Localization is discussed separately in Subsection L.5; it is not reported here as a global analytic invariant-set proof. The computed JT gap is reported as a diagnostic for the solved gravity specification, not as a standalone certificate that the baseline CES global CDCP logic carries over unchanged.

diagnostics for the solved single-agent subproblems at the calibrated specification. They should not be read as theorem-level guarantees that the baseline CES certification logic carries over unchanged unless the relevant monotone or single-crossing structure is separately verified.

Subsection L.5 records the localization diagnostic qualitatively. Table 19 summarizes the calibrated numerical diagnostics that support the preferred gravity baseline and its leading robustness checks.

L.7 HQ-Distance Extension

The baseline gravity layer models the *site-to-market* leg of serviceability through $\psi_{fn}(I_f)$, but collapses the *headquarters-to-site* leg into the scalar firm wedge g_f . A firm headquartered in Seattle deploying at a São Paulo site faces coordination frictions (time zones, compliance, engineering pipelines) that a firm headquartered in São Paulo would not face; without an explicit HQ channel, any such friction is absorbed into g_f . This subsection records a small extension that restores symmetry with the site-to-market gravity object, along with the falsification diagnostics it was designed to support.

Let $d_{\text{HQ}(f),i}$ denote the great-circle distance in kilometers between firm f 's HQ city and site i , and define the HQ-friction object

$$\text{HQfric}(f, i) = \log\left(1 + \frac{d_{\text{HQ}(f),i}}{1000}\right),$$

which mirrors the log-distance primitive used in the site-to-market gravity score. The

extension introduces two HQ-gravity channels, each loading on $\text{HQfric}(f, i)$:

Channel F (fixed cost): $\phi_{fi} = \phi_i \cdot g_f \cdot w_i \cdot \exp(\xi_f \cdot \text{HQfric}(f, i))$,

Channel V (variable cost): $a_{fi} = a_f \cdot \exp(-\xi_v \cdot \text{HQfric}(f, i))$.

Channel F loads an HQ-distance wedge onto the deployment fixed cost; Channel V follows the Head–Mayer (2019) *Brands in Motion* mechanism, in which intangible productivity decays with distance from the headquarters and enters the Fréchet cost index as a site-dependent productivity shifter. The HQ coordinates are {Seattle, Redmond, Mountain View, Austin, Hangzhou} for {AWS, Microsoft, Google, Oracle, Alibaba}.

Identification and grid. With four US-headquartered firms and a single Chinese-headquartered firm, ξ_f is identified primarily off Alibaba’s footprint asymmetry. We therefore do not estimate ξ_f off a thin cross-section. Instead we report a plausibility grid, fixing ξ_v at the Head–Mayer preferred distance elasticity ($\xi_v = 0.15$) and at the null ($\xi_v = 0$), and sweeping $\xi_f \in \{-0.10, 0, +0.10\}$. At each grid point, (ϕ_i, g_f) are re-calibrated by the masked routine used for the baseline, so the extension affects the reported g_f vector and cell fit but keeps every other moment and calibration target unchanged. This is deliberately a falsification design rather than a point estimate: if the mechanism claim depends on g_f absorbing an HQ-distance channel, the AB-vs-CES contrast should move materially across the grid.

Readout. Table 20 records the grid. Three observations:

(i) *The AB-vs-CES mechanism is robust to the HQ extension.* Across all six grid points, the CES counterfactual loses between 19 and 29 active firm–market pairs and reduces aggregate profit by 30.2%–32.3%. This matches the baseline gravity CF2 verdict ($\Delta \text{ active} = -28, \Delta \text{ profit} = -33.8\%$, Subsection L.1) and the main-text cloud quantification (−30.1%). Endogenous markups sustain entry regardless of which HQ channel is loaded.

(ii) *The Head–Mayer variable-cost channel modestly improves in-sample fit.* Turning on $\xi_v = 0.15$ raises correct cells from 198 to 199 out of 210 (at $\xi_f = 0$) and reduces the fit-objective by roughly 10%. The improvement is small because the baseline already fits 198 of 210 cells and because the HQ-distance object is effectively a single-contrast (US vs. China) regressor at $F = 5$.

Table 20: HQ-Distance Extension: Grid of (ξ_v, ξ_f) Re-calibrations

ξ_v	ξ_f	Cell fit	Active (AB)	Δ active (CES–AB)	Δ profit (CES/AB)	g_{BABA}
0	–0.10	198/210	127	–27	–31.4%	0.638
0	0.00	198/210	127	–25	–31.6%	0.517
0	+0.10	197/210	124	–19	–32.3%	0.445
0.15	–0.10	199/210	128	–29	–30.2%	0.763
0.15	0.00	199/210	128	–26	–30.4%	0.677
0.15	+0.10	199/210	126	–23	–30.8%	0.619

Notes: At each grid point, (ϕ_i, g_f) are re-calibrated via the masked routine used for the baseline gravity specification, with $\text{HQfric}(f, i) = \log(1 + d_{\text{HQ}(f),i}/1000)$ and the HQ coordinates reported above. ξ_v is fixed externally at the Head–Mayer (2019) distance elasticity ($\xi_v = 0.15$) or at the null ($\xi_v = 0$). ξ_f is not estimated: it is swept on a plausibility grid because the single US/China HQ split gives thin cross-sectional identification. Cell fit is out of $210 = 5 \times 42$. Active pairs, Δ active, and Δ profit refer to the AB-vs-CES counterfactual holding gravity and calibrated fixed costs fixed. g_{BABA} is the deployment wedge of the sole non-US-headquartered firm.

(iii) g_{BABA} and ξ_v are partially substitutes. Under the null ($\xi_v = \xi_f = 0$), $g_{\text{BABA}} = 0.517$. Turning on the Head–Mayer variable-cost channel ($\xi_v = 0.15, \xi_f = 0$) raises g_{BABA} to 0.677: part of Alibaba’s estimated deployment wedge is reallocated to the variable-cost HQ-decay channel once that channel is present in the calibration. The fixed-cost channel ξ_f moves g_{BABA} in the opposite direction—increasing ξ_f tightens the HQ wedge directly and shrinks g_f to compensate—which is consistent with each channel entering the Fréchet-indexed fixed-cost block multiplicatively.

Verdict. The HQ-distance extension does not overturn the mechanism claim in CF2. The AB-vs-CES profit decline remains tightly clustered around 30–32%, and the active-pair response stays in the –19 to –29 range, including under the Head–Mayer preferred distance elasticity. The extension does, however, partially relabel g_{BABA} : roughly a third of the Alibaba wedge is absorbed by the Head–Mayer variable-cost channel once it is included. A single HQ split is too thin to sharpen ξ_f into a point estimate—the grid is best read as a robustness sweep rather than a regression—and we flag this as an identification caveat rather than attempt to report a profile-likelihood confidence interval from a five-firm cross-section.

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